

Connectivity Bounds in Graphs and Beyond

Erdős Problem 915, from Proof to Search

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Foreword

This thesis studies how many edges or arcs a network can contain when no pair of vertices is allowed to have too many independent routes between them. The starting point is Erdős Problem 915, an extremal question about edge-disjoint paths in simple graphs. The work is told as a journey. It begins with the variants whose answers can be proved cleanly by hand, watches those proofs grow longer and more delicate as the model changes, and at the point where a single argument no longer suffices it turns to a unified program: one representation for every variant, an exact connectivity checker built on maximum flow (a standard routine for finding the most independent routes between two points), a certificate engine that turns finite checks into proofs, and a temperature-guided search that hunts for extremal graphs in the space of all graphs.

I am grateful to my supervisor, Prof. Stijn Cambie, and my daily advisor for their guidance, corrections, and mathematical feedback throughout the project. I thank my promotor, Prof. Jan Goedgebeur, whose suggestion of the `nauty` generation pipeline sharply accelerated the directed enumeration. I also thank the people who discussed early versions of the arguments, checked examples, or helped clarify the presentation. Any remaining mistakes are my own.

Contribution Statement

The author was responsible for the literature study, the formulation of the variants, the design and implementation of the unified program, the construction and verification of the examples, the writing of the proofs included in the thesis, and the final $\text{\LaTeX}$ implementation. Results from the literature are cited explicitly, in particular the theorems of Mader, Leonard, Sørensen-Thomassen, Menger, Gomory-Hu, Baranyai, the hypergraph connectivity results of Bérczi, Chandrasekaran, Király and Kulkarni, and the probabilistic connectivity results used for random graphs.

The connectivity checker computes local connectivity exactly through maximum flow. The certificate engine yields a genuine upper-bound proof for a fixed size whenever it reports an optimal solution with zero gap over a finite domain. The temperature-guided search is a discovery tool: it produces constructions and therefore lower bounds, proposes conjectures, and rediscovers known extremal graphs, but it never certifies an upper bound on its own. Every general statement in the thesis rests on a hand proof or a cited theorem. Computational searches are supporting evidence, not substitutes for proofs. AI language models, including Claude Code, ChatGPT, and Gemini, were used as tools for literature search assistance, $\text{\LaTeX}$ editing, proofreading, and exploring proof drafts. Every mathematical statement and proof included in the thesis was checked by the author, who takes full responsibility for the content.

The following new results are contributed by this thesis. The hypergraph vertex problem is solved completely at $m = 2$ and $m = 3$ for all uniformities r and all n (Theorems A.34 and A.37): the vertex-disjoint and edge-disjoint extremal values coincide, equalling $\lfloor (m-1)(n-1)/(r-1) \rfloor$. The $m = 3$ proof rests on a new rank bound for incidence graphs, derived via Tutte's triconnected decomposition (Lemma A.36). For the directed multigraph problem at $m = 3$, an unconditional attachment lemma (Lemma A.26) and an odd-step theorem (Theorem A.28, Remark A.29) reduce the entire problem for all n to two finite machine-checkable statements about multigraphs on seven vertices, without requiring the unproved minimum-degree conjecture.

Short Summary

Erdős Problem 915, posed by Bollobás and Erdős [4], asks for the maximum number of edges in a graph on n vertices such that no two vertices are connected by m edge-disjoint paths. In modern notation this asks for

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1\}.$$

Mader proved that, for simple graphs and $n \geq m \geq 2$,

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

This thesis revisits this theorem and studies the corresponding extremal problem under three independent changes: the graph model (simple graphs, multigraphs, and hypergraphs), the path-separation notion (edge-disjoint or internally vertex-disjoint), and the presence or absence of directions. Sampled random graphs $G(n, p)$ run alongside throughout as a lens on the typical case rather than as a separate model.

The thesis is organised as a journey from hand proofs to computation. The early variants are settled rigorously: the multigraph value $L_m(n) = (m-1)(n-1)$, the random-graph threshold $p^* = m/n$, the equality of the edge and vertex problems for $m \leq 4$, and the first divergence at $m = 5$, where Sørensen and Thomassen prove $k_5(n) = \lfloor 8n/3 \rfloor - 3$. The directed case forces the turn: the value $\ell_2^{\text{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved by a long induction, and for $m \geq 3$ the hand proof gives way to a conjecture supported by an explicit counterexample. The centerpiece is then a unified program. One multiplicity-matrix representation models almost all twelve variants, an exact maximum-flow checker measures local connectivity, a cut-counting optimisation certifies upper bounds for small sizes, and a simulated-annealing search guided by edge sensitivity discovers extremal graphs. The program rediscovers the double star, the hub, the one-directional bipartite digraph, and the augmented-bipartite construction as extremal graphs, certifies the directed multigraph values $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$ for $n \leq 6$, and frames the remaining open problems on the directed frontier.

The interplay then pays back in the other direction. The hypergraph vertex problem is settled completely at $m = 2$ and $m = 3$: $k_m^{(r)}(n) = \lfloor (m-1)(n-1)/(r-1) \rfloor$ for every uniformity r , the $m = 3$ case through a new rank bound on incidence graphs proved by way of Tutte's triconnected decomposition. On the directed side, an unconditional attachment lemma and a conditional odd-step theorem reduce the entire directed multigraph problem at $m = 3$, value and extremal characterisation for all n , to two finite, machine-checkable statements about multigraphs on seven vertices, and directed hypergraphs receive their first bounds, quadratic in n .

Abbreviations and Symbols

GH Gomory-Hu

w.h.p. with high probability

$G = (V, E)$	graph with vertex set V and edge set E
$D = (V, A)$	directed graph with vertex set V and arc set A
n	number of vertices
m	forbidden-connectivity parameter, with all local connectivities at most $m - 1$
$\lambda_G(u, v)$	maximum number of edge-disjoint u - v paths
$\lambda_G^{\max}$	maximum local edge-connectivity over all vertex pairs
$\kappa_G(u, v)$	maximum number of internally vertex-disjoint u - v paths
$\kappa_G^{\max}$	maximum local vertex-connectivity over all vertex pairs
$\ell_m(n)$	simple undirected edge-disjoint extremal function
$L_m(n)$	undirected multigraph edge-disjoint extremal function
$k_m(n)$	simple undirected vertex-disjoint extremal function
$\ell_m^{\text{dir}}(n)$	simple directed arc-disjoint extremal function
$L_m^{\text{dir}}(n)$	directed multigraph arc-disjoint extremal function
$k_m^{\text{dir}}(n)$	simple directed vertex-disjoint extremal function
$\mu(u, v)$	multiplicity of an edge or arc between u and v
$\sigma(e)$	edge sensitivity $\lambda^{\max}(G) - \lambda^{\max}(G - e)$
$G(n, p)$	binomial random graph

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Chapter 1

The Problem and Its Base Cases

A network is useful not only because it connects, but because it connects in more than one way. A single road between two towns is a liability, and a second independent road is insurance. The mathematical content of that intuition is the number of routes between two points that share no segment, and the question this thesis circles is how many connections a network can hold before some pair of points is forced to have too many such routes. This chapter builds the question from nothing, settles the variants whose proofs are short enough to read in an afternoon, and then follows the same question along three axes until the proofs stretch past breaking. By the end the case for a machine has made itself.

1.1 Graphs and routes

To state the question precisely we need only a few ideas. A *graph* G is a finite set of points, called *vertices*, together with a set of links, called *edges*, each joining two of the vertices. One pictures the vertices as dots and the edges as lines between them.

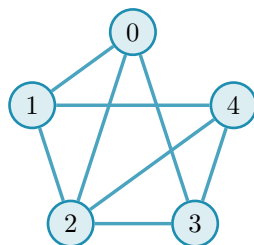


Figure 1.1. A simple graph on five vertices: K_5 with the two edges $\{0, 4\}$ and $\{1, 3\}$ removed, leaving eight of the original ten edges. The five dots are the vertices and the lines between them are the edges. Since at most one line joins any given pair and no edge loops back to its own endpoint, this is a simple graph.

A graph is the bare skeleton of a network: the vertices are the towns or the devices, and the edges are the roads or the links between them. We write n for the number of vertices, and the *degree* of a vertex is the number of edges meeting it. Unless we say otherwise our graphs are *simple*, meaning that at most one edge joins any given pair of vertices and its endpoints are distinct.

A *path* from a vertex u to a vertex v is a sequence of distinct vertices that starts at u , ends at v , and joins each consecutive pair by an edge. It is the formal version of a route through the network. Two paths between the same endpoints are *edge-disjoint* if no edge belongs to both, so that the

loss of a single edge can break at most one of them. The largest number of pairwise edge-disjoint paths between u and v is their *local edge-connectivity* $\lambda_G(u, v)$, the count of genuinely independent routes the network offers to that pair. The pair with the most independent routes sets the ceiling for the whole graph:

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v),$$

the largest local edge-connectivity anywhere in G . A theorem of Menger [15], recalled in [Chapter A](#), gives this quantity a second face: $\lambda_G(u, v)$ also equals the least number of edges one must remove to cut every route from u to v . Many independent routes between a pair is exactly the same thing as a costly cut between them.

1.2 Erdős Problem 915

The constraint at the heart of this thesis is a ceiling on connectivity. Requiring $\lambda_G^{\max} \leq m - 1$ says that no pair of vertices is joined by as many as m edge-disjoint paths, or equivalently that every pair can be severed by removing fewer than m edges. The question is how many edges a graph can hold while staying under that ceiling.

Question 1.1 (Erdős Problem 915 [5]). How many edges can a graph on n vertices have if no two of its vertices are joined by m edge-disjoint paths?

Writing the answer as an extremal function,

$$\ell_m(n) = \max\{ |E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1 \},$$

the problem asks for $\ell_m(n)$, the most edges possible under the ceiling. Mader settled the simple-graph case completely.

Theorem 1.2 (Mader [12]). *For all $n \geq m \geq 2$,*

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

An elegant proof by way of the Gomory–Hu tree, found independently of Mader, is given in full in [Chapter A](#), together with a characterisation of the graphs that attain equality ([Theorem A.11](#)). We single out the one identity that makes it work, because the same idea reappears when the program of [Chapter 2](#) stores connectivities as a tree (graph without cycles): a Gomory–Hu tree of G is a weighted tree on the same vertex set in which $\lambda_G(u, v)$ equals the lightest edge on the tree path from u to v . All $\binom{n}{2}$ local connectivities are therefore encoded by $n - 1$ numbers, and $\lambda_G^{\max}$ is simply the heaviest edge of the tree.

The whole argument ([Chapter A](#)) is one act of double counting on that tree. Each of the $n - 1$ tree edges stands for a minimum cut of G , and an original edge of G is charged once for every tree edge lying between its endpoints, so summing the tree-edge weights counts every edge of G

at least once and, crucially, counts most of them at least twice. The only edges charged a single time are those joining tree-adjacent vertices, and a *simple* graph has at most $n - 1$ of those, one per tree edge. Every other edge is charged twice. Since each tree weight is a local connectivity and so at most $m - 1$, the two counts meet at $2|E(G)| - (n - 1) \leq (m - 1)(n - 1)$, which rearranges to the floor. The factor of two, and with it the gap between Mader's $\lfloor m(n - 1)/2 \rfloor$ and the multigraph value $(m - 1)(n - 1)$ proved below, is exactly the dividend of simplicity, the one place the argument uses that no pair carries a parallel edge.

1.3 Three axes of variation

Mader's theorem is the floor we build on, and everything afterwards is a variation on it. There are three independent ways to change the rules, and they generate the landscape this thesis traverses.

The first axis is the *model*, drawn in Figure 1.2. A simple graph allows at most one edge between two vertices. A *multigraph* allows parallel edges, recorded by a multiplicity $\mu(u, v)$, so a single pair can already carry several independent routes. A *hypergraph* lets an edge join more than two vertices at once: a hyperedge is a set of vertices rather than a pair, and a route may pass through it between any two of the vertices it contains. These three models are the three rows of every comparison figure in this thesis.

The second axis is the *direction*: arcs may be one-way, and then a network can be lopsided in a manner no undirected graph can imitate. Although it might seem a generalisation of the undirected setting, they are distinct in complexity and methodology: the arcs $u \rightarrow v$ and $v \rightarrow u$ offer a round trip while the edge uv does not. This distinction fades when we no longer distinguish paths by their edges, as in our next choice of interpretation.

The third axis is the *separation*: routes may be required to avoid sharing an edge, as above, or the stricter *internally vertex-disjoint*, sharing no intermediate vertex, which gives the local vertex-connectivity $\kappa_G(u, v)$ and its maximum $\kappa_G^{\max}$. Whitney's inequality $\kappa_G(u, v) \leq \lambda_G(u, v)$ [20]

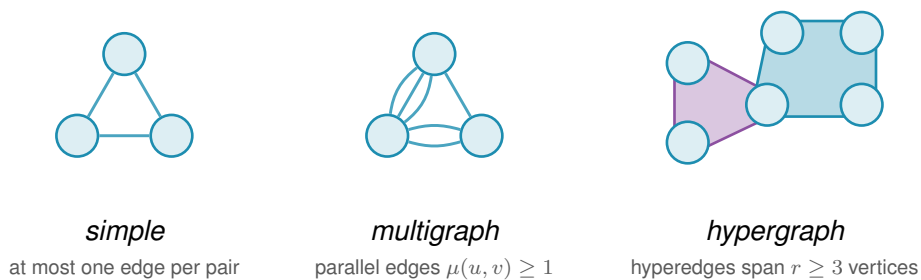


Figure 1.2. The first axis of variation, the *model*. In a *simple* graph each pair of vertices is joined by at most one edge. A *multigraph* permits parallel edges, here a triple and a double, so the multiplicity $\mu(u, v)$ counts how many copies join a pair. A *hypergraph* replaces pairwise edges by hyperedges: the violet region spans three vertices and the blue region spans four, and they share a single vertex where the two colours blend, illustrating hyperedges of different sizes that may overlap. When all hyperedges have the same size r we call it an r -uniform hypergraph. The same connectivity question, how many independent routes a pair can be forced to share, is asked of all three.

records that avoiding shared vertices is at least as demanding as avoiding shared edges, and [Figure 1.3](#) shows the gap on a graph where two routes are edge-disjoint yet forced through one common vertex.

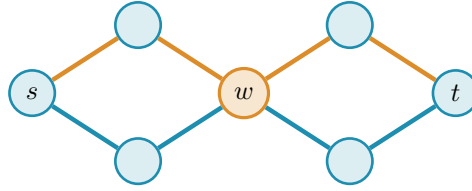


Figure 1.3. The third axis, the *separation*. The two s – t routes drawn here, one orange and one blue, share no edge, so they count as two edge-disjoint paths and $\lambda(s, t) = 2$. Both pass through the single vertex w , however, so they are *not* internally vertex-disjoint, and since every s – t route must cross w we have $\kappa(s, t) = 1$. This is Whitney’s inequality $\kappa \leq \lambda$ made concrete: the gap between the two separations is exactly what the vertex-disjoint problem measures, and the rest of the chapter shows it stays shut until $m = 5$.

Three models, two directions, two separations: twelve versions of one question. The thesis is the attempt to answer as many of them as honest mathematics allows, and to build a machine for the rest. Some answers are short and we prove them here. Some answers are long, and their length is the reason [Chapter 2](#) exists. The rest of this chapter collects the variants whose proofs are short enough to belong in the body, so that the reader meets the easy slope before the climb.

1.4 Cases with short proofs

Allowing parallel edges removes the parity subtlety that produces the floor in Mader’s theorem, and the answer becomes exactly linear. Write $L_m(n)$ for the multigraph analogue of $\ell_m(n)$.

Theorem 1.3. *For multigraphs on n vertices, $L_m(n) = (m - 1)(n - 1)$.*

Proof. For the lower bound, take a spanning tree of K_n , that is, a connected cycle-free subgraph spanning all n vertices, which necessarily has $n - 1$ edges, and give every tree edge multiplicity $m - 1$. The result has $(m - 1)(n - 1)$ edges. Any two vertices are joined by a unique path in the tree, of some length $k \geq 1$. The minimum edge cut between them is the lightest tree edge on that path, of capacity $m - 1$, so $\lambda(u, v) = m - 1$ and $\lambda^{\max} = m - 1$.

For the upper bound, suppose $\lambda_G^{\max} \leq m - 1$ and build a Gomory–Hu tree T of G ([Theorem A.2](#)). By that theorem, each tree edge e corresponds to a minimum cut of G whose capacity $c(e) = \lambda_G(u, v) \leq m - 1$ for the pair (u, v) it represents. Every original edge $\{x, y\}$ of G crosses the cut corresponding to the minimum-weight tree edge on the tree path from x to y , so summing cut capacities over the $n - 1$ tree edges counts each original edge at least once. Hence $|E(G)| \leq \sum_{e \in T} c(e) \leq (m - 1)(n - 1)$. $\square$

The vertex-disjoint question is just as quick at the first nontrivial value. Forbidding two internally vertex-disjoint paths is a strong constraint: it forbids cycles.

Proposition 1.4. *For simple graphs, $k_2(n) = n - 1$, attained exactly by trees.*

Proof. If G contains a cycle, then any two vertices on that cycle are joined by the two paths of the cycle, which are internally vertex-disjoint, so $\kappa_G^{\max} \geq 2$. Hence $\kappa_G^{\max} \leq 1$ forces G to be acyclic, and an acyclic graph on n vertices has at most $n - 1$ edges, with equality for trees. A tree indeed has $\kappa^{\max} = 1$, since removing the single intermediate vertex on any path disconnects its endpoints. $\square$

Here $k_m(n)$ denotes the vertex-disjoint extremal function, the counterpart of $\ell_m(n)$ for $\kappa^{\max}$. We have just seen $k_2(n) = n - 1 = \ell_2(n)$. The agreement of the edge and vertex problems is no accident at small m , and the moment it breaks is the first place the proofs begin to strain.

1.5 The first divergence: vertex-disjoint paths

Replacing edge-disjoint by internally vertex-disjoint routes changes nothing at first. Leonard proved that the edge and vertex problems give the same answer while m stays small.

Theorem 1.5 (Leonard [11]). *For simple graphs and $m \leq 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

For $m \leq 4$ the extremal graphs for the two problems coincide: Whitney's inequality $\kappa \leq \lambda$ is tight where it matters and the harder vertex constraint costs nothing (Figure 1.4, the three overlapping blue curves). One is tempted to expect this forever. It fails at the very next value, and the failure is sharp.

Theorem 1.6 (Sørensen-Thomassen [18]). *For simple graphs and all $n \geq 10$,*

$$k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 3 > \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n).$$

The threshold $n \geq 10$ is where the inequality first holds with strict positivity: at $n = 10$ the vertex formula gives 23 and the edge formula gives 22, while at $n = 7$ they tie at 15. For $n \leq 9$ one must verify the formula directly against the structural classification Sørensen-Thomassen supply. The asymptotic formula equals or overestimates $k_5(n)$ for small values, so the gap between k_5 and ℓ_5 is not yet guaranteed by the formula alone.

At $m = 5$ the vertex problem genuinely diverges from the edge problem, visible in Figure 1.4 as the shaded gap that opens between the dashed blue and solid orange curves. The reason is structural. Leonard's method leans on the extremal graphs being assembled from cliques glued along shared cliques. A k -tree is the graph built by starting from a complete graph K_{k+1} and repeatedly adding a new vertex adjacent to exactly the vertices of some k -clique already present. These chordal scaffolds are exactly the building blocks the method uses. Through $m = 4$ those scaffolds are forced, and counting their edges gives the bound. At $m = 5$ they are no longer the unique extremal shape. Additional configurations appear, and the clean count breaks. Sørensen

and Thomassen recover the exact value by a delicate structural induction, which we cite rather than reproduce. For $m \geq 6$ even that much is unknown: the vertex-disjoint problem has stood open since 1974, and the extremal structure is genuinely not understood. It is the one direction this thesis does not attempt to force, and [Chapter 4](#) explains why.

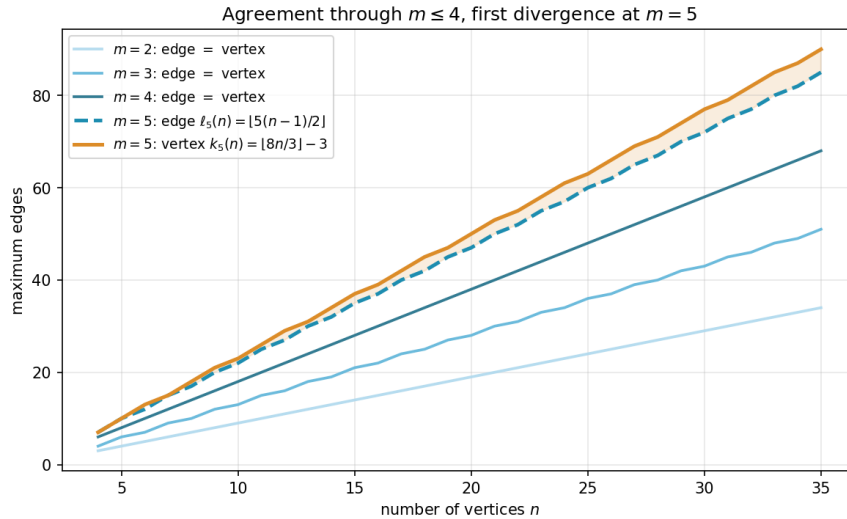


Figure 1.4. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). Through $m = 4$ the two problems give identical answers, and at $m = 5$ the vertex formula $k_5(n) = \lfloor 8n/3 \rfloor - 3$ separates above the edge formula $\ell_5(n) = \lfloor 5(n-1)/2 \rfloor$, with the gap shaded.

1.6 The directed case and the quadratic phenomenon

Direction breaks a symmetry that undirected graphs never had. In an undirected graph the number of edges is tied to the connectivity through every vertex's degree. In a digraph a vertex can have many arcs in and none out, and a whole graph can lean entirely one way. Send every arc from one half of the vertices to the other and nothing comes back: between any two vertices there is at most one route, yet the number of arcs is quadratic.

Construction 1.7 (One-directional bipartite digraph). Split V into parts A and B with $|A| = \lfloor n/2 \rfloor$ and $|B| = \lceil n/2 \rceil$, and include every arc from A to B . All arcs run from A to B , and none within A , within B , or from B back to A . Since B has no outgoing arcs, a vertex in A cannot reach any other vertex of A , and two vertices of B cannot communicate at all. For any $a \in A$ and $b \in B$ the only directed path from a to b is the single arc $a \rightarrow b$, so $\lambda(a, b) = 1$ and $\lambda^{\max} = 1$. The arc count is $|A| \cdot |B| = \lfloor n^2/4 \rfloor$.

[Figure 1.5](#) draws the construction. Every arrowhead lands in B and none leaves it, so the picture is a wall of one-way traffic: from any a on the left there is exactly one way to reach any b on the right, the single arc between them, and there is no way at all to travel between two vertices on the same side. The arc count $|A| \cdot |B|$ is quadratic, yet no pair carries more than one route. This is the lopsidedness direction buys, and it has no undirected analogue. It competes with a

second construction built around a single centre.

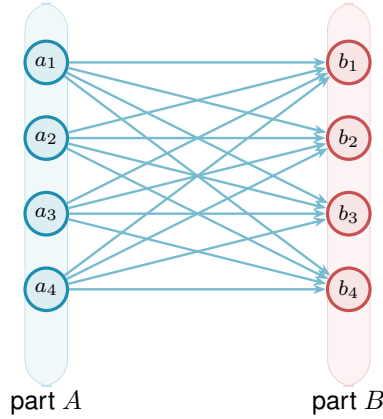


Figure 1.5. The one-directional complete bipartite digraph $B_{k,k}$ of Construction 1.7, drawn here at $n = 8$ with $|A| = |B| = 4$. Every one of the $|A| \cdot |B| = \lfloor n^2/4 \rfloor$ arcs runs left to right, from A to B . There are no arcs inside A , inside B , or back from B to A . Because B has no outgoing arcs and A no incoming ones, the only route from any a_i to any b_j is the single arc $a_i \rightarrow b_j$, so every pair has local edge-connectivity 1 while the arc count grows quadratically.

In a digraph each vertex has two distinct degree counts. The *out-degree* $d^+(v)$ is the number of arcs leaving v , and the *in-degree* $d^-(v)$ is the number entering v . A vertex with high out-degree fans many arcs outward, and one with high in-degree receives many. Keeping both counts balanced is the structural idea behind the next construction.

Before the two competing constructions, it helps to see the directed measure on one concrete digraph. Figure 1.6 shows a pair joined by two arc-disjoint routes, the smallest arc-cut that separates them, and the degree bound that limits both. This is the object every directed argument later in the thesis reasons about.

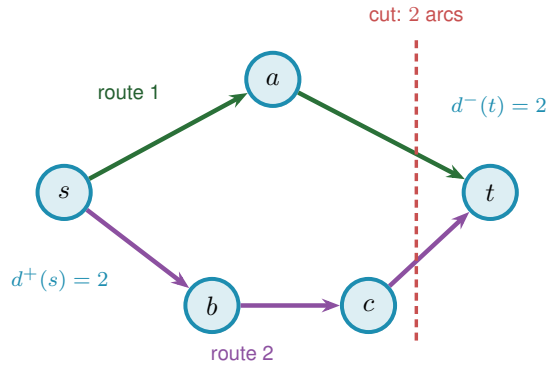


Figure 1.6. The directed measure on one small digraph. The pair (s, t) carries two arc-disjoint directed routes, $s \rightarrow a \rightarrow t$ (green) and $s \rightarrow b \rightarrow c \rightarrow t$ (purple). Deleting the two arcs that enter t separates s from t , so by Menger (Theorem A.1) the local connectivity $\lambda(s, t) = 2$ equals the size of that minimum arc-cut (red dashed). The degree bound $\lambda(s, t) \leq \min(d^+(s), d^-(t))$ is tight here, since both out-degree of s and in-degree of t equal 2. Arc-disjoint routes counted against the smallest arc-cut are the whole of what the directed problems measure.

Construction 1.8 (Directed hub). For $n \geq m \geq 2$, take a hub vertex h and label the remaining $N = n - 1$ vertices $0, 1, \dots, N - 1$. Add the $2N$ hub arcs $h \rightarrow i$ and $i \rightarrow h$ for every i , and on

the non-hub vertices add the circulant arcs $i \rightarrow i + j \pmod{N}$ for $j = 1, \dots, m - 2$. The total is $2N + (m - 2)N = m(n - 1)$ arcs. Every non-hub vertex has $d^+ = d^- = 1 + (m - 2) = m - 1$, and any ordered pair of vertices contains a non-hub endpoint, so the directed degree bound (Lemma A.3) gives $\lambda^{\max} \leq m - 1$.

At $m = 2$ this construction is easiest to picture: the circulant layer is empty and what remains is the *bidirected star* of Figure 1.7, a hub joined to each spoke by a matched pair of opposite arcs. Every spoke can reach every other only by going in to the hub and back out, two hops, so no pair is ever joined by two independent routes, and the arc total is exactly $2(n - 1)$. For $m \geq 3$ not all $m(n - 1)$ arcs can touch the hub, since a simple digraph has at most $2(n - 1)$ arcs incident to one vertex: the surplus lives in the circulant layers among the spokes, tuned so that no degree exceeds the budget. The two constructions compete on different scales, linear against quadratic, and which one wins depends on n . At $m = 2$ the competition resolves exactly.

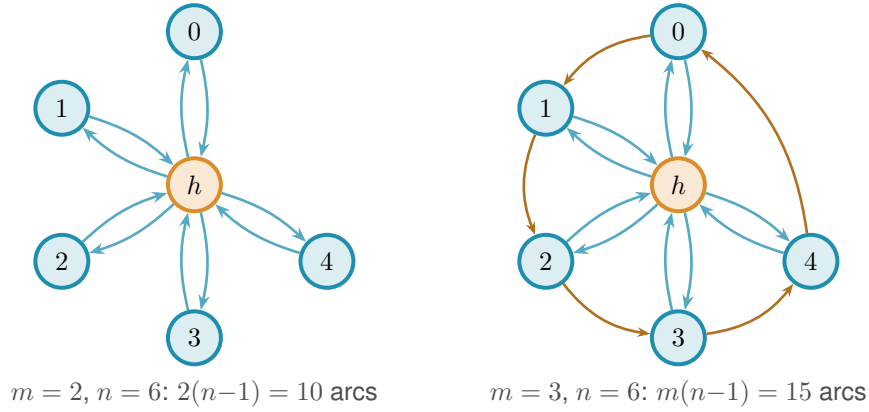


Figure 1.7. The directed hub at $n = 6$ for two values of m . *Left* ($m = 2$): the bidirected star. The hub h is joined to each spoke by one arc in each direction, giving $2(n - 1) = 10$ arcs. No circulant layer is added, so all spoke communication routes through h , so $\lambda^{\max} = 1$. *Right* ($m = 3$): a single circulant layer is added among the five spokes. The blue arcs are the same hub-spoke pairs as on the left. The orange arcs are that one circulant layer, the cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$, so each spoke has $d^+ = d^- = m - 1 = 2$. The total is $m(n - 1) = 15$ arcs and $\lambda^{\max} = 2$. For $m \geq 3$ the spokes are no longer isolated from one another, and tracing all the routes becomes the argument that fills the rest of the chapter.

Theorem 1.9 (Directed arc value at $m = 2$). *For simple digraphs,*

$$\ell_2^{\text{dir}}(n) = \max\left(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

The hub construction leads for $n < 7$, the one-directional bipartite construction leads for $n > 7$, and the two tie at $n = 7$, where both reach 12 arcs.

The lower bound is the better of the two constructions. The upper bound is where the trouble starts.

Idea of the upper bound at $m = 2$. The value $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is a contest between a linear term and a quadratic one, and which leads is pure arithmetic: $2(n-1) \geq \lfloor n^2/4 \rfloor$ precisely for $n \leq 7$, with equality at $n = 7$, where both give 12. So M follows the bidirected star up to the crossover and the bipartite digraph past it, and $n = 7$ is the single value where the two constructions tie. The proof that no digraph beats $M(n)$ is an induction on n (Chapter A). Delete a vertex of *minimum* total degree: since the degrees sum to twice the arc count, an averaging argument bounds that degree by $2a/n$, where a is the number of arcs, and the remaining digraph on $n-1$ vertices still respects the ceiling, so the inductive bound applies to it. Feeding the two facts together gives $a \leq \frac{n}{n-2} \lfloor (n-1)^2/4 \rfloor$, and a short parity check shows the right-hand side never exceeds $\lfloor n^2/4 \rfloor$ once $n \geq 8$: for even n it lands exactly on the bipartite value, and for odd n a sub-unit slack is swallowed by the fact that a is an integer. The averaging closes the step with no separate hub case, which is why the difficulty is not depth but bulk. Nothing in the argument is deep. It is a careful walk through two parity regimes, with the base cases $n \leq 7$ settled by exhaustive computer search rather than by hand. That experience of a correct but long and almost entirely mechanical case-check is exactly what motivates the chapters that follow. One genuinely wants to hand the case-checking to a machine.

The vertex-disjoint version of the same question needs no separate machinery at $m = 2$. Vertex-disjoint routes are at most as numerous as arc-disjoint ones, and the two constructions above keep $\kappa^{\max} = 1$ as readily as $\lambda^{\max} = 1$, so the value transfers exactly.

Theorem 1.10 (Directed vertex value at $m = 2$). *For simple digraphs,*

$$k_2^{\text{dir}}(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right) = \ell_2^{\text{dir}}(n).$$

1.7 The failure of direct proof for $m \geq 3$

For $m \geq 3$ the induction no longer closes, and the natural guess is wrong. The natural initial conjecture was that the two constructions above continued to be the whole story, giving $\max(m(n-1), \lfloor n^2/4 \rfloor)$. A single graph on ten vertices refutes it.

Remark 1.11 (A thirty-arc counterexample). Take $A = \{a_1, \dots, a_5\}$ and $B = \{b_1, \dots, b_5\}$ with all 25 arcs from A to B , and add a directed five-cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B . This digraph has 30 arcs. Every vertex of B has exactly one extra in-arc from inside B , so two arc-disjoint routes from a_i to b_j exist: the direct arc $a_i \rightarrow b_j$, and the detour $a_i \rightarrow b_{j-1} \rightarrow b_j$ using the 5-cycle predecessor of b_j (indices mod 5). For the matching upper bound, the only in-arcs of b_j reachable on a path from a_i are the direct arc $a_i \rightarrow b_j$ and the cycle arc $b_{j-1} \rightarrow b_j$. Removing both disconnects a_i from b_j , while neither alone does, so the minimum cut has size 2. Hence $\lambda(a_i, b_j) = 2$. Routes between two vertices of B never leave B and give $\lambda(b_i, b_j) = 1$. Hence $\lambda^{\max} = 2$, so the graph is feasible at $m = 3$, yet $30 > \max(3 \cdot 9, 25) = 27$.

Figure 1.8 draws this thirty-arc digraph and traces the mechanism. The twenty-five faint arcs are the complete one-directional bipartite layer from A to B , the construction we already

understand. On top of them the orange five-cycle gives each vertex of B a single extra in-arc from inside B . That one extra arc is exactly what doubles the connectivity. For the marked pair a_1, b_2 the two heavy routes show how: the blue arc is the direct hop $a_1 \rightarrow b_2$, and the green path $a_1 \rightarrow b_1 \rightarrow b_2$ borrows the cycle arc $b_1 \rightarrow b_2$ for its second step. The two share no arc, so $\lambda(a_1, b_2) = 2$, and the same pair of routes exists for every a_i, b_j . Five extra arcs lift the whole digraph from one independent route per pair to two, and the thirty arcs overshoot the old prediction of twenty-seven.

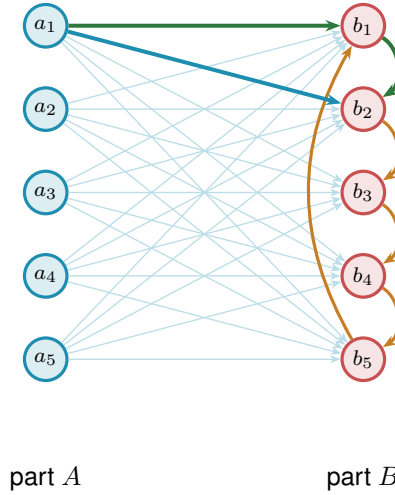


Figure 1.8. The thirty-arc counterexample of Remark 1.11: the augmented bipartite digraph (Construction 1.12) at $m = 3$, $n = 10$, $|A| = |B| = 5$. The 25 faint arcs are the complete one-directional layer from A to B , and the 5 orange arcs form the directed cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B , one extra in-arc per vertex of B . The two heavy arcs and the green path show the pair a_1, b_2 carrying 2 arc-disjoint routes: the direct arc $a_1 \rightarrow b_2$ (blue) and the detour $a_1 \rightarrow b_1 \rightarrow b_2$ (green) through the cycle predecessor b_1 . Hence $\lambda^{\max} = 2$, feasible at $m = 3$, while $30 > \max(3 \cdot 9, 25) = 27$.

The counterexample generalises: the five-cycle gave every vertex of B exactly one extra in-arc from inside B , and the budget for such arcs is $m - 2$ per vertex, not one. Figure 1.8 is the instance $m = 3$, where that budget is one and a single cycle suffices. For larger m each b_j collects $m - 2$ cyclic predecessors and the same picture acquires more concentric chords inside B .

Construction 1.12 (Augmented bipartite digraph). For $m \geq 2$ and $n \geq m$, set $|B| = \lceil (n + m - 2)/2 \rceil$ and $|A| = n - |B| = \lfloor (n - m + 2)/2 \rfloor$, include every arc from A to B , and inside B give each vertex b_j the $m - 2$ in-arcs from its cyclic predecessors $b_{j-1}, \dots, b_{j-(m-2)}$ (indices mod $|B|$). The total is $\lfloor (n + m - 2)^2/4 \rfloor$ arcs. For the connectivity, fix $a \in A$ and $b \in B$: the direct arc and the $m - 2$ two-step detours $a \rightarrow b' \rightarrow b$ through the in-neighbours b' of b inside B are $m - 1$ arc-disjoint routes, and they are all there are, because the only in-arcs of b reachable from a are exactly those $m - 1$ arcs, which therefore form a cut. Pairs inside B are limited by the in-degree $m - 2$, pairs inside A and pairs from B to A have no routes at all, so $\lambda^{\max} = m - 1$. At $m = 2$ the inside of B is empty and the construction reduces to Construction 1.7, with the balanced partition $(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$. At $m = 3$ the formula gives $|B| = \lceil (n + 1)/2 \rceil$, which coincides

with $\lceil n/2 \rceil$ at odd n but gives $|B| = n/2 + 1$ at even n . Both the balanced and the shifted partition achieve the same arc count $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$ in this case, so the two are interchangeable at $m = 3$. The shift to a strictly larger B is structurally significant only at $m \geq 4$, where the balanced partition is suboptimal. At $m = 3$, $n = 10$ it is the thirty-arc counterexample of [Remark 1.11](#), drawn in [Figure 1.8](#) (using the equivalent balanced partition $|A| = |B| = 5$).

This is the corrected shape, and it leads to the standing conjecture for the directed arc problem.

Conjecture 1.13. *For simple digraphs and $m \geq 2$,*

$$\ell_m^{\text{dir}}(n) = \max \left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \right).$$

For $m = 2$ this is [Theorem 1.9](#). For $m = 3$ the formula $\lfloor (n+1)^2/4 \rfloor$ equals $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$, so the conjecture reduces to the previous formula for $m = 2$ and $m = 3$. For $m = 3$, $n = 10$ the second branch gives $\lfloor 121/4 \rfloor = 30$, matching [Remark 1.11](#). For $m \geq 4$ the new formula and the old balanced-partition count $\lfloor n^2/4 \rfloor + (m-2) \lceil n/2 \rceil$ no longer agree: the quadratic branches already differ (for $m = 4$ the new one is larger by exactly one at every even n), but the hub term $m(n-1)$ dominates both at small n , so the full conjectured value first overtakes the old one at $n = 12$ for $m = 4$. The construction of [Construction 1.12](#) with the shifted partition witnesses the difference. The lower bound holds for all m , witnessed by [Construction 1.8](#) on the first branch and [Construction 1.12](#) on the second. The upper bound is open, and [Chapter 4](#) returns to it with what structural progress the machine and the hand can jointly supply.

1.8 The hypergraph variant

The third model deserves its own short treatment, not because its base case is hard but because it is the one variant whose objects are stored and manipulated differently from all the rest, and it is worth meeting that difference now rather than at the moment of computation. A hypergraph keeps no multiplicity matrix, only the list of its hyperedges, each a set of r vertices. We fix the edge size r , following the extremal literature, for a structural reason: under a connectivity ceiling a small edge is always at least as efficient per route as a large one, so a hypergraph free to mix sizes would maximise its edge count by using pairs only, and the question would collapse back to the graph case. Fixing r is what makes the hypergraph question a new question. The route between two vertices is a *Berge path*, which alternates vertices and hyperedges so that each consecutive pair of vertices lies in the hyperedge listed between them, and two such routes are edge-disjoint when they share no hyperedge. Concretely, in a three-uniform hypergraph a route from u to w might be the single step $u, \{u, x, w\}, w$ that crosses one hyperedge, or a longer alternation $u, \{u, x, y\}, y, \{y, z, w\}, w$ that switches hyperedge at the intermediate vertex y . With routes redefined this way the same Gomory–Hu machinery that proved the multigraph value carries over, and the base case has the same linear shape. [Figure 1.9](#) draws such a

two-hyperedge route.

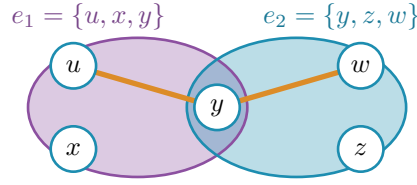


Figure 1.9. A Berge path from u to w in a three-uniform hypergraph, crossing two hyperedges. Each hyperedge (shaded in its own colour) is a set of three vertices, and they meet at the shared vertex y where the two colours blend, and the route (orange) enters e_1 at u and leaves it at y , then enters e_2 at y and leaves it at w . The two steps use distinct hyperedges, so this counts as one Berge path. A second route edge-disjoint from it would have to avoid both e_1 and e_2 entirely. Unlike an ordinary edge, a single hyperedge touches all $\binom{3}{2} = 3$ pairs of its vertices at once, which is the bookkeeping the checker of [Chapter 2](#) must handle.

A single hyperedge may serve only one route, so two Berge routes that share no hyperedge are edge-disjoint, the hypergraph counterpart of two edge-disjoint paths in a graph. [Figure 1.10](#) shows a larger hypergraph in which one pair carries two such routes at once.

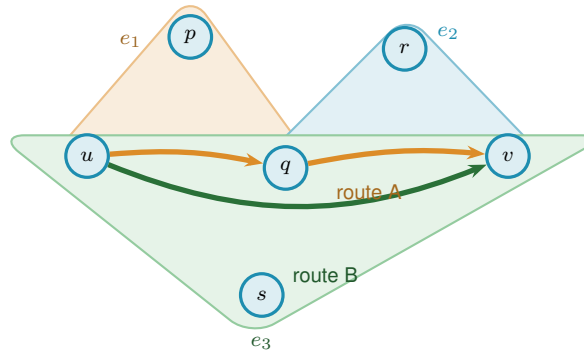


Figure 1.10. Two Berge routes between u and v that share no hyperedge. Route A (orange) threads e_1 then e_2 , hopping hyperedge at q , and route B (green) crosses the single hyperedge e_3 . The two use disjoint hyperedge sets $\{e_1, e_2\}$ and $\{e_3\}$, so they are Berge edge-disjoint and $\lambda(u, v) \geq 2$. Any third route would have to reuse one of the three hyperedges. This is the hypergraph form of edge-disjoint routes, with a whole hyperedge, not a single edge, consumed by each route.

Proposition 1.14 (Hypergraph edge bound). *An r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m - 1$ has at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. A simple hypergraph attains the bound whenever $m - 1 \leq \binom{n-2}{r-2}$, and a multihypergraph attains it whenever $(r - 1) \mid n - 1$.*

The proof, given in [Chapter A](#), runs the multigraph argument through the hypergraph Gomory–Hu theorem of Bérczi, Chandrasekaran, Király and Kulkarni [2], charging each hyperedge to the at least $r - 1$ tree cuts it crosses. The star hypertree attains the $m = 2$ case, a hub joined to disjoint blocks of $r - 1$ vertices, with Berge connectivity one everywhere and $\lfloor (n - 1)/(r - 1) \rfloor$ hyperedges, exactly as a tree attained the multigraph bound. One genuine surprise is worth recording here: for $r \geq 3$ the bound is attained by *simple* hypergraphs, with no repeated hyperedge

(Theorem A.32). In the graph case simplicity is exactly what separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$. One edge size higher, it costs nothing.

What the proposition does not say is how to *measure* the Berge connectivity of an arbitrary hypergraph, and that measurement is genuinely different from the graph case. The difference is not the one-route-per-edge rule, which graphs share, but what an edge touches: an ordinary edge joins one pair, so edge-disjointness is policed pair by pair, while a single hyperedge touches all $\binom{r}{2}$ pairs among its r vertices at once, and two routes may legitimately want to enter it between different pairs. The measurement must let any one of those pairs use the hyperedge while barring all the others, simultaneously, for every hyperedge. That bookkeeping is the one new piece of machinery the next chapter has to build, and it is the reason the hypergraph rejoins the family only once the checker is in place.

1.9 Random graphs as intuition

The short proofs above hinted that the edge and vertex problems travel together for a while, agreeing through $m = 4$ before the split at $m = 5$. That separation is a statement about extremal graphs, the densest ones the ceiling allows, and it is worth asking whether the same split can be seen in ordinary graphs drawn at random, with no extremal tuning at all. The random model is the natural place to build intuition, because it shows the qualitative behaviour of the connectivity notions before any construction is pushed to its limit, and we will return to it whenever a picture of the typical graph clarifies a result. It is not a detour from the program to come, either: a sampled $G(n, p)$ is an ordinary graph, measured by the very same connectivity checker the next chapter builds for every other variant, so the random model is simply the one pipeline turned to *observe* the typical case rather than to prove or to discover the extremal one.

Each of the $\binom{n}{2}$ possible edges of $G(n, p)$ is included independently with probability p , and on the resulting graph we read off both binding connectivities, the edge version $\lambda_G^{\max}$ and the vertex version $\kappa_G^{\max}$, from the very same sample. Whitney's inequality $\kappa_G^{\max} \leq \lambda_G^{\max}$ guarantees the vertex value never exceeds the edge one, but it is silent on how often the two actually come apart. Figure 1.11 looks directly at the distributions, fixing $n = 16$ and histogramming both connectivities over many samples at three densities. Two things are visible at once. As the density rises the whole distribution marches rightward, from a binding connectivity around five at $p = \frac{1}{4}$ to around thirteen at $p = \frac{3}{4}$, so by $n = 16$ the typical graph already sits well past the $m = 4$ at which the proofs agreed. And comparing the edge row to the vertex row, the vertex distribution sits at or just to the left of the edge one, never to its right, which is Whitney's inequality seen across a whole distribution rather than a single graph. The two pull apart on a fraction of samples that shrinks as the density grows, from about a fifth of the samples at $p = \frac{1}{4}$ to only a handful by $p = \frac{3}{4}$. The reason is structural. For $\kappa_G^{\max}$ to fall below $\lambda_G^{\max}$ some pair must carry more edge-disjoint routes than internally vertex-disjoint ones, which happens only when several of those routes are forced through one shared intermediate vertex. In a sparse graph routes are scarce and a single low-degree vertex can be exactly such a bottleneck, but as p grows every

vertex gains degree and the graph offers many alternative intermediate vertices, so independent routes almost never need to reuse one. In the dense limit the graph approaches the complete graph, where $\kappa^{\max} = \lambda^{\max} = n - 1$ for every pair and the gap closes entirely. The divergence the thesis records at $m = 5$ is therefore not an artefact of the extremal construction. It is already there, in miniature, in the random model.

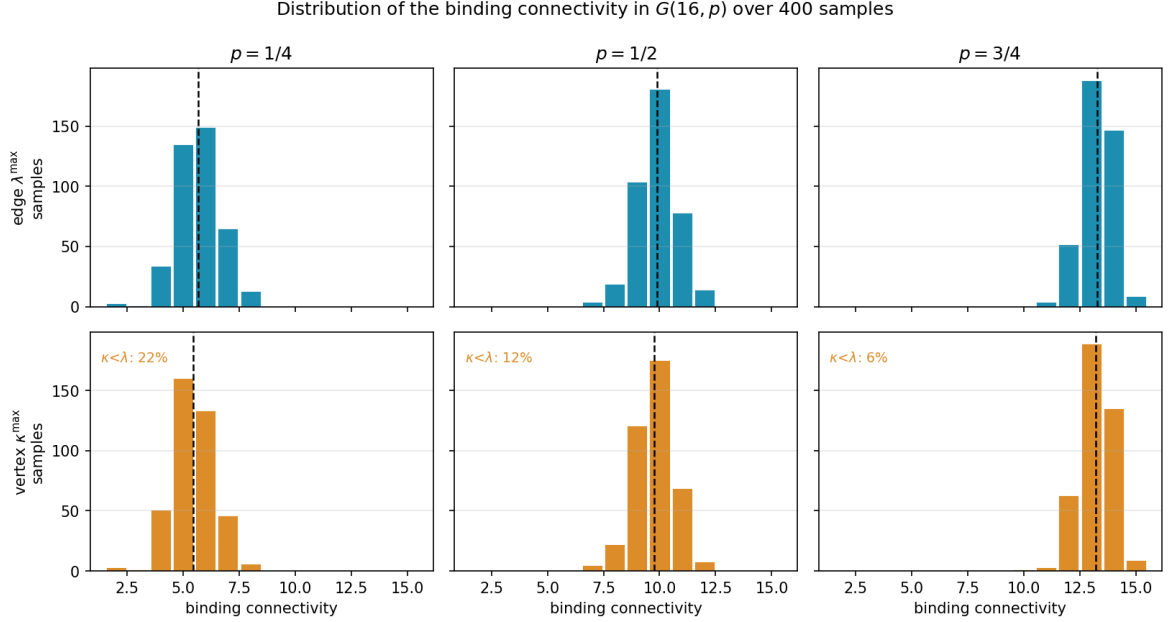


Figure 1.11. Distributions of the binding connectivity in $G(16, p)$ over 400 samples, at densities $p = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. The top row histograms the edge-connectivity $\lambda^{\max}$, the bottom row the vertex-connectivity $\kappa^{\max}$, measured on the same samples, with the mean marked by a dashed line and the fraction of samples on which $\kappa^{\max} < \lambda^{\max}$ printed on each vertex panel. Reading a column top to bottom, the vertex distribution sits at or to the left of the edge one, which is Whitney's inequality, and the gap is largest at the sparsest density (22% of samples) and closes as p grows (6%). All six panels share one axis range.

The random model also supplies a landmark of a different flavour. Below a sharp density the forbidden configuration is almost surely absent, and above it almost surely present, and the location of that threshold is the same for every one of the connectivity notions above.

Theorem 1.15 (Appearance threshold). *Let $m = m(n) \geq 2$ grow with n so that $m / \ln n \rightarrow \infty$ and $m = o(n)$, and let $c > 0$ be a constant. In the binomial random graph $G(n, p)$ with $p = c \cdot m/n$,*

$$\lim_{n \rightarrow \infty} \Pr[\lambda^{\max} \geq m] = \begin{cases} 1 & \text{if } c > 1, \\ 0 & \text{if } c < 1, \end{cases}$$

so the threshold for the appearance of a pair with local edge-connectivity at least m is $p^ = m/n$. The same threshold governs local vertex-connectivity and the directed analogues (Remark A.42).*

The proof is a concentration argument with two halves (Chapter A). Above the threshold the edge count concentrates past Mader's bound, so Theorem 1.2 itself forces a high pair into

existence. Below it the maximum degree stays under m , and the degree bound caps every local connectivity. Both halves lean on the same growth condition $m/\ln n \rightarrow \infty$, which is what lets a union bound over the n vertices close. The collapse of complexity is what deserves emphasis: in the probabilistic limit the distinctions between edges and vertices, directions and models all wash out, and the single number m/n decides the matter, a universality the synthesis of [Chapter 4](#) records again.

One limitation must be stated plainly, lest the threshold be read as evidence about the problem this thesis actually studies. The growth hypothesis $m/\ln n \rightarrow \infty$ forces m to climb with n , so the theorem says nothing whatever about a fixed small m – the regime of every exact value above, where m is 2, 3, or 4 and n runs to infinity around it. There the union bound is vacuous and the appearance of a dense pair is governed by rare low-degree configurations that need finer, Poisson-type tools. The random model therefore enters as contrast and not as evidence: it shows how cleanly the question collapses in one limit, which throws into relief how stubbornly it resists in the fixed- m limit the rest of the thesis must attack by hand and by machine.

1.10 Where this work sits

It is worth pausing, before the machine, to place the question and the method against the two traditions they descend from: the extremal graph theory that posed the problem and the computational mathematics that will help settle it.

The mathematical lineage is the older of the two. The problem belongs to the family of extremal connectivity questions opened by Bollobás and Erdős, who asked how dense a graph can be while every pair of vertices stays below a fixed connectivity [4], logged today as Erdős Problem 915 [5]. Mader’s theorem [12] closed the simple undirected edge case with the clean floor $\lfloor m(n-1)/2 \rfloor$ recalled in [Theorem 1.2](#), and it is the floor every variant in this thesis is measured against. The vertex-disjoint branch has a longer and rougher history: Leonard [11] showed the edge and vertex problems coincide through $m \leq 4$, and Sørensen and Thomassen [18] found the first divergence at $m = 5$ with $k_5(n) = \lfloor 8n/3 \rfloor - 3$, after which the exact values $k_m(n)$ remain unknown. That gap between a settled edge case and a stalled vertex case is exactly the terrain [Chapter 1](#) has been mapping, and the directed and hypergraph axes extend it further. The thesis adds to this lineage not a single new closed form but a uniform way of pushing each axis as far as exact computation allows.

The computational tradition this thesis’s pipeline belongs to is the practice of settling combinatorial questions by machine in a way a human can audit, and it splits into a few schools worth contrasting with the certify-then-prove loop built next. The oldest is *exhaustive generation*: enumerate every object of a given size up to isomorphism and check each one. McKay and Piperno’s *nauty/geng* toolkit [13] is the standard engine for this, and its flagship results include exact small Ramsey numbers such as $R(4, 5) = 25$ [14]. The enumeration of [Chapter 2](#) is the same idea, run on the twelve variants of Problem 915. The second school is *SAT-assisted proof*, in which a question is encoded in propositional logic and a solver emits a machine-checkable

certificate: Heule, Kullmann and Marek’s resolution of the Boolean Pythagorean triples problem [9], whose verified proof runs to some two hundred terabytes, and Heule’s determination of Schur number five [8] are the canonical examples of certified solver output at scale. The third is *flag algebras*, Razborov’s framework [17] that converts asymptotic extremal bounds into semidefinite-programming certificates, the dominant tool for density problems in the large- n limit. Closest of all to the method here is the fourth, *LP/ILP-based certification*, where a feasibility or optimality argument over the rationals or integers is exhibited as a linear program whose optimal dual is itself the proof. This is precisely the form the cut-counting certifier of Chapter 2 takes.

Against that backdrop, what is distinctive here is consolidation rather than a new solver. A single checker measures connectivity exactly across all twelve problem variants on one data model, so search and certification share one codebase rather than living in separate tools. The certificates it emits are exact rational or integral objects, not floating-point estimates, so an upper bound it reports is a proof and not evidence. And because the same program both hunts for dense graphs and certifies the bounds they suggest, the discover step and the prove step are two modes of one loop, which is the arrangement the next chapter builds. What makes the consolidation honest is that it never blurs three kinds of knowledge: a value can be *proved* (by hand, or by exhausting every graph of a size), *certified* (by the same optimisation, reaching a vertex or two further), or only *conjectured* (witnessed by a dense graph the search exhibits, with no matching upper bound). This trichotomy, named in Section 3.6 and obeyed by the summary of Chapter 4, is the through-line of everything that follows.

Computer note

We are now stuck in three different ways: a fifty-year-old open problem at $m \geq 6$ for vertices, a long proof we would rather not check by hand at $m = 2$ for arcs, and a conjecture with no proof at all at $m \geq 3$. The base cases of the long induction were verified by exhaustive search, and the thirty-arc counterexample was found the same way. The next chapter builds a single program that models all of these cases at once, measures their connectivity exactly, and proves the small-case upper bounds outright.

Chapter 2

Certifying Bounds by Machine

The last chapter ended in three different kinds of stuckness, and all three share a shape. The objects are graphs of one flavour or another, the constraint is always a ceiling on connectivity, and the work is always either measuring a given graph or checking many small ones. That shared shape is what a single program can exploit. This chapter builds it, and then uses it for one purpose only: to *prove*. It gives every variant one representation, measures connectivity exactly through a max-flow checker, checks small cases by honest enumeration, and turns the finite checks the proofs of [Chapter 1](#) relied on into a single auditable certifier. The hunt for dense examples, which proves nothing on its own, waits until [Chapter 3](#).

A word on what the program is for. It is a microscope, not a source of final answers. It measures connectivity exactly, so what it reports about a given graph is a fact. It enumerates graphs of a fixed size, so for that size it can prove a theorem. It will later discover dense graphs, which are only lower bounds, but not here. Each of these has a different epistemic weight, and the program is built so that the three never get confused. [Figure 2.1](#) draws this spine: a single *model* holds every variant, one exact *measure* reads connectivity off it, and from that measurement a *prove* step bounds a fixed size from above, with the *discover* step of the next chapter and a final *observe* step for the random model added later. Every routine introduced from here on is shown as a small card carrying its name, its inputs and output, a one-line account of how it works, and a coloured badge naming its place in the spine. The function bodies are not reproduced, because the point is the interface, not the code.

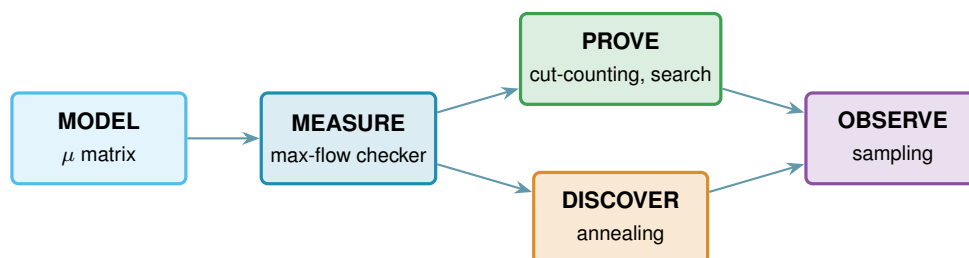


Figure 2.1. The architecture as one pipeline. A single data model, the multiplicity matrix μ , feeds an exact measurement, the max-flow checker. That measurement feeds the two bound-producing roles: a prover for upper bounds (the cut-counting optimisation and the exhaustive search), built in this chapter, and a discoverer for lower bounds (simulated annealing), built in [Chapter 3](#). A final sampling role studies the random model. The badge colours on the code cards below match the boxes here.

2.1 The solver

Everything in this chapter is a single function, `solve`, and it helps to hold its most basic version in mind from the start, because every later idea is a refinement of it and nothing is ever a separate program for a separate case. Given n and m , the basic strategy is as plain as it sounds. Walk through every graph on n vertices, measure the largest local connectivity $\lambda^{\max}$ of each with a max-flow computation, and keep the densest graph that stays at or below the ceiling $m - 1$. Because it has looked at every graph, the count it returns is the exact answer, not an estimate.

That one loop already settles the smallest cases, and the rest of the chapter is built on top of it without ever leaving `solve`. We first make the measurement step trustworthy and quick, via the connectivity checker and a Gomory–Hu shortcut for undirected graphs. We then make the enumeration step skip whole families of hopeless graphs, via a pruned search. Finally, for the directed multigraph case, we replace enumeration altogether with one small optimisation that proves every m at once, which is the certifier. Each of these is the same `solve` doing less work for the same guarantee, and the choice among them is made automatically from the booleans it receives.

```
solve(n, m, directed, simple, separation, exhaustive, maxSeconds)
```

DRIVER

in the variant booleans `directed` and `simple`, the separation (edge or vertex), n , m , a mode, and a time budget

out a value with an honest label: exact, lower bound, or upper bound

how picks the method the case allows: brute-force enumeration, the pruned search, or the cut-counting certifier when proving, or the temperature search when discovering ([Chapter 3](#)). It always respects the budget, and it is the one function the rest of the thesis ever calls.

2.2 A single representation for all variants

Two of the three axes from [Chapter 1](#), the model and the direction, are properties of the object, and a single data structure holds them all. Of the three structural models, simple graphs and multigraphs fit the matrix directly, while the hypergraph keeps the separate storage promised in [Chapter 1](#) and rejoins at the end of this chapter. The hypergraph could in principle be pressed into the same shape as a higher-order tensor, an r -dimensional array marking which r -sets are present, but that array is almost entirely zero and costs n^r entries, so the program keeps the compact list of hyperedges instead and pays nothing for the empty cells. Sampled $G(n, p)$ graphs are ordinary simple graphs and therefore use this same representation without any change. A graph or digraph on n vertices is stored as an integer matrix μ , where $\mu(u, v)$ is the multiplicity of the edge or arc from u to v . A simple graph is the case $\mu \in \{0, 1\}$, an undirected graph satisfies $\mu = \mu^\top$, and a multigraph allows larger entries. The third axis, edge against vertex separation, is a property of the question rather than the object, so it lives in how we measure μ , not in μ itself.

Figure 2.2 illustrates the encoding.

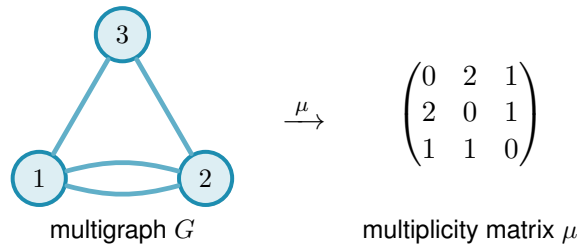


Figure 2.2. A multigraph on three vertices and its multiplicity matrix μ . Entry $\mu(u, v)$ records the number of parallel edges between u and v . The double edge between vertices 1 and 2 gives $\mu(1, 2) = 2$. A simple graph is the case $\mu \in \{0, 1\}$, an undirected graph satisfies $\mu = \mu^\top$, and a directed graph allows $\mu(u, v) \neq \mu(v, u)$.

Graph(<i>n</i> , <i>directed</i> , <i>simple</i>)		MODEL
in	the vertex count n and the two booleans	
out	a graph whose whole state is the matrix μ	
how	one $n \times n$ integer matrix is the object, an undirected graph keeps $\mu = \mu^\top$, and a simple graph caps every entry at one.	

Keeping a single representation is what lets one measurement routine, one certifier, and one search serve every case. Nothing downstream asks which of the twelve problems it is solving. It asks only the two booleans (directed or undirected, simple or multi) and the matrix. Everything that builds or edits a graph is a thin wrapper around μ that respects those booleans, and these operations are so routine that they need no individual exposition, and Table 2.1 lists them together.

Operation	Meaning
addEdge(<i>G</i> , <i>u</i> , <i>v</i> , <i>k</i>)	add k parallel edges or arcs $u \rightarrow v$ (a simple graph saturates at one)
removeEdge(<i>G</i> , <i>u</i> , <i>v</i> , <i>k</i>)	remove k copies, never below zero
setMultiplicity(<i>G</i> , <i>u</i> , <i>v</i> , <i>c</i>)	set $\mu(u, v) = c$ directly
multiplicity(<i>G</i> , <i>u</i> , <i>v</i>)	read $\mu(u, v)$
hasEdge(<i>G</i> , <i>u</i> , <i>v</i>)	whether $\mu(u, v) > 0$
degree(<i>G</i> , <i>v</i>)	total degree of v (in- plus out-degree if directed)
edgeCount(<i>G</i>)	number of edges or arcs, counted with multiplicity
edges(<i>G</i>), vertices(<i>G</i>)	iterate the present edges, and the vertex set
copy(<i>G</i>)	an independent duplicate

Table 2.1. The routine operations on the graph model. Each is a one-line wrapper that reads or writes the matrix μ while keeping an undirected graph symmetric and a simple graph binary. They are bookkeeping, and the chapter’s attention belongs to the routines that follow.

2.3 The connectivity checker built on maximum flow

Menger’s theorem turns the connectivity question into a flow computation. A maximum flow is the standard way to ask how much can be pushed from one point to another through a network

in which every link has a limited capacity. Menger's theorem says the maximum number of edge-disjoint $u-v$ paths equals the minimum $u-v$ edge cut, and that number is exactly the value of such a maximum flow once each multiplicity $\mu(u, v)$ is read as the capacity of that link. The routine that runs this flow and reports the route count is the connectivity *checker*. It is the single most important routine in the program: it is the only place graph theory meets computation, and every later claim about connectivity passes through it.

`localEdgeConnectivity(G, s, t)  $\rightarrow \mathbb{N}$`

MEASURE

in a graph G and an ordered pair (s, t)

out $\lambda_G(s, t)$, the maximum number of edge-disjoint $s-t$ paths

how a single maximum flow on the network whose arc capacities are the multiplicities μ , so by Menger the flow value is the path count.

`maxEdgeConnectivity(G)  $\rightarrow \mathbb{N}$`

MEASURE

in a graph G

out $\lambda^{\max}(G)$, the largest local edge-connectivity over all pairs

how one max-flow per pair (ordered if directed), take the largest. This is the quantity the edge problem caps at $m - 1$.

The vertex-disjoint version follows the same idea after one construction: split each vertex v into an in-copy v^{in} and an out-copy v^{out} joined by a single unit-capacity arc (Figure 2.3). Every arc $u \rightarrow v$ of the original graph becomes $u^{\text{out}} \rightarrow v^{\text{in}}$ in the split network. Routing through v now consumes the one unit of capacity on its internal arc, so two paths sharing v as an intermediate vertex compete for that unit and cannot coexist in any maximum flow. A max-flow from u^{out} to v^{in} in the split network therefore equals the number of internally vertex-disjoint $u-v$ paths in the original graph.

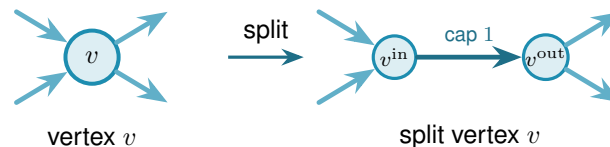


Figure 2.3. Vertex splitting for the vertex-disjoint checker. Incoming arcs attach to v^{in} and outgoing arcs leave from v^{out} , connected by a single unit-capacity arc. Any directed path that uses v as an intermediate vertex passes through this bottleneck, and two such paths would need two units of capacity and are therefore separated.

Figure 2.4 applies the split to a whole digraph. Two arc-disjoint routes are forced through one shared vertex w , so on the split network both must cross the single unit-capacity arc inside w , which caps the number of internally vertex-disjoint routes below the edge-disjoint count.

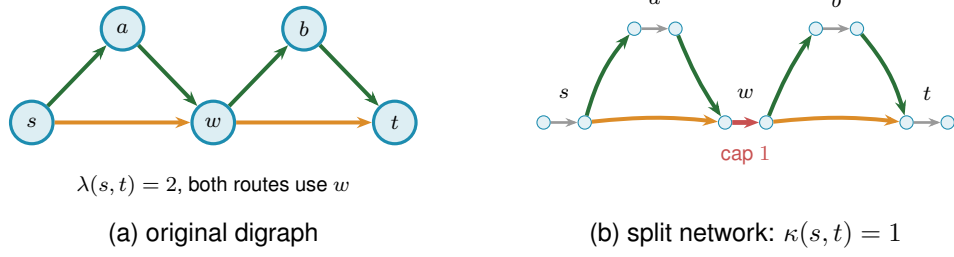


Figure 2.4. The vertex-split checker on a larger example. **(a)** A digraph in which the two arc-disjoint s – t routes, $s \rightarrow w \rightarrow t$ (orange) and $s \rightarrow a \rightarrow w \rightarrow b \rightarrow t$ (green), both pass through w , so $\lambda(s, t) = 2$. **(b)** The split network replaces every vertex V by an arc $V^{\text{in}} \rightarrow V^{\text{out}}$ of capacity one. Both routes must now traverse the single unit arc inside w (red), which carries only one unit, so at most one internally vertex-disjoint route gets through and $\kappa(s, t) = 1$. Splitting every vertex this way turns the vertex-disjoint count into an ordinary maximum flow.

`maxVertexConnectivity(G) → ℕ`

MEASURE

in a graph G

out $\kappa^{\max}(G)$, the largest local vertex-connectivity over all pairs

how the same sweep as the edge checker, but each flow runs on the split network of [Figure 2.3](#), so the unit internal arcs count internally vertex-disjoint paths. Parallel edges are given capacity one here, so they never raise κ .

The two checkers are the entire interface between graph theory and computation in this thesis. As a sanity check, applied to the directed double star on n vertices, the edge checker reports $\lambda^{\max} = 1$, and the arc count $2(n - 1)$ matches the proved values $\ell_2^{\text{dir}}(n) = 2, 4, 6, 8$ for $n = 2, 3, 4, 5$.

The same checker answers two different questions. Throughout this chapter it runs in its exact-value mode, returning $\lambda^{\max}$ or $\kappa^{\max}$ on the nose, and every proved or reported number rests on that exact reading. The search of [Chapter 3](#) needs only the cheaper yes-or-no question “is the binding connectivity above the bound?”, so there the same flow runs in a capped mode that stops as soon as it has seen one route too many. The capped mode never changes a reported value: it is a faster way to ask for a single bit, and [Chapter 3](#) shows that the search built on it reproduces the exact search step for step.

The Gomory–Hu tree from [Chapter 1](#) reappears here as an efficiency, and it is the honest kind. For undirected graphs all $\binom{n}{2}$ edge-connectivities are read off a single weighted tree, so $\lambda^{\max}$ is the heaviest tree edge rather than the maximum over $\binom{n}{2}$ separate flows. The speed-up is real, but it is bought with a theorem, not a heuristic.

`maxEdgeConnectivityViaTree(G) → ℕ`

MEASURE

in an undirected graph G

out $\lambda^{\max}(G)$, identical to the pairwise sweep

how build one Gomory–Hu tree, then return its heaviest edge, which costs $n - 1$ flows in place of $\binom{n}{2}$ and cross-checks the checker.

The hypergraph is the one model that never used the multiplicity matrix, since it is stored as a plain list of hyperedges, each one a set of r vertices. The checker still has to read it, and the trick is to do the same flow computation as before on a slightly larger network that we build on the spot. The difficulty is that one hyperedge ties several vertices together at once and, by the definition of a Berge route in [Chapter 1](#), it may be used by only one route at a time, so we cannot simply treat it as a bundle of ordinary links. Instead, for every hyperedge we add one extra helper point and connect each of that hyperedge’s member vertices to it ([Figure 2.5](#)). A route that wants to use the hyperedge now enters through one member vertex, passes through the helper point, and leaves through another. By letting only a single route pass through each helper point, we capture exactly the rule that one hyperedge serves one route. Counting routes between two vertices is then the very same maximum flow as before, run on this enlarged network, and when a hyperedge happens to hold just two vertices the helper point collapses back to an ordinary link, so the answer agrees with the graph case. That the flow on the enlarged network counts exactly the hyperedge-disjoint Berge routes is the hypergraph version of Menger’s theorem, proved in [Chapter A](#) ([Theorem A.30](#)), so the checker rests on a theorem rather than an analogy.

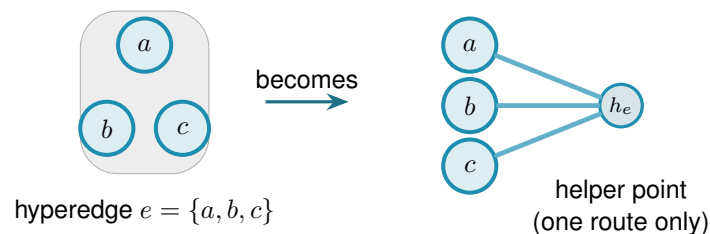


Figure 2.5. Measuring Berge connectivity by maximum flow. A hyperedge e joining several vertices (left) is replaced by a helper point h_e joined to each of its members (right). Because at most one route may pass through h_e , the helper point enforces that the hyperedge serves only one route at a time. Counting independent routes between two vertices is then an ordinary maximum flow on this enlarged network, and a two-vertex hyperedge collapses to a single ordinary edge.

[Figure 2.5](#) shows one hyperedge in isolation. [Figure 2.6](#) carries the same idea to a whole small hypergraph, where a Berge route that switches hyperedge partway becomes a single flow path threading two helper points in turn.

One way to picture the helper point: think of each hyperedge as a room with several doors. Entering through any door counts as using that room. Because only one path may pass through the helper node, at most one traveller can occupy the room at once, so the second traveller must find a route that never enters it. The same flow computation that counts edge-disjoint paths through ordinary links now counts routes that avoid sharing any room, which is exactly what

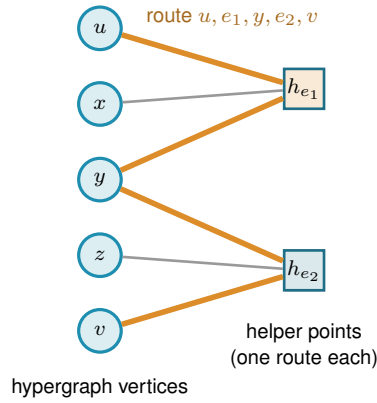


Figure 2.6. The same construction on a two-hyperedge hypergraph $e_1 = \{u, x, y\}$, $e_2 = \{y, z, v\}$. The Berge route that enters e_1 at u , switches hyperedge at the shared vertex y , and leaves e_2 at v becomes the single flow path $u \rightarrow h_{e_1} \rightarrow y \rightarrow h_{e_2} \rightarrow v$ (orange), entering each hyperedge through one member and leaving through another. Because each helper carries only one unit of flow, a second $u-v$ route would have to avoid both helpers, which is exactly the rule that one hyperedge serves one route.

Berge edge-disjointness means.

<code>maxHyperedgeConnectivity(H) → ℕ</code>		MEASURE
in	an r -uniform hypergraph H	
out	$\lambda^{\max}(H)$, the largest Berge edge-connectivity over all pairs	
how	build the enlarged network in which every hyperedge becomes a helper point that only one route may pass through, then take the same maximum flow as the graph checker. On two-vertex hyperedges it matches ordinary edge-connectivity.	

On the complete graph K_n the hypergraph checker returns $n - 1$, and on the complete three-uniform hypergraph $K_4^{(3)}$ it returns 3, larger than the two hyperedges through each pair because longer Berge routes also carry flow. The self-test confirms both of these values, so the construction is checked rather than merely asserted. The same helper-point network answers the other three hypergraph questions without alteration: splitting each ordinary vertex as in [Figure 2.3](#) counts internally vertex-disjoint Berge routes, and reading a hyperedge as a single tail feeding its remaining heads gives a directed version, so `solve` measures all four hypergraph variants through this one construction. With the checker in place the proposed value of [Proposition 1.14](#) can be tested on any small hypergraph the way every other variant is, which is the subject of the next section.

2.4 Checking every graph of a fixed size

The checker measures one graph. To bound every graph of a given size, the most direct method is the one the base cases of [Chapter 1](#) already needed: enumerate them all. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ on each off-diagonal cell, a multigraph a choice in

$\{0, \dots, m-1\}$, and a digraph varies every ordered pair rather than only the upper triangle. Walking that product, measuring each candidate with the checker, and keeping the densest feasible one settles the value exactly, and because it has looked at every graph it is a genuine upper bound, not evidence.

This is the first thing `solve` does when it is asked to prove a small case. With its exhaustive flag set it walks that product, measures each candidate with the checker, and keeps the densest feasible one. The value it returns is exact and exhausted, a true upper bound for that n , but only while the count of graphs stays small. No new routine is involved, only `solve` choosing to enumerate.

Run on the small cases this confirms the proved values directly, with no induction and no cleverness. [Table 2.2](#) collects a spread of variants where the enumeration finishes: in every row the machine returns exactly the value [Chapter 1](#) proved by hand. This is the first payoff of the uniform representation. One driver, `solve`, pointed at six different problems, settles all six.

Variant	m	n	<code>solve</code>	Theory
simple undirected, edge	2	5	4	$\lfloor m(n-1)/2 \rfloor = 4$
simple undirected, edge	3	5	6	$\lfloor m(n-1)/2 \rfloor = 6$
multigraph undirected, edge	3	5	8	$(m-1)(n-1) = 8$
simple undirected, vertex	2	6	5	$k_2(n) = n-1 = 5$
simple directed, arc	2	5	8	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 8$
simple directed, arc	2	6	10	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 10$

Table 2.2. Variants that converge under plain enumeration. For each small case `solve`, in its exhaustive mode, walks every graph of the variant, measures it with the checker, and returns the densest feasible one. Every value matches the hand proof of [Chapter 1](#) exactly.

The honesty of the method is also its limit. The number of graphs it must visit grows as $2^{\binom{n}{2}}$ for simple graphs and $2^{n(n-1)}$ for digraphs, and a cap of m on multiplicities raises the base to m . The enumeration is exact wherever it finishes, and it finishes only for the smallest sizes. [Chapter 3](#) opens with exactly how fast this wall arrives. For now the point is that the wall arrives precisely where [Chapter 1](#) needed help, at the directed base cases $n = 6, 7$, so the rest of this chapter is about reaching a little further than blind enumeration can.

2.5 Reaching further: a pruned search for the directed base cases

The base cases $n \leq 7$ of the directed $m = 2$ theorem ([Theorem 1.9](#)) are the finite checks the induction of [Chapter A](#) depends on, and they sit just past where blind enumeration of digraphs is tractable. They can be reached by enumerating more cleverly. Feasibility is monotone: adding an arc can never lower $\lambda^{\max}$, so once a partial digraph is infeasible every completion of it is infeasible too and the whole branch is abandoned at once. A counting bound prunes further, dropping any branch that cannot beat the best feasible digraph found so far. What remains is a branch and bound over simple digraphs with the connectivity checker as the feasibility test, exact and complete at these sizes where the blind walk is not.

Here `solve` changes its implementation without changing its interface. For a simple digraph it runs this branch and bound in place of the blind walk, still with the connectivity checker as the feasibility test, and so reaches the base cases $n \leq 7$ that blind enumeration cannot. The booleans alone decide which of the two it uses, and the caller never asks.

Run on the directed $m = 2$ problem `solve` returns the complete sequence 2, 4, 6, 8, 10, 12 for $n = 2, \dots, 7$, each proved complete at its size. These are exactly the base cases the induction requires, and with them the long proof of [Theorem 1.9](#) rests on a single auditable search rather than on anything done by hand. The exhaustive-search transcript is in [Chapter A](#).

2.6 Generating the directed cases faster

Both enumerations so far are written from first principles. The blind walk lists every labelled graph, and for the directed multigraph base cases that the $m = 3$ result reduces to ([Remark A.29](#)) the program runs a depth-first search over multiplicity matrices, pruning on feasibility and on a counting bound and keeping one representative per isomorphism class through a canonical relabelling. The canonical form holds the memory in check, but the cost that remains is time spent walking labelled partial graphs, and it is time, not memory, that sets the ceiling. The full classification at $n = 5$ costs this in-house search about 456 seconds.

A faster route was suggested by Prof. Jan Goedgebeur, an author of the `nauty` generators [13]. Rather than search, one generates. The tool `geng` lists the non-isomorphic undirected graphs on n vertices, and the program then decorates each one edge by edge with a directed multiplicity pair, so that the underlying support graph already carries the isomorphism reduction. The orientation tools `directg` and `watercluster2` are not used directly here: each produces a single orientation per edge and so cannot express the bidirected arcs and the repeated multiplicities the problem needs (the extremal graphs are themselves bidirected), which is exactly why the multiplicity decoration is layered in-house on top of `geng`. The expensive part, listing one representative per isomorphism class, now happens inside `geng`, where `nauty` is heavily optimised, rather than inside the search.

The two routes agree to the last graph. The generated pipeline reproduces the in-house enumeration exactly at every settled size, two non-isomorphic extremal families at $n = 4$ and three at $n = 5$, and it is substantially faster, because the only prunings it never has to repeat are the isomorphism reductions `geng` has already done. One difference matters for what comes next: where the in-house search leans on a conjectured counting bound past $n = 6$, the generated enumeration applies only proved bounds, so it is a sound complete search at every n . That makes it the practical route to the single finite case the $m = 3$ argument still waits on, the degree-bounded enumeration at $n = 7$ that is fact (b) of [Remark A.29](#).

2.7 Proving every m at once

The pruned search proves one (n, m) at a time. For the directed multigraph arc problem `solve` has a sharper implementation, one that proves every m at a fixed n in a single pass, and it is worth the extra machinery because it closes the directed multigraph case outright for the sizes it reaches.

The key is that m can be scaled out of the problem completely, so that it never appears in the computation at all. Take any directed multigraph that is feasible at level m , meaning $\lambda^{\max} \leq m-1$, and divide every multiplicity by $m-1$. The result is a matrix of weights $w(u, v) \in [0, 1]$, and since dividing all capacities by $m-1$ divides every flow value by the same factor, the maximum flow between each pair is now at most one. Its total weight is $\sum_{u \neq v} w(u, v) = |A|/(m-1)$, so a feasible multigraph with many arcs becomes a feasible weight matrix with large total weight, and vice versa. Writing $M^*(n)$ for the largest total weight any such matrix can carry,

$$L_m^{\text{dir}}(n) = (m-1) M^*(n), \quad M^*(n) = \max \left\{ \sum_{u \neq v} w(u, v) : \text{maxflow}(s, t; w) \leq 1 \text{ for all } s, t \right\}.$$

There is no m on the right-hand side. One solve for $M^*(n)$ therefore fixes $L_m^{\text{dir}}(n)$ for every m at once. The scaling down always gives the upper bound $L_m^{\text{dir}}(n) \leq (m-1)M^*(n)$, and the matching lower bound holds because the heaviest matrix the solve finds is the *double star*: the bidirected star, with weight one on both arcs $h \rightarrow v$ and $v \rightarrow h$ between the hub and every other vertex. Its weights lie in $\{0, 1\}$, so multiplied back by $m-1$ it is an honest integer multigraph for each m , not a fractional one: the directed multigraph carrying every arc of the bidirected star at full multiplicity $m-1$, with $2(n-1)(m-1)$ arcs and $\lambda^{\max} = m-1$. So the two bounds meet for every m together. The two halves are what close the gap jointly: the optimisation ranges over fractional weights in $[0, 1]$, yet its optimum is realised by this $\{0, 1\}$ matrix, so the continuous relaxation is tight and the reported $M^*(n)$ is at once the largest fractional total weight and the densest integer multigraph divided by $m-1$, never an overestimate of the latter.

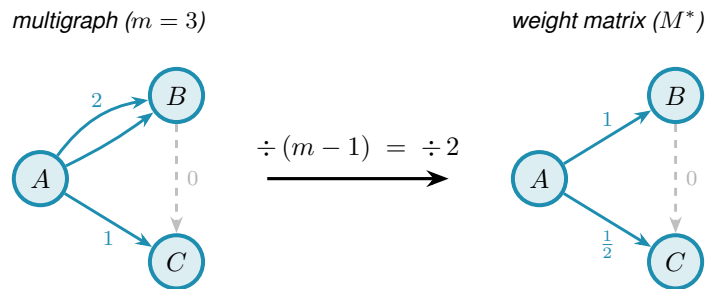


Figure 2.7. The scaling step for $m = 3$. Dividing each multiplicity by $m-1 = 2$ maps the integer multigraph (left) to a weight matrix with all values in $[0, 1]$ and max-flow at most one between every pair (right). The double arc $A \rightarrow B$ (multiplicity 2) becomes weight 1, the single arc $A \rightarrow C$ (multiplicity 1) becomes weight $\frac{1}{2}$, and the absent arc $B \rightarrow C$ (multiplicity 0, shown dashed) stays absent. Crucially, m drops out on the right, so a single optimisation over all weight matrices pins $L_m^{\text{dir}}(n) = (m-1) M^*(n)$ for every m simultaneously.

What remains is to compute $M^*(n)$, and the difficulty is that the constraint “maximum flow at

most one” is itself the answer to a flow problem rather than a plain inequality on w . The max-flow / min-cut theorem removes it. The maximum s – t flow equals the smallest capacity of any *cut* separating s from t , where a cut splits the vertices into a source side holding s and a sink side holding t , and its capacity is the total weight of the arcs crossing from the source side to the sink side. So $\text{maxflow}(s, t) \leq 1$ holds exactly when there is at least one such cut of capacity at most one. We can drop the flow entirely and instead let the optimisation choose, for every ordered pair (s, t) , one cut and require that its capacity stay at or below one.

A cut is described by a yes/no label x_u^{st} on each vertex, equal to one when u is on the source side, with s fixed on the source side and t on the sink side. An arc $u \rightarrow v$ crosses the cut exactly when u is on the source side and v is not, and a helper variable p_{uv}^{st} then picks up its weight $w(u, v)$. Forcing the crossing weights to sum to at most one for every pair forces every pair to admit a cut of capacity one, hence maximum flow at most one. This is the exact feasible region, not a relaxation of it, and that is what turns the result into a proof. The optimisation reports both the best total weight it has found and a value it has shown no matrix can exceed, and when those two meet the gap is zero and $M^*(n)$ is certified.

For the directed multigraph arc problem, then, `solve` runs this cut-counting optimisation in place of any enumeration: it chooses the minimum cut for every ordered pair, and a zero gap makes the reported $M^*(n)$ a proved upper bound, from which $L_m^{\text{dir}}(n) = (m - 1)M^*(n)$ delivers every m at once.

`solve` assembles this optimisation from n alone, and a reader content to take it on the strength of the cut argument just given can skip to [Theorem 2.1](#) without loss. Written out it reads more heavily than it behaves, so we first name every symbol in plain words, then show the one inequality that looks cryptic is a simple yes-or-no switch.

The optimisation is a *Mixed-Integer Linear Program*, or MILP for short. “Linear” means every constraint and the objective are linear functions of the variables: no products, no squares, no exponentials. “Mixed-integer” means the variables split into two kinds: most are ordinary real numbers that may take any value in a continuous range, but a few are forced to be whole numbers, either 0 or 1. The reason for the split is conceptual. Arc weights w are naturally continuous: they are scaled multiplicities and can sit anywhere in $[0, 1]$. But the cut assignment for a vertex is a hard binary choice: a vertex is either on the source side of a given cut or on the sink side, with no meaningful notion of being “half on each side”. Allowing the cut labels to be fractional would describe a different, weaker condition and could report an optimistic value that no integer-weight multigraph can actually achieve. Keeping those labels in $\{0, 1\}$ is what keeps the optimal gap honest, and it is what makes the program’s result a proof rather than an estimate.

Three quantities appear, and only three. The first is already familiar.

★ $w(u, v) \in [0, 1]$ is the *scaled multiplicity* of the arc $u \rightarrow v$, the amount of arc weight the matrix places on $u \rightarrow v$ after the division by $m - 1$ of the previous section. It is what we are maximising the total of. This is a continuous variable: it may be any real number in $[0, 1]$.

- ★ $x_u^{st} \in \{0, 1\}$ is the *side label* for vertex u with respect to the ordered pair (s, t) . The optimisation chooses one cut for each pair, and x_u^{st} records whether u is on the source side (1) or the sink side (0). The two endpoints are pinned: $x_s^{st} = 1$ and $x_t^{st} = 0$. This is the integer variable: a vertex cannot straddle the cut.
- ★ $p_{uv}^{st} \geq 0$ is the *crossing weight*, the part of $w(u, v)$ that the pair (s, t) 's chosen cut is forced to count. It equals $w(u, v)$ when the arc $u \rightarrow v$ crosses from source to sink, and is otherwise left at zero. This is a continuous variable.

With those three named, the program is short:

$$\begin{aligned}
& \text{maximise} && \sum_{u \neq v} w(u, v) \\
& \text{subject to} && p_{uv}^{st} \geq w(u, v) + x_u^{st} - x_v^{st} - 1 && \text{for all } (s, t), (u, v), \\
& && \sum_{u \neq v} p_{uv}^{st} \leq 1 && \text{for all } (s, t), \\
& && w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1 && \text{for all } (s, t), \\
& && d(v_0) \leq d(v_1) \leq \dots \leq d(v_{n-1}), && \text{with } d(v) = \sum_{u \neq v} (w(u, v) + w(v, u)), \\
& && w \in [0, 1], \quad x \in \{0, 1\}.
\end{aligned}$$

The first line is the only one that looks cryptic, and it is just an indicator written as an inequality. Its right-hand side $w(u, v) + x_u^{st} - x_v^{st} - 1$ takes one of four shapes according to which side the cut puts the two endpoints u and v on, and [Table 2.3](#) works all four out. Because $w(u, v) \leq 1$, the right-hand side is positive in one case only, the crossing case, where it equals $w(u, v)$ and so forces $p_{uv}^{st} \geq w(u, v)$. In the other three cases it is zero or negative, so the constraint reads $p_{uv}^{st} \geq (\text{something} \leq 0)$ and leaves p_{uv}^{st} free to stay at zero, which the maximisation is happy to let it do.

x_u^{st}	x_v^{st}	meaning	$w + x_u^{st} - x_v^{st} - 1$	forces
1	0	u source side, v sink side: arc <i>crosses</i>	$w(u, v)$	$p_{uv}^{st} \geq w(u, v)$
1	1	both on the source side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	0	both on the sink side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	1	u sink side, v source side	$w(u, v) - 2 \leq 0$	$p_{uv}^{st} \geq 0$

Table 2.3. The first constraint, case by case. This is where the bound $w \leq 1$ earns its keep: it makes the three non-crossing rows land at or below zero, so the helper p_{uv}^{st} is pushed up to the arc weight in exactly one case, the crossing one. The constraint is therefore an exact indicator, “count $w(u, v)$ when the arc crosses this cut, and nothing otherwise”, with no slack and no fudge.

[Figure 2.8](#) shows one chosen cut and the weights it counts. The second line of the program now reads plainly. It says the total crossing weight of this pair's cut, $\sum_{u \neq v} p_{uv}^{st}$, is at most one. By max-flow / min-cut a cut of capacity at most one exists for (s, t) exactly when $\text{maxflow}(s, t) \leq 1$, so the two lines together say “every ordered pair admits a cut of capacity at most one”, which

is precisely the feasibility we want. This is the true feasible region, not a relaxation of it, so an optimal solution reported with zero gap is a proof and not merely evidence.

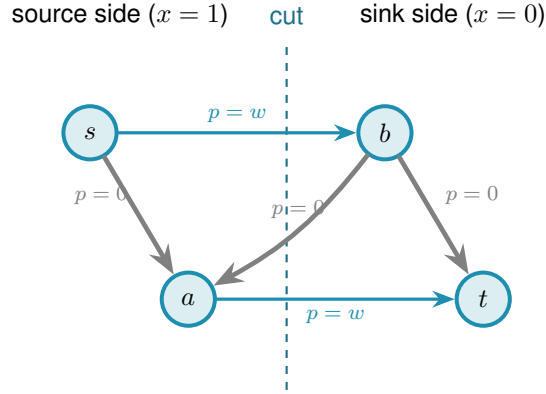


Figure 2.8. One ordered pair's chosen cut. The dashed line splits the vertices into the source side, where $x_u^{st} = 1$, and the sink side, where $x_u^{st} = 0$. Only arcs running from source side to sink side cross the cut, and the first constraint forces their helper p_{uv}^{st} up to the arc weight w (the two thick arcs). Arcs that stay on one side, and arcs that run the other way across the line, contribute nothing. The second constraint then caps the total counted weight, the thick arcs, at one.

The last two lines change no answer. They only help the solver finish in time. The third is a redundant shortcut: the direct arc $s \rightarrow t$ and one two-step detour $s \rightarrow x \rightarrow t$ through each intermediate vertex x already form that many routes between s and t , so $w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1$ must hold of any feasible matrix anyway, and stating it outright tightens the continuous picture the solver reasons from.

The $\min$ in that line is not a linear function, so it cannot appear in the MILP as written. To keep the program linear, each $\min(w(s, x), w(x, t))$ is replaced by a fresh helper variable z_x^{st} together with a binary selector $b_x^{st} \in \{0, 1\}$. The selector records which of the two arguments is smaller: $b_x^{st} = 1$ means $w(s, x)$ is no larger than $w(x, t)$. Four linear inequalities then pin z_x^{st} exactly to the minimum. Two upper bounds, $z_x^{st} \leq w(s, x)$ and $z_x^{st} \leq w(x, t)$, prevent z from exceeding either argument. Two lower bounds, one of the form $z_x^{st} \geq w(s, x) - (1 - b_x^{st})$ and the other $z_x^{st} \geq w(x, t) - b_x^{st}$, each become active only when the selector chooses that argument: at any integer value of b exactly one lower bound bites and forces z up to the chosen weight. Together with the upper bounds this pins $z_x^{st} = \min(w(s, x), w(x, t))$ exactly. This technique is called the McCormick linearisation of a minimum. It adds one continuous variable z and one binary variable b per intermediate vertex per ordered pair, all of them appearing only in the tightening constraint and nowhere else, so the proof structure of the program is unchanged. The fourth line orders the vertices by weighted degree, which is allowed because renaming vertices cannot change any count, and it spares the solver the many relabelled copies of each graph that differ only in vertex names.

Theorem 2.1 (Directed multigraph value, small n). *For $n \leq 6$ and all $m \geq 2$, $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$, attained by the double star. Equivalently $M^*(n) = 2(n - 1)$.*

It is worth being clear about what “for all m ” rests on, because the theorem proves infinitely

many values from finitely many solves. It is not that each m was tested. In this mode `solve` never sees m at all. It proves the single m -free number $M^*(n)$, and the identity $L_m^{\text{dir}}(n) = (m-1)M^*(n)$ then carries that one fact to every m at once. The only quantity that is checked case by case, and the only genuine limit on how far the result reaches, is n .

`solve` proves [Theorem 2.1](#) by returning $M^*(3) = 4$, $M^*(4) = 6$, $M^*(5) = 8$, and $M^*(6) = 10$, each optimal with zero gap, so that $M^*(n) = 2(n-1)$ over this range. The first three finish quickly, $M^*(3)$ and $M^*(4)$ in well under a second and $M^*(5)$ in a few seconds, while $n = 6$ takes about twenty minutes. At $n = 7$ the same encoding does not close, and we do not pretend otherwise: it is a finite obstacle of scale, not a new idea, and it marks the current edge of certified knowledge for this variant. The solver transcript is in [Chapter A](#).

2.8 What the solver may claim

Now that the pieces inside `solve` are built, it is worth stating plainly, because the rest of the thesis depends on it, what each is allowed to claim. The checker *measures*: its connectivity values are exact. Its three proving implementations, the exhaustive enumeration, the pruned search, and the cut-counting, *prove*: an exhausted enumeration or an optimal, zero-gap solve is an upper bound for that fixed n . The search of [Chapter 3](#) will *discover*: it produces concrete graphs, hence lower bounds, and never an upper bound. These three claims are exactly what `solve` labels its answer with, so that an exact value, a lower bound, and an upper bound can never be quietly confused with one another.

With the microscope built and aimed at proof, the variants with small answers are settled and the directed base cases the hand proof needed are in hand. What proof by enumeration cannot do is reach the sizes where the answers are only conjectured, because the number of graphs explodes long before the interesting cases arrive. The next chapter measures that explosion exactly, and then turns the same program loose to *discover* the dense graphs that blind enumeration can never reach.

2.9 Reproducibility

Every claim in this thesis that rests on computation can be re-run from a single Python driver. All of the program's logic lives in `program/erdos915_unified.py`, a self-contained Python program whose core needs only `numpy` and `scipy`: every connectivity measure it reports is one `scipy` integer maximum flow on a capacity matrix, so the model, the checker, the search, the enumeration, and the self-test all run on those two libraries alone. The certifier adds one library, `pulp`, which builds the cut-counting optimisation once for any solver and runs it on the bundled CBC by default or on Gurobi by a one-line switch. Two further libraries are optional and used only for presentation: `matplotlib` renders the figures, and `networkx` supplies the Gomory–Hu tree behind one figure. An optional compiled helper, `_erdos_fast.so`, accelerates small hot-path max-flow and canonical-form calls when built, but the pure-Python path gives the same answers. Running

`python erdos915_unified.py` executes the built-in self-test, the suite of invariant checks that confirms the connectivity values quoted throughout this chapter, including the sanity checks on the directed double star and on K_n and $K_4^{(3)}$. Every figure in the thesis is regenerated by its companion driver `program/make_figures.py`, which imports that same file and fixes every random seed, so the pictures are reproduced exactly rather than merely resembled. The table in `program/README.md` records which routine produces which figure. The solver transcripts behind the certified values of this chapter, the pruned-search log for [Theorem 1.9](#) and the cut-counting certificate for [Theorem 2.1](#), are reproduced verbatim in [Chapter A](#), so a reader may check the proofs without running anything at all.

Chapter 3

Discovering Bounds by Search

The previous chapter proved what could be proved by looking at every graph of a fixed size. That method is honest and exact, and it stops almost at once. This chapter begins by measuring exactly how fast it stops, because the speed of that collapse is the whole reason a second kind of tool is needed. Then it builds that tool. Where the certifier walks all graphs, the search walks a few of them well, cooling like a physical system toward the densest feasible shape. What it returns is never a proof. It is a concrete graph, hence a lower bound and a conjecture, a stepping stone toward an upper bound that a later proof or certificate must supply. The chapter ends by asking the honest question about any such tool: how good is it, and how far from the truth might its answers be.

3.1 The wall: how enumeration explodes

The brute force of [Chapter 2](#) visits every graph of a given size. The trouble is how many that is. A simple undirected graph on n vertices is a yes-or-no choice on each of the $\binom{n}{2}$ possible edges, so there are $2^{\binom{n}{2}}$ of them. A digraph chooses on each of the $n(n-1)$ ordered pairs, giving $2^{n(n-1)}$, and allowing multiplicities up to a cap c raises the base from two to $c+1$. These are not numbers that merely grow. They detonate. [Figure 3.1](#) draws the growth on a logarithmic scale, and the message is the same in every panel: the count passes any reachable budget while n is still in the single digits.

Three readings of the figure matter for what follows. Adding vertices is the sharpest cost: each new vertex multiplies the count by an exponential in n , so a single extra vertex can move a problem from seconds to centuries. Raising the connectivity budget is the next cost, lifting the base of the exponent rather than its height, which is why the multigraph curves sit above the simple one in every panel. Direction is the third: a digraph has twice as many cells to fill as the undirected graph on the same vertices (and a directed 3-uniform hyperedge, a tail with two heads, has three times as many), so the directed panel climbs fastest of all, and those are precisely the variants whose answers [Chapter 1](#) could not prove. The separation, by contrast, costs nothing here, because edge and vertex disjointness search the same graphs and differ only in what the checker measures, which is why a single pair of panels suffices for all of them. The certifier of [Chapter 2](#) reaches a few vertices further than blind enumeration, because its cut-counting optimisation is polynomial in the size of the encoding rather than in the number of graphs, but solving that encoding is itself hard and the gain is measured in single vertices, not orders of magnitude. Past that, no exhaustive method reaches the interesting sizes. Something

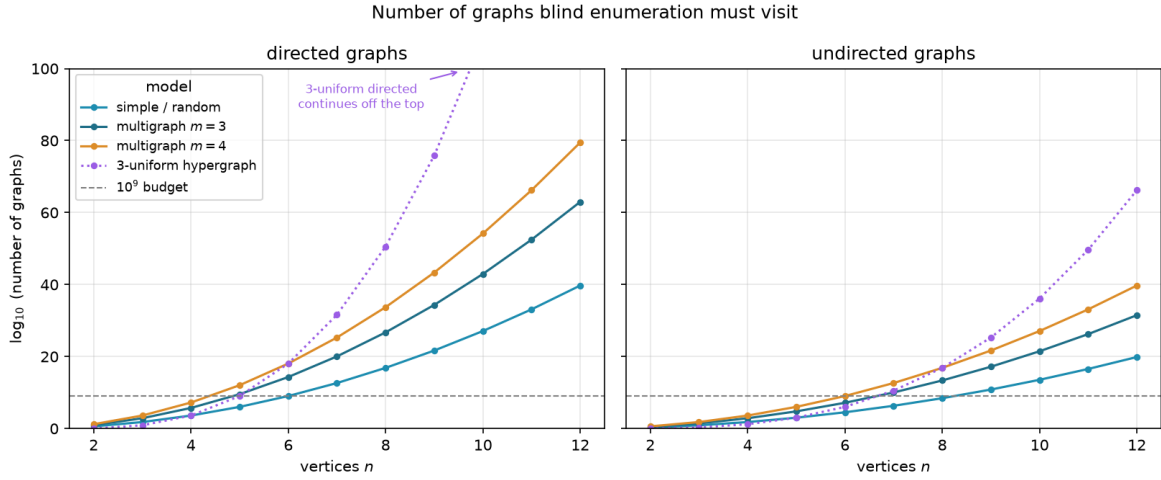


Figure 3.1. The number of graphs blind enumeration must visit, on a logarithmic vertical axis, for directed graphs (left) and undirected graphs (right). Each panel shows the simple model, two multigraph connectivity caps, and the three-uniform hypergraph (dotted), with a dashed line at a generous budget of 10^9 candidates, and every curve crosses that budget while n is still in the single digits. Writing E for the number of off-diagonal cells a model may fill, a simple graph offers 2^E candidates, a multigraph with multiplicity cap $m - 1$ (the connectivity ceiling) offers m^E (one choice per cell from $\{0, \dots, m - 1\}$), and the 3-uniform hypergraph offers $2^{\binom{n}{3}}$ undirected or $2^{n \binom{n-1}{2}}$ directed, since a directed hyperedge is a tail together with two heads. The count depends only on the model and the direction, never on whether one later asks for edge- or vertex-disjoint routes, so a single pair of panels covers all of those variants at once. Direction multiplies the number of cells to fill, by two for graphs (ordered versus unordered pairs) but by three for the 3-uniform hypergraph, so the directed panel sits far above the undirected one, and the directed cases, exactly the ones [Chapter 1](#) left open, are the first to go.

that does not look at every graph is required.

3.2 The temperature search

We want, for fixed n and m , a graph with as many edges as possible subject to $\lambda^{\max} \leq m - 1$ (or $\kappa^{\max} \leq m - 1$ for the vertex variants). The space of graphs on n vertices is enormous and its feasible region is not convex: adding an edge can be harmless until, together with edges added long before, it suddenly creates a forbidden pair. A greedy walk that only ever accepts an improving move stalls at the first such collision, in a graph that is locally maximal but globally far from extremal.

The classical method for this kind of landscape is *simulated annealing*, named after the metallurgical process of cooling a heated material slowly so that its atoms settle into a low-energy crystal rather than a disordered glass. Applied to graphs, annealing escapes local optima by sometimes accepting a worsening move, and by accepting fewer of them as the run proceeds. Each graph G is a state with energy

$$E(G) = -|E(G)| + \Lambda \cdot \max(0, \lambda^{\max}(G) - (m - 1)),$$

Algorithm 1: Temperature search for a dense feasible graph

Input: vertices n , value m , start T_0 , cooling α , penalty Λ , steps S
Output: the densest feasible graph encountered

```
1  $G \leftarrow$  empty graph on  $n$  vertices;  
2  $T \leftarrow T_0$ ;  
3 for  $S$  steps do  
4   propose  $G'$  by toggling one adjacency, biasing removals toward low-sensitivity edges;  
5    $\Delta \leftarrow E(G') - E(G)$ ;  
6   if  $\Delta \leq 0$  or  $\text{rand}() < e^{-\Delta/T}$  then  
7      $G \leftarrow G'$ ;  
8   if  $G$  is feasible and denser than the best so far then  
9     record  $G$  as the best;  
10   $T \leftarrow \alpha T$ ;  
11 return the best recorded graph;
```

which rewards edges and penalises any excess connectivity. The penalty is a soft constraint rather than a hard one, so the walk may pass briefly through infeasible graphs rather than being walled inside the feasible region, and we return below to why that freedom is worth having. A move toggles one adjacency. The Metropolis rule accepts an improving move always and a worsening move with probability $e^{-\Delta E/T}$, and the temperature T falls geometrically. [Algorithm 1](#) is the whole loop, the one place in this chapter where a body is worth showing, because the loop *is* the method.

This loop is what `solve` runs in its discovery mode. With the exhaustive flag off it anneals instead of enumerating, following the Metropolis walk of [Algorithm 1](#) under geometric cooling, removals biased toward low-sensitivity edges, and returns the densest feasible graph it encountered together with the full temperature trace. What it returns is a witness, hence a lower bound, never a proof.

It is fair to ask why the walk is allowed above the ceiling at all. Reaching every feasible graph does not require it. Removing an edge can only lower $\lambda^{\max}$ and $\kappa^{\max}$, never raise them, so every subgraph of a feasible graph is again feasible, and the empty graph is feasible trivially. From any feasible target we can therefore delete edges one at a time down to the empty graph without ever crossing the ceiling, and reading that path backward builds the target up the same way. The feasible region is connected through the empty graph by single-edge moves that stay inside it, so a walk that never went infeasible could in principle visit every feasible graph there is.

We let it go infeasible anyway, because reachability and efficiency are different things. A feasible graph can be locally maximal, meaning every edge we might add creates a forbidden pair, and yet still sit far below the densest feasible graph. To improve such a graph the walk has to add an edge and then remove a different one, a swap whose first half is infeasible. A walk confined to feasible graphs could only find that better graph by retreating most of the way back to the empty graph and climbing a different slope, a detour so long that the cooling schedule usually

ends first. Allowing a brief excursion above the ceiling lets the walk tunnel straight across instead, trading one load-bearing edge for a better one in two moves rather than dozens. The soft penalty also turns the search into a single smooth landscape with no hard walls for the Metropolis rule to bounce off, and the same machinery then carries over unchanged to variants whose feasible region is not so conveniently connected. Going above the bound is never the goal, then. It is the shortcut the annealer takes on its way back down to it.

3.3 Temperature and edge sensitivity

The phrase “biasing removals toward low-sensitivity edges” in [Algorithm 1](#) is where the temperature acquires its meaning, and it is the heart of the method. For an edge e define its *sensitivity*

$$\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e),$$

the amount by which deleting e relaxes the binding connectivity. A load-bearing edge sits on a tightest cut and has $\sigma(e) > 0$. Removing it would drop $\lambda^{\max}$ and so loosen the very constraint that limits how many edges the graph may hold. A free edge has $\sigma(e) = 0$: it can be removed without easing the binding pair, which usually means it can be put to better use elsewhere.

`edgeSensitivity( $G, e$ )  $\rightarrow \mathbb{N}$`

MEASURE

in a graph G and one of its edges e

out $\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e)$

how delete e and let the checker remeasure, where a positive value marks a load-bearing edge. This single number is the dial the cooling turns.

When the search proposes a removal it draws the edge with probability proportional to $e^{-\sigma(e)/T}$. The dial is now explicit. At high temperature these weights flatten and the walk toggles edges regardless of how load-bearing they are, exploring widely and willing to dismantle essential structure. As the temperature falls the weights concentrate on $\sigma = 0$, so the walk almost never disturbs a load-bearing edge and instead rearranges only the free ones, until the graph crystallises onto an extremal shape. This is exactly the behaviour one wants, and it is the behaviour the formulation asks for: the temperature tracks how strongly removing an edge changes the bound. [Figure 3.2](#) shows one such run settling onto a certified optimum. The other variants behave identically, and their traces are collected in [Section B.2](#).

The same sensitivities also explain a finished graph. Colouring a graph’s arcs by σ shows at a glance which arcs are structural and which are slack. [Figure 3.3](#) illustrates all four sensitivity levels on one directed multigraph: the darkest arcs carry the most flow and lower $\lambda^{\max}$ the most when removed, the cool arc contributes nothing. The picture is not decoration: it is the certificate of tightness made visible, and it tells the search exactly which arcs to leave alone as the temperature falls.

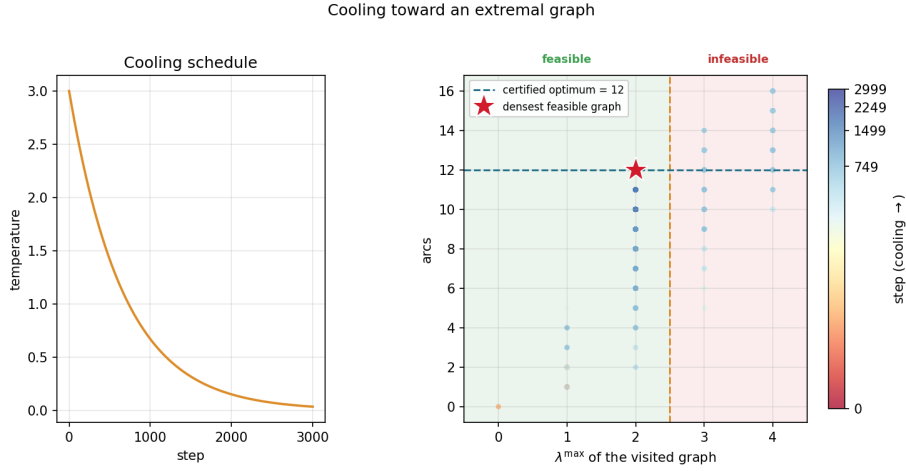


Figure 3.2. A single cooling run on the directed multigraph problem at $n = 4$, $m = 3$. *Left:* the temperature schedule, cooling geometrically from its starting value. *Right:* every visited graph plotted by its binding connectivity $\lambda^{\max}$ (horizontal) against its arc count (vertical), coloured by the step at which it was visited. Annealing uses a soft penalty rather than a hard wall, so while hot (dark points) the walk strays into the infeasible region $\lambda^{\max} > m - 1 = 2$, picking up extra arcs it is not allowed to keep. As the temperature falls (bright points) the penalty drives the walk back across the feasibility boundary and onto the densest graph that stays feasible, the certified optimum of 12 arcs at $\lambda^{\max} = 2$ (star).

A single cooling schedule can still settle into a local optimum, so in discovery mode `solve` runs several independent schedules from different starting seeds and keeps the densest result across all of them. Because the restarts share no state and none depends on the outcome of another, they run in parallel: the program dispatches them across the available processor cores, so the wall-clock cost of k restarts is a small multiple of a single run rather than k times it, bounded below by the slowest restart and by the cost of starting the worker processes. A handful of parallel restarts is enough to reach the known optima at the sizes we examine. This is a reliability measure, not a change of method, and the parallelism is invisible to every number the thesis reports: the same best graph would be found by running the schedules sequentially, just more slowly.

3.4 Keeping the checker cheap inside the search

The energy of [Algorithm 1](#) reads the binding connectivity $\lambda^{\max}(G)$, and a literal reading of the loop would call the checker afresh on the whole graph at every step. That checker is the most expensive routine in the program, one maximum flow per ordered pair, and the walk runs for thousands of steps with several restarts, so such a search would spend nearly all of its time remeasuring graphs that barely changed from the step before. Two elementary facts remove almost all of that work without altering a single number the search reports.

Proposition 3.1 (Monotonicity of the binding connectivity). *Let G be a graph or directed multi-graph, and let G' be obtained from G by changing one adjacency by a single unit of multiplicity. Then*

capacity crosses any fixed cut at most once, so every cut has capacity at most one greater in G'' than in G . The smallest cut capacity thus grows by at most one, giving $\lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$, and together with part (1) we have $\lambda_G(s, t) \leq \lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$. The maximum over pairs gives $\lambda^{\max}(G'') \leq \lambda^{\max}(G) + 1$.

For $\kappa^{\max}$ the argument is identical on the vertex-split network of [Figure 2.3](#), where changing an adjacency changes the capacity of its single arc $u^{\text{out}} \rightarrow v^{\text{in}}$ in the same way. Parallel copies leave that capacity at one and so change $\kappa^{\max}$ not at all, which only strengthens both parts. $\square$

These two facts settle almost every step without a flow. The energy depends on $\lambda^{\max}$ only through the excess $\max(0, \lambda^{\max} - (m - 1))$, which is zero exactly when the graph is feasible. So a removal applied to a feasible graph leaves it feasible by part (1), with excess still zero, and needs no measurement at all. An addition made while the graph sits strictly below the bound, at $\lambda^{\max} \leq m - 2$, stays feasible by part (2), again with excess zero and no measurement. Only an addition made right at the boundary $\lambda^{\max} = m - 1$ can tip the graph over, and only there does the search run a single flow, capped so that it stops the instant it finds one route too many. The exact connectivity value is recomputed only on the rare step whose proposal is genuinely infeasible, where the penalty term needs it.

To know which case applies, the search carries a running upper bound on $\lambda^{\max}$, raised by one whenever it accepts an addition (part (2)) and left untouched when it accepts a removal (part (1)), and resynchronised to the exact value at the cadence where it already pays for flows to refresh the sensitivity map. The bound can only ever sit at or above the truth, so any step it clears is certainly feasible.

The key point is that the energy returned on every step is, by [Proposition 3.1](#), exactly the value the naive implementation would have computed. Every accept-or-reject decision, every random draw, and the graph the walk finally returns are therefore unchanged, and the speed-up is invisible to every number the thesis reports. This is verified rather than asserted. The regression test `test_fast_path_matches_exact` runs both implementations, the fast one and a reference that measures exactly at every step, across all four graph variants and both separations, and checks that the two agree step for step in energy, in every accept-or-reject decision, and in the final graph, and that the returned graph is feasible under the exact checker. Nothing about the optimisation hides inside the bound: the witness the search reports is always certified by an exact measurement, exactly as in the slower search.

It is worth noting where this method earns its keep. For undirected graphs one could resynchronise the exact value cheaply from the Gomory–Hu tree of [Chapter 2](#), which encodes every pairwise edge-connectivity in $n - 1$ flows. The directed and vertex variants admit no such tree, so there the monotone bound and the capped predicate are the only inexpensive handle on feasibility, and they are what let one annealer serve every variant at speed.

Construction	n	m	Search	Theory
spanning tree (simple undirected, $m=2$)	6	2	5	5
multiplicity- $(m-1)$ star (multi undirected, $m=3$)	6	3	10	10
bidirected star (simple directed, $m=2$)	4	2	6	6
bidirected star (simple directed, $m=2$)	6	2	10	10
hub-bipartite tie (simple directed, $m=2$)	7	2	12	12
double star (multi directed, $m=3$)	4	3	12	12
double star (multi directed, $m=3$)	5	3	16	16

Table 3.1. Validation by rediscovery. For each variant the temperature search, run from scratch with a handful of restarts, recovers exactly the extremal value proved or certified in the earlier chapters. The “Search” column is an honest lower bound found by annealing, and the “Theory” column is the proved or certified value.

3.5 Rediscovering the known extremisers

A search method is only worth trusting if it recovers what is already known before it is pointed at what is not. Run from the empty graph on the cases whose answers [Chapter 1](#) and [Chapter 2](#) settled, and told only to pack in edges without breaching the connectivity ceiling, the cooled search arrives unprompted at the named constructions. [Table 3.1](#) reports the densest graph it found against the value the theory predicts, and in every case the two agree.

What the table hides is that the graphs themselves are the named constructions, not merely graphs of the right size. The simple undirected run at $m = 2$ settles on a spanning tree, the only shape with $n - 1$ edges and no cycle. The undirected multigraph run settles on a star with every edge at multiplicity $m - 1$, an instance of the multi-tree extremiser behind [Theorem 1.3](#). The directed multigraph runs settle on the double star of [Chapter 2](#), both directions of every spoke at full multiplicity. The simple directed runs find the bidirected star at $n = 4$ and, at $n = 7$, land on twelve arcs, the value at which the hub and the one-directional wall tie, exactly the crossover [Theorem 1.9](#) predicts. The cooled search does not know these names. It rediscovers the structures because, under the connectivity ceiling, there is nowhere denser to go.

Some extremisers are past the size at which a quick search lands reliably, but the exact checker of [Chapter 2](#) settles them immediately by measuring their connectivity. The one-directional bipartite digraph on eight vertices has 16 arcs at $\lambda^{\max} = 1$, and 16 already exceeds the linear hub value $2(n - 1) = 14$: the smallest size at which the quadratic phenomenon strictly wins, made concrete. It is also a measured illustration of the search’s limit: repeated cooling runs at $n = 8$, $m = 2$ stall at the hub value of 14, because dismantling a finished hub and rebuilding a one-directional wall is far more than the single-arc moves of the walk can tunnel through. The known answer there comes from the construction and the checker, not from the search, which is exactly the division of labour this chapter argues for. The augmented bipartite digraph on ten vertices at $m = 3$ has 30 arcs at $\lambda^{\max} = 2$, the thirty-arc counterexample of [Remark 1.11](#), confirmed pair by pair against the naive bound of 27 it refutes. The directed double star on seven vertices at $m = 4$ has 36 arcs at $\lambda^{\max} = 3$, matching $2(n - 1)(m - 1)$. None of these is a proof of optimality.

Each is an exactly measured feasible graph, a lower bound the checker stamps as genuine.

3.5.1 Simulated annealing versus tabu search

The annealer is not the only way to walk this landscape. A natural alternative, suggested by Jan Goedgebeur, is *tabu search*, which keeps the energy and the neighbourhood of [Algorithm 1](#) but replaces the cooling schedule with a deterministic best-improving step. At each step it evaluates every single-edge move from the current graph and takes the one that lowers the energy most, then forbids the pair it just changed for a fixed number of steps, the *tabu tenure*, so the walk cannot immediately undo its last move and stall in a two-step cycle. When no improving move is left and the search stalls, it perturbs the incumbent with a few random moves and carries on. There is no temperature and no accept-worse coin: apart from the perturbation kicks the walk is deterministic. Where annealing explores by occasionally stepping uphill and trusts the cooling to settle it, tabu climbs greedily and trusts the tabu list to keep it from retracing its path.

The two engines rediscover the same values, but at very different speeds, and on the harder directed cases the difference is decisive. [Table 3.2](#) runs them head to head at an equal budget of six seconds each, both restarted within that budget. On the simple undirected problem both reach the optimum at once. On the two directed problems past the certified range tabu reaches the optimum within the budget while the annealer plateaus below it, assembling the eighteen-arc simple-directed extremiser and the full twenty-four-arc double star at $n = 7$ where the annealer stalls at sixteen and twenty. The cause is the one already met at the $n = 8$, $m = 2$ hub stall above. The annealer cools onto whatever dense shape it first reaches and then cannot dismantle it within the budget, whereas tabu's best-improving step follows the steepest density gradient and the tabu list stops it sliding back, so it assembles the structured extremiser the cooled walk struggles to find.

Variant	n	optimum	best in 6 s		time to optimum	
			SA	tabu	SA	tabu
simple undirected	7	9	9	9	1.3 s	0.1 s
simple directed	7	18	16	18	—	2.6 s
directed multigraph	7	24	20	24	—	2.5 s

Table 3.2. Simulated annealing against tabu search at an equal six-second budget per method, the annealer restarted from fresh seeds and tabu restarted on stalls. Both reach the simple undirected optimum. On the two directed problems past the certified range tabu reaches the optimum while the annealer plateaus below it. The last two columns give the wall-clock time each method first reached the optimum on the reference machine, with a dash where it did not reach it in the budget.

[Figure 3.4](#) shows the divergence directly, plotting the densest feasible graph each engine has found against wall-clock time on the two directed multigraph cases. Tabu climbs in a few decisive steps and holds at the optimum, while a single annealing schedule rises quickly and then flattens a few arcs below it. The annealer's independent restarts recover the optimum on the smaller case, as [Table 3.1](#) records, so a single plateau is not the whole story there. But at $n = 7$ the plateau persists within the budget, and one tabu schedule already reaches what the restarted

annealer does not.

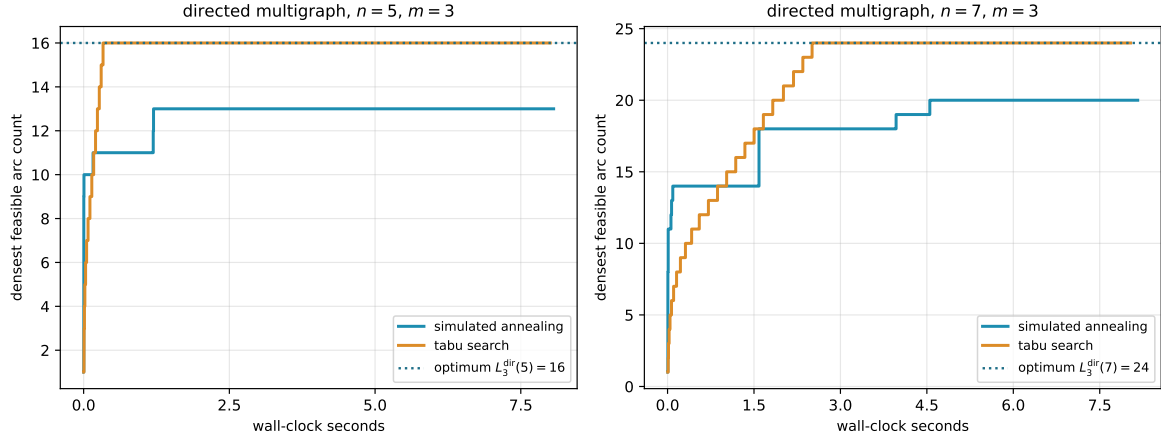


Figure 3.4. Densest feasible graph found against wall-clock time, for one simulated-annealing schedule and one tabu schedule on the directed multigraph at $n = 5$ and $n = 7$, $m = 3$. Tabu climbs in a few best-improving steps to the optimum (dotted) and holds, while a single annealing schedule rises quickly and then flattens below it. The time axis is wall-clock on the reference machine, so unlike the seed-reproducible figures elsewhere this one is a representative timed run.

This advantage comes at a cost the equal-time comparison already absorbs but is worth naming: each tabu step scans the whole neighbourhood, so it is far more expensive than the annealer’s single random proposal, and tabu takes many fewer but better-aimed steps in the same wall-clock. Simulated annealing remains the program’s default, because it reaches every value this thesis reports, the rediscovery of Table 3.1 and the conjectural lower bounds alike, and because its independent restarts parallelise trivially across cores. Tabu is the stronger engine for the directed problems past the certified range, and it is the natural tool for any future attempt to push those lower bounds further by machine.

3.6 The trichotomy: proved, certified, conjectured

It is worth stopping to lay out, for every variant, the three kinds of knowledge the computation can hold about it, because the whole value of the pipeline is that they are kept apart and never quietly merged. About any given value the program can do one of three things. It can *prove* the value, closing a fixed size from above by exhausting every graph of that size, an upper bound with the standing of a hand proof. It can *certify* it, the same kind of upper bound but obtained by the cut-counting optimisation rather than blind exhaustion, reaching a vertex or two further. Or, where proof and certificate both run out, it can only *conjecture* the value, exhibiting a concrete dense graph that pins down a lower bound and leaving the matching upper bound to a later argument.

Proof and certificate bound the answer from above. The search bounds it only from below. Where a closed *formula* exists it can tie the two together, threading through the certified small cases and the conjectured large ones as a single curve. This is the discipline the whole thesis runs on, and it is worth stating as one: the search proposes a lower bound, a certificate or a hand proof disposes the upper bound, and only when both meet is a value called settled.

We begin with the single variant where all three kinds of knowledge are non-trivial and visibly distinct, the directed simple arc problem at a fixed $m \geq 3$, and then sweep the whole family. For that variant the knowledge splits cleanly by the size of the graph.

For small n the answer is not conjectured at all. The pruned exhaustion inside `solve` walks every simple digraph on n vertices and returns the exact maximum, complete and certain, an upper bound with the standing of a hand proof. At $m = 3$ it certifies 6, 9, 12 for $n = 3, 4, 5$, and there the exhaustion stops within a reasonable budget, the pruned walk no longer closing $n = 6$ in tractable time. That is the certified region, finite and closed.

Beyond it the ceiling is gone and only lower bounds remain, each witnessed by a concrete feasible graph. The search finds the extremal 15 arcs at $n = 6$ on its own. Past that size it falls behind: across six restarts it reaches 17, 17, 20, 23 for $n = 7, \dots, 10$, while the named constructions of [Chapter 1](#), measured exactly by the checker, supply 18, 21, 25, 30, the last being the thirty-arc graph of [Remark 1.11](#). The reported lower bound at each size is the better of the two, exactly as the rediscovery section combined search and checker by hand. These are conjectures with an honest floor and no matching ceiling. That is the searched region, unbounded and forever one-sided.

The two regions are tied together by a single closed form. The conjectured value

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right)$$

of [Conjecture 1.13](#) is not a fresh guess bolted on past the certified range. It is one formula that already passes through every certified value at small n and then continues, without a seam, through every searched value at large n . The directed-arc panels of [Figure 3.5](#) show this directly, the certified squares at the small sizes and the open lower-bound circles beyond both sitting on the one dashed conjecture curve, proof below the formula and witnessed graphs above it. The formula is the through-line that makes the small certified cases and the large conjectured ones a single statement rather than two.

Two features of those directed-arc panels are worth naming. The crossover from the linear hub term to the quadratic bipartite term happens at $n = 9$, where the second argument of the maximum first wins, and that is past the last certified size. The shape of the answer therefore changes in exactly the region where only the search can see it, which is precisely why a discovery tool was needed and a certifier alone would not do. And at $n = 10$ the formula reads 30, the count of the very counterexample that overturned the natural initial conjecture, so the closed form is not a smooth fit to easy data but one that survives the case that broke the previous guess.

Asking the same three questions of every variant, the answers sort into three regimes, and [Figure 3.5](#) shows all twelve at once, one panel per variant, in exactly this proved, conjectured, and guessed language.

In the first regime the closed form is already a theorem for every n , proved by hand in [Chapter 1](#) or cited, and the computation only confirms it. The undirected families live here, edge and

vertex alike, together with the directed problems at $m = 2$. For these the certified small cases and the rediscovered constructions of [Table 3.1](#) are validation, not new knowledge: the formula was never in doubt, and the program's role is to check that the pipeline reproduces it. That parallel edges do not help for vertex-connectivity, and that the directed vertex multigraph reduces to the simple digraph, are settled by hand, so several rows here need no computation of their own at all.

In the second regime the three columns genuinely differ and the computation is load-bearing. This is the regime of the worked example above. The directed simple arc and vertex problems at $m \geq 3$ are certified only at the small n the exhaustion reaches and conjectured by the search beyond, with the formula of [Conjecture 1.13](#) the through-line.

The directed multigraph arc problem sits here too, with one extra economy: in its cut-counting mode `solve` proves the m -free number $M^*(n) = 2(n - 1)$ outright for $n \leq 6$, and the scaling identity $L_m^{\text{dir}}(n) = (m - 1) M^*(n)$ of [Chapter 2](#) carries that one certified number to every m at once, so the closed form $2(n - 1)(m - 1)$ is proved across the whole m axis at those sizes. Past the crossover at $n = 7$ the conjectured value switches to the bipartite branch $(m - 1) \lfloor n^2/4 \rfloor$, where the even sizes are proved conditional on the odd ones and only a single mixed-regime minimum-degree statement keeps the induction from closing ([Chapter A](#)).

In every row of this regime the formula agrees with proof where proof reaches and with the search where it does not, exactly the linkage the worked example displayed.

In the third regime there is no formula at all, and the honesty is in drawing the guess as a guess. The undirected vertex problem at $m \geq 6$ has been open since 1974 with its extremal structure genuinely unknown, and the hypergraph variants beyond the proved edge case are unexplored, the directed ones needing a one-tail, many-head reading of a Berge edge that this thesis defines itself only so that `solve` can measure it at all. Here the program still computes exact values at the few sizes that finish, and its search still exhibits feasible graphs at a handful more, but no closed form is in sight. The panels of [Figure 3.5](#) carry, instead of a dashed conjecture, a densely dotted line that simply interpolates the machine points in straight segments, trusted no further than the eye. We draw it *through* the points rather than fitting a smooth curve above them: the points are honest lower bounds, the truth lies on or above them, and a line riding above the points would claim more than the search ever found.

Two cautions govern how to read the dotted curves. At small n the complete graph or hypergraph is itself feasible, its binding connectivity being only $n - 1$, so the earliest points merely trace the trivial maximum and reveal nothing of the open behaviour: the undirected-vertex points run up $\binom{n}{2}$ until n passes m , and only then bend toward the linear growth the proved edge value predicts.

Past that bend the dotted line is only as strong as the quick search behind it. Exactly as the search stalled on the quadratic wall in the rediscovery above, the lower bound it reports climbs only linearly where it cannot build a denser construction. The clearest case is the directed hypergraph, whose bipartite family is quadratic, reaching $\sim (m - 1)n^2/(4(r - 1))$ hyperarcs by

Proposition A.39: there the search slips below that construction at larger n , so the dots understate the truth rather than tracing it. We therefore read the dotted trends as honest lower bounds and not as fitted laws, and the one closed-form hypothesis we record, the quadratic for the directed hypergraph, comes from the construction and not from the shape of the dots.

So to the question of whether one formula links what is proved to what is conjectured, the answer is three-valued and depends on the variant, and the grid of [Figure 3.5](#) shows all three at a glance. Where the problem is already solved the curve is solid, a theorem the computation merely agrees with. Where the problem is open but tractable the curve is dashed, a conjecture the certified squares confirm below and the searched circles confirm above, and that is the case the program was built for. Where the problem is genuinely hard the curve is dotted, a guess interpolated through the search points and an open question of what, if anything, the dots truly trace. [Figure 3.6](#) redraws the same grid at $m = 6$: the proved curves climb linearly with m exactly as their formulas predict, while the open variants' search points and their dotted interpolation climb higher without a proved curve to meet, so raising the threshold does not change which regime a variant lives in, only how far the dashed and dotted curves have travelled.

The same grid answers the harder question the trichotomy raises, which is what the search is worth where no upper bound exists to check it against. It says two things at once. First, wherever an upper bound exists, proved or certified, the search lands exactly on it and never above it at every plotted size, which is strong evidence that the search is not leaving edges on the table, with the one instructive exception recorded above: the directed $m = 2$ run at $n = 8$, where it stalls on the hub and the bipartite optimum must be supplied by the construction instead.

Second, on the directed arc problem at $m \geq 3$, where no upper bound is known past the certified range, the plotted lower bounds land exactly on the conjectured value $\max(m(n - 1), \lfloor (n + m - 2)^2/4 \rfloor)$ at every reachable n , and the certified points agree with it as far as they go, but past the first open size it is the constructions, not the cooled walk, that carry the bounds there. Run from scratch the search confirms the conjecture at $n = 6$ and then plateaus a few arcs short (17, 17, 20, 23 against 18, 21, 25, 30 at $n = 7, \dots, 10$, across six restarts), for the same reason as the $m = 2$ stall recorded earlier: the extremal shapes past the crossover are bipartite walls, and a walk that has crystallised onto a hub cannot rebuild itself into a wall by single-arc moves within the cooling budget.

So to the question “do we believe the reported lower bounds are best possible,” the grid answers: wherever a proof or a certificate can check them, yes, and on the open sizes they sit on the only value anyone has conjectured, with the search confirming the first open size and the constructions carrying the rest. The gap between those lower bounds and what is proved is exactly the gap between the conjecture and the missing upper bound, which [Chapter 4](#) names precisely. The search has not closed that gap, and it cannot. What the pipeline supplies is reproducible evidence for which side of the gap the truth lies on.

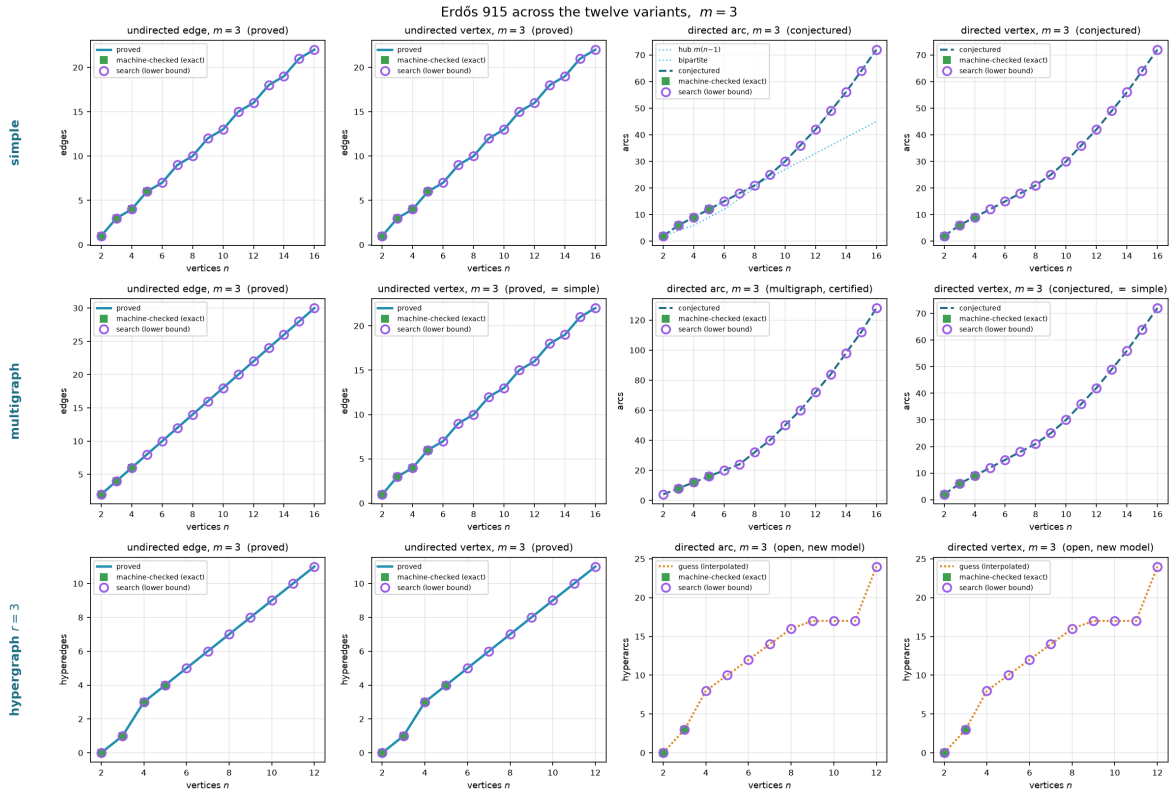


Figure 3.5. All twelve variants at $m = 3$, edges against n , laid out by model (rows: simple, multigraph, 3-uniform hypergraph) and by separation and direction (columns: undirected edge, undirected vertex, directed arc, directed vertex). A *solid* curve is a value proved for all n , a *dashed* curve a conjectured formula, and a *densely dotted* curve a bare interpolation of the machine points where no formula is known. Filled squares are the sizes `solve` proved or certified exactly. Open circles are lower bounds at larger n , the better at each size of the search's densest find and a named construction measured by the checker, reported as a running maximum so a feasible graph at one size is never lost at the next. On the open variants the dotted curve is the piecewise-linear interpolation *through* those circles, never above them, and the axes are scaled to the data rather than to a loose certain interval.

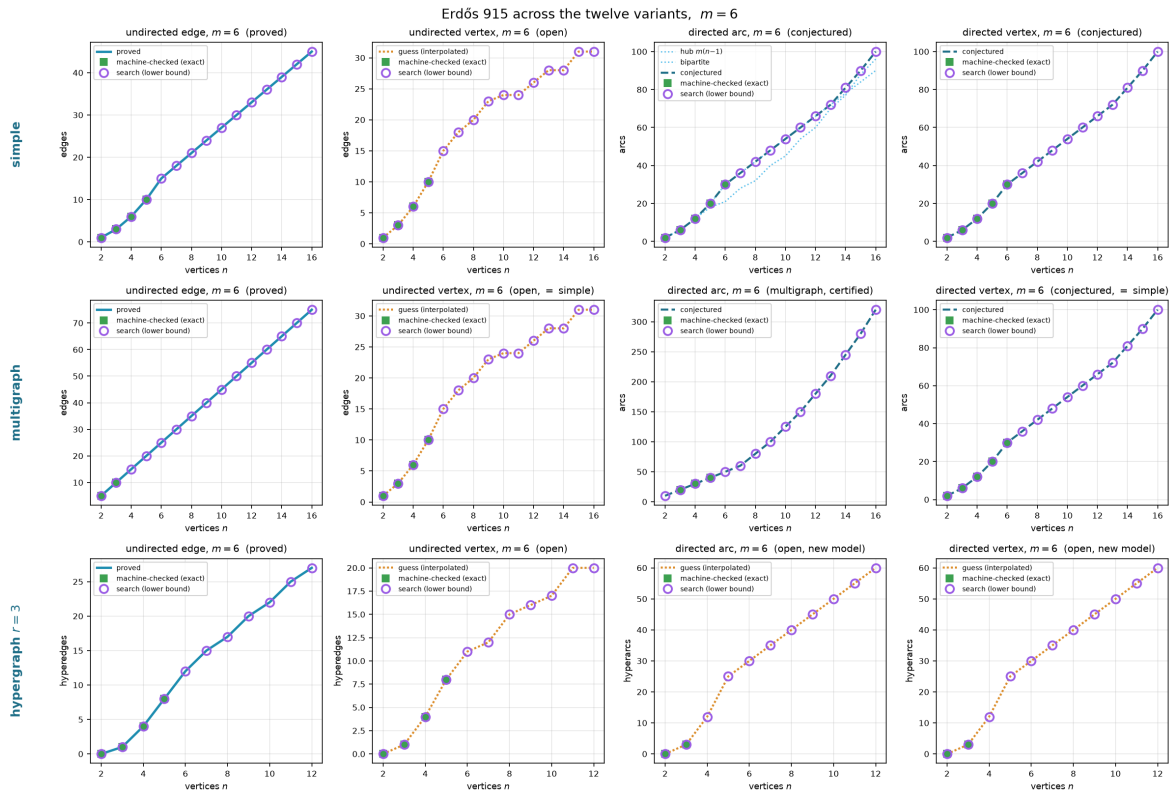


Figure 3.6. The same twelve variants at $m = 6$. The undirected edge and multigraph edge cases remain proved (Mader's theorem holds for all m), while the undirected vertex separation is the famous open problem at $m \geq 5$ and the directed cases carry the same conjectured formula scaled to $m = 6$. Comparing this figure with the $m = 3$ version above shows that the proved cases grow linearly with m as the formulas predict, while on the open cases the search circles and their dotted interpolation climb steadily without a proved curve to meet.

Honesty

A number found by the search is reported as a lower bound, and is marked as certified or proved only when a separate cut-counting optimisation or a hand proof supplies the matching upper bound. No extremal value is ever asserted on the strength of the search alone. The figures in this chapter use fixed random seeds, so every search point is reproducible.

Chapter 4

Synthesis, Results, and Open Problems

The journey set out from one clean theorem and followed it across twelve variants, proving what proved cleanly, building a machine where the proofs ran out, certifying the small cases the machine could close, and using the search to discover the dense graphs the proofs had singled out and to point at the ones still open. This closing chapter draws the threads together. It maps the directed frontier, where proof and certificate both still fall short of the full answer, names the two genuine surprises the landscape contains, gathers the twelve answers into one picture, and says plainly what the program changed and where it stops.

4.1 The directed frontier and the single open lemma

Two of the open directions are directed, and they share a structure worth stating exactly rather than hiding. The directed arc problem is a race between two constructions on different scales. The directed hub ([Construction 1.8](#)) reaches $m(n-1)$ arcs and grows linearly. The augmented bipartite construction ([Construction 1.12](#)) packs $\lfloor (n+m-2)^2/4 \rfloor$ arcs and grows quadratically. Which leads depends on n , and the two cross at an order linear in m , near $n = 9$ at the $m = 3$ of [Figure 4.1](#). [Conjecture 1.13](#) asserts that together they are the whole story.

[Conjecture 1.13](#) is proved as a lower bound for all m , and it is a theorem at $m = 2$ by [Theorem 1.9](#). Past $m = 2$ it sits squarely in the second regime of the trichotomy of [Section 3.6](#): certified by exhaustion at the small sizes `solve` can close for $m = 3$, conjectured by the temperature search beyond, and never beaten anywhere the search reaches. What is missing is a general upper bound for $m \geq 3$, and the obstacle is structural rather than computational.

The route to the upper bound is to show that an extremal digraph, once it is past the hub regime, must look like the augmented bipartite construction: its vertices split into a part A and a part B with every arc running $A \rightarrow B$, and B thickened only mildly. Three structural facts, each proved in [Chapter A](#), push in this direction. Out-neighbourhoods are mutually unreachable, second out-neighbourhoods are disjoint, and a complete $A \rightarrow B$ partition forces each vertex of B to absorb at most $m-2$ arcs from inside B . With the partition free to vary, the total arc count $|B|(|A| + m - 2) = |B|(n - |B| + m - 2)$ is maximised at $|B| = \lceil (n + m - 2)/2 \rceil$, which gives $\lfloor (n + m - 2)^2/4 \rfloor$, precisely the conjectured upper bound.

The reduction is clean except at a single point, and it is worth naming the point exactly rather

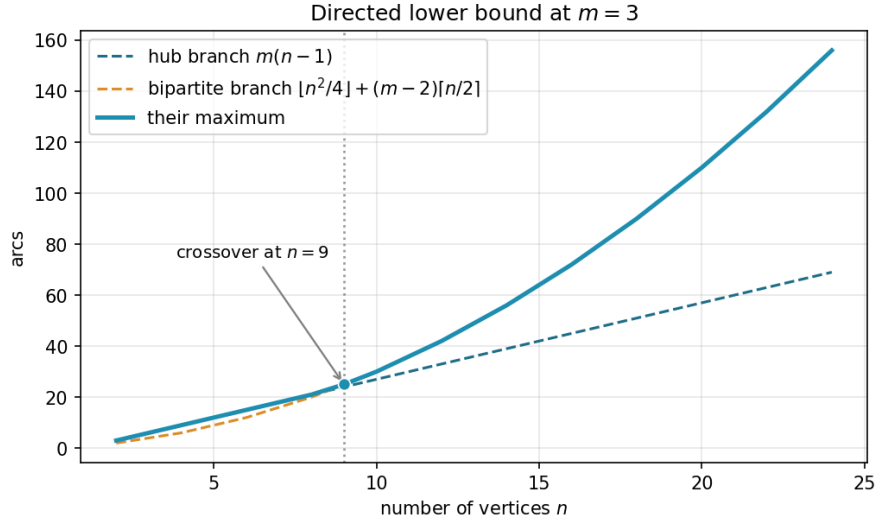


Figure 4.1. The two directed lower bounds at $m = 3$. The linear hub branch leads for small n . The quadratic augmented-bipartite branch overtakes it near $n = 9$ and leads thereafter. [Conjecture 1.13](#) asserts their maximum is the exact value.

than hiding it. The count goes through provided the extremal digraph has *no arc from B back to A* : under that hypothesis a route between two vertices of B can never escape B , so the connectivity inside B is governed entirely by B 's own arcs and the budget applies. The bare existence of a complete $A \rightarrow B$ partition is not enough on its own, because a small example shows a backward arc can manufacture an extra route. So the whole upper bound for $m \geq 3$ now rests on one missing lemma: that an extremal non-hub digraph has no backward arc. The natural delete-and-compensate exchange that would prove it is not monotone, which is the precise reason it remains open.

The machine has been useful here in the negative direction too: a proposed counterexample to the conjecture, a backward arc bolted onto the thirty-arc graph, was refuted by the checker, which found the third route the construction's author had missed.

The directed vertex-disjoint problem needs no separate machinery. Vertex-disjoint routes are at most as numerous as arc-disjoint ones, and the constructions that achieve the arc bounds also achieve them for vertices, so the two directed problems share both their constructions and their conjecture, and the $m = 2$ value transfers exactly in [Theorem 1.10](#). For $m \geq 3$ the vertex value is conjectured equal to the arc value, and it will follow from the arc case the moment the missing backward-arc lemma is supplied. The directed multigraph problem has its own two-branch story, parallel to the simple one. The conjectured value is

$$L_m^{\text{dir}}(n) = (m-1) \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right),$$

where the double star realises the linear branch and the one-directional bipartite multigraph, every arc at multiplicity $m-1$, realises the quadratic one.

The quadratic branch here uses the balanced partition $|A| = |B| = \lfloor n/2 \rfloor$, giving $(m-1)$

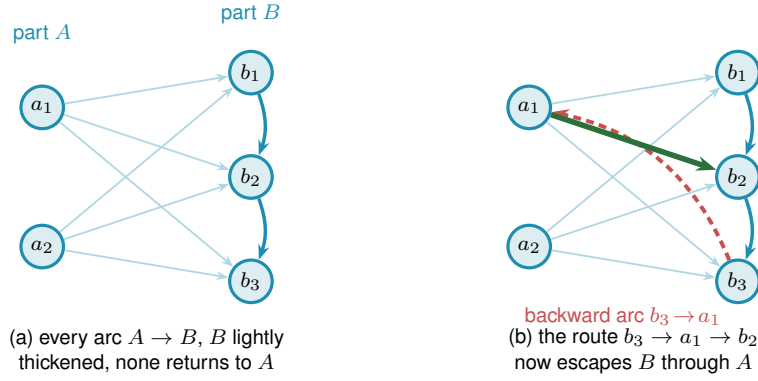


Figure 4.2. The directed frontier reduction and the single obstruction. **(a)** The structure the upper bound wants. Every arc runs from A to B (faint), B carries only a light internal thickening (the $m - 2$ budget, dark blue), and no arc returns to A , so a route between two vertices of B never leaves B and the count $|A||B| + (m - 2)|B|$ is complete. **(b)** A single backward arc $b_3 \rightarrow a_1$ (red) breaks this. The route $b_3 \rightarrow a_1 \rightarrow b_2$ (red leg then green leg) now escapes B through A , an extra route the budget never counted. Ruling out backward arcs in an extremal digraph is the one missing lemma behind the $m \geq 3$ upper bound, and the checker refuted a proposed counterexample by finding exactly such a hidden route.

1) $\lfloor n^2/4 \rfloor$ arcs. This is distinct from the simple-directed conjecture above, whose optimal partition shifts to $|B| = \lceil (n + m - 2)/2 \rceil$ at $m \geq 4$ but whose single-multiplicity arcs cannot benefit from scaling by $m - 1$. The two formulas agree for $m \leq 3$ and diverge only at $m \geq 4$, where the multigraph's multiplicity advantage dominates the partition shift. The branches cross at $n = 7$ for every m , because the factor $m - 1$ cancels, unlike the simple case, where the crossing point grows with m and already sits near $n = 9$ at $m = 3$.

On the linear side the certifier closes the problem, proving $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$ for $n \leq 6$ and every m at once (Theorem 2.1), and at $m = 2$ the whole problem collapses onto the simple digraph theorem for every n , because two parallel arcs are already two routes (Theorem A.24). On the quadratic side, a scaling-and-deletion induction proves the bipartite bound for even $n \geq 8$ conditional on the bound one vertex below, and that odd case once reduced to a single minimum-degree statement (Conjecture A.22), confirmed by the program in the purely saturated and purely unsaturated regimes but open in the mixed one, where multiplicities strictly between 0 and $m - 1$ defeat the integrality the two pure regimes lean on (Chapter A).

A sharper route now bypasses that conjecture at $m = 3$, and the way it does so is the cleanest instance in the thesis of the proof handing its remaining work to the machine. An *attachment lemma* (Lemma A.26) bounds the degree of a vertex glued onto the bipartite extremiser, and a *conditional odd step* built on it (Theorem A.28, for $m \in \{3, 4\}$) carries the value and extremal characterisation from each even level to the next odd one. At $m = 3$ that characterisation propagates through both parities (Remark A.29), so the entire problem for all $n \geq 8$ collapses onto two finite, integral statements at $n = 7$:

- (a) $L_3^{\text{dir}}(7) = 24$, and
- (b) the only 24-arc feasible multigraphs on 7 vertices with maximum degree at most 8 are $2 \cdot B_{3,4}$,

$2 \cdot B_{4,3}$, and the doubled bidirected path P_7 .

[Conjecture A.22](#) is thereby off the critical path for $m = 3$, replaced by two machine-checkable targets: a MILP infeasibility check for fact (a), and an exhaustive enumeration of degree-bounded multigraphs for fact (b). Both stay within memory but remain runtime-limited, and neither has closed yet.

The linear branch turns out to be richer than the double star first suggested. Every doubled bidirected spanning tree is extremal, confirmed exhaustively at $n = 4$ and 5 , and several distinct extremisers tie at 24 arcs for $n = 7$. Sifting and classifying them is precisely what the enumeration must do.

For $m \geq 4$ the conditional step still yields the value but uniqueness no longer propagates through odd levels, and for $m \geq 5$ even the value step stalls. These residual gaps are stated precisely among the open problems below. They are finite, identified targets rather than mysteries, and natural goals for the next round of certified computation.

4.2 Two principal phenomena

Across all the variation, two phenomena stand out as the ones a first guess would not have predicted.

The first is the divergence at $m = 5$. The edge and vertex problems agree for $m \leq 4$, and it would be natural to expect them to agree forever. Instead they part company at the very next value, with $k_5(n) = \lfloor 8n/3 \rfloor - 3$ pulling strictly above $\ell_5(n)$, and beyond it the vertex problem becomes genuinely hard and has stayed open since 1974.

The second is the quadratic bipartite phenomenon. In an undirected graph every edge contributes to the degree of both its endpoints, and Mader's theorem turns that symmetry into a hard ceiling: $\lfloor m(n-1)/2 \rfloor$ edges, linear in n . A digraph breaks the symmetry. Place all n vertices into two equal parts A and B and draw every arc from A to B , one direction only. The result has $\lfloor n^2/4 \rfloor$ arcs, quadratic in n , yet $\lambda^{\max} = 1$: every pair has at most one arc-disjoint route, because every route must cross the A -to- B wall exactly once and no arc crosses the other way to offer a second path. A graph with $\Theta(n^2)$ arcs can sit at connectivity one, something no undirected graph can do past a handful of vertices.

The initial conjecture for the directed case was linear, matching the undirected intuition. The one-directional bipartite construction refutes it outright, already at $m = 2$: for $n = 8$ the wall holds 16 arcs while the linear hub holds only 14. At $m = 3$ the refutation is the thirty-arc augmented bipartite graph of [Remark 1.11](#), which packs 30 arcs at $\lambda^{\max} = 2$ against the naive prediction of 27. The construction generalises to every m via [Construction 1.12](#), thickening the larger part B slightly so that each vertex of B absorbs $m - 2$ extra in-arcs from inside B : those extra arcs add exactly $m - 2$ short detours from A to each $b \in B$, bringing $\lambda^{\max}$ from 1 to $m - 1$ while the arc count climbs to $\lfloor (n + m - 2)^2/4 \rfloor$.

The quadratic term is what makes the directed cases the deepest in the thesis. The upper bound argument must explain why no configuration beats the bipartite wall, which requires ruling out backward arcs and controlling the internal structure of the large part B , and that structural task is exactly what the backward-arc lemma of [Chapter 1](#) has not yet resolved for $m \geq 3$. The undirected problem has no analogous obstacle, because Mader’s degree-counting argument closes it in a few lines. The quadratic phenomenon is therefore not just a curiosity: it is the geometric reason the directed and undirected problems live on different levels of difficulty.

That same scarcity of extremal graphs is the thread through the remaining figures. [Figure 4.3](#) shows where the frontier runs: it plots every labelled graph at small n as a single point, its binding connectivity $\lambda^{\max}$ on the horizontal axis and its edge count on the vertical, so the whole population of graphs is visible at once. The points crowd into the low-connectivity, low-edge corner and thin out upward, where the blue extremal envelope, an integer staircase carrying one dot per connectivity level, traces the densest graph available at each level. The empty region above that staircase is precisely the part the main theorems forbid a graph to occupy. The extremal problem is the question of where that staircase runs, and the figure shows it is a thin frontier with almost nothing pressed against it from below.

[Figure 4.4](#) steps inside each graph to the level of individual pairs. For every graph in the full enumeration it records the connectivity of every vertex pair and pools all the observations, so the histogram counts not graphs but (graph, pair) observations. Each bar is split and stacked by the connectivity of the graph the pair came from: blue for pairs drawn from the lower-connectivity graphs, red for pairs from the higher-connectivity ones. The boundary between the two is set per variant at the midpoint of the connectivity range that variant’s enumeration can reach, since the twelve variants span very different ranges and a single fixed boundary would leave some panels entirely one colour. A pair whose own connectivity exceeds the boundary can only come from a graph above it, so the bars to the right of the dashed line are wholly red, while to the left the two colours mix, since a high-connectivity graph still carries many low-connectivity pairs. The blue mass sits almost entirely at the lowest levels: high-connectivity pairs are rare even inside the graphs that have one.

[Figure 4.5](#) then shows how the same graphs distribute by sheer edge count, with the same per-variant stacked colouring. At each edge count the blue part counts the lower-connectivity graphs and the red part the higher-connectivity ones, so the bar height is the total number of graphs there and the split shows how the two populations separate. The red mass sits to the right, since denser graphs tend to carry higher connectivity, and the dotted line marks the densest graph still on the low-connectivity side, landing far out in the sparse tail. The densest low-connectivity graphs, the ones the extremal problem is about, are exceptional in every variant.

Finally, [Figure 4.6](#) lifts the whole study onto the (n, m) grid at once, drawing the best feasible edge count as a bar surface across every variant. The three proved variants rise as smooth linear or quadratic sheets that the formulas describe exactly, while the open variants show a visible step between the machine-proved blue bars and the search’s purple lower bounds, so the

Separation	Model	Value	Status
<i>Undirected</i>			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (cited GH)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$), $\lfloor 8n/3 \rfloor - 3$ ($m = 5$)	cited, open $m \geq 6$
vertex	multi	parallel edges do not help	proved
vertex	hyper	$\lfloor \frac{(m-1)(n-1)}{r-1} \rfloor$ ($m \leq 3$), $\geq$ that for all m	proved $m \leq 3$ (Theorems A.34 and A.37), open $m \geq 4$
<i>Directed</i>			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$)	proved, conj. $m \geq 3$
arc	multi	$(m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$	proved $n \leq 6$ and $m=2$, cond. $m=3$
arc	hyper	quadratic in n , exact value open	lower bd. (Proposition A.39)
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$)	proved, conj. $m \geq 3$
vertex	multi	reduces to simple digraph	proved
vertex	hyper	quadratic in n , exact value open	lower bd. (Proposition A.39)

Table 4.1. The twelve structural variants of Problem 915 and their status. The threshold result $p^* = m/n$ applies uniformly across all models and separations and is discussed separately in Chapter 1 and Chapter 4 as a sampling observation. “Certified” means proved by the cut-counting optimisation for the stated sizes. “conj.” means a conjecture with a proved matching lower bound. “cond.” means proved conditional on a finite, identified statement (for $m = 3$ directed multigraphs, the two $n = 7$ computations of Remark A.29). “lower bd.” means only a proved lower bound is known, with the exact value open.

surface makes the proved-versus-open divide of the trichotomy a single readable shape. Reading along the m axis recovers the linear scaling in the threshold, and reading along n recovers each variant’s growth rate, which is the whole of Table 4.1 compressed into one object.

4.3 Summary of the twelve variants

Table 4.1 records the status of all twelve structural variants. Each entry is marked by how it is known: proved by hand here or in Chapter A, cited to the literature, certified by the cut-counting optimisation for the stated sizes, or conjectured with a proved lower bound. The distinction is the honesty contract of Chapter 2 carried through to the summary.

4.4 Contributions and limitations of the program

The program turned a per-case craft into a uniform pipeline. Before it, each variant came with its own one-off computation and its own hand argument. After it, one representation holds every variant, one exact checker measures connectivity, one certifier proves small-case upper bounds, and one cooled search discovers extremisers and the lower bounds they witness. The rediscovery of Chapter 3 is the evidence that the pipeline is faithful: pointed at solved cases, it returns the solved answers.

Its limits are as clear as its reach, and stating them is part of the honesty the thesis insists on. The search proves no upper bounds. It only ever exhibits graphs. The certifier proves upper bounds, but only for a fixed size, and the directed multigraph encoding closed unconditionally only up to $n = 6$. The deletion-cut strengthening made the two $m = 3$ computations at $n = 7$ memory-

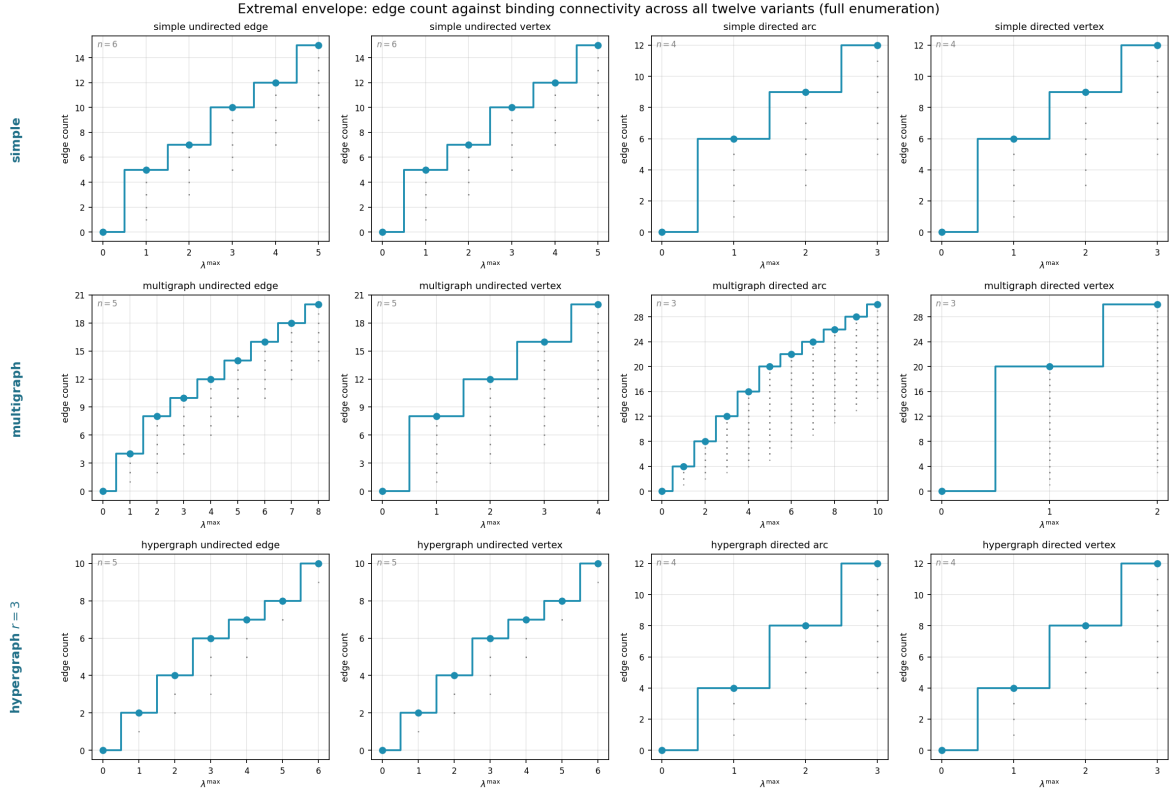


Figure 4.3. The complete landscape of all labelled graphs at small n (see panel annotations for each variant's n): every point is one graph, plotted at its binding connectivity $\lambda^{\max}$ horizontally and its edge count vertically. The blue staircase marking the densest graph at each connectivity level is the extremal envelope, the frontier the extremal problem asks about, with one dot per level so its exact value is legible. Graphs crowd the low-connectivity, low-edge region. The extremal graphs live at the top of the staircase. The emptiness above the envelope is exactly the region the main theorems forbid. The horizontal reach differs by panel because the connectivity that fits in a variant's enumeration n differs: arc-disjoint multigraph routes grow with edge multiplicity (the multigraph directed arc panel reaches $\lambda^{\max} = 10$ at $n = 3$), whereas vertex-disjoint routes are capped by the available internal vertices, so the multigraph directed *vertex* panel reaches only $\kappa^{\max} = 2$ at the same $n = 3$ (with a single internal vertex, a pair has at most the direct route plus one detour). This is a property of the separation, not of the search.

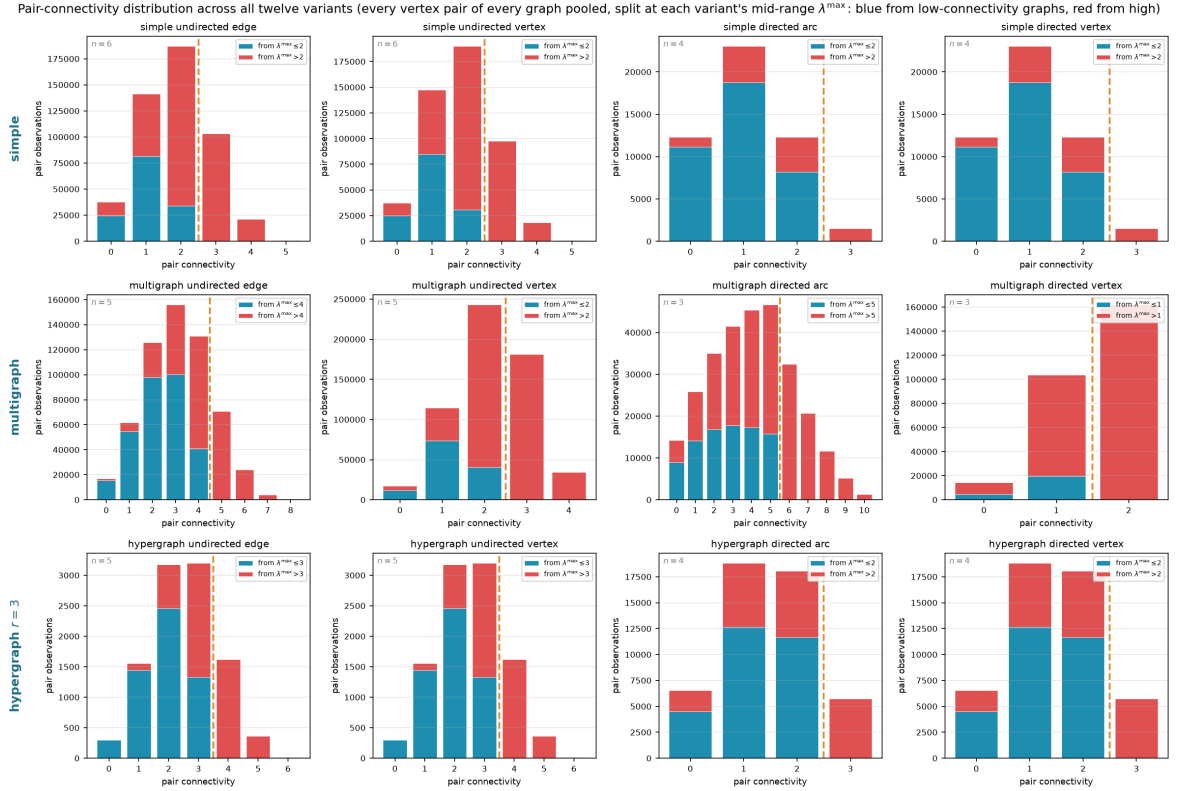


Figure 4.4. Pair-connectivity distribution across all twelve variants, pooling every vertex pair of every graph in the full enumeration. The horizontal axis is the connectivity of a single pair, and a bar's height is the number of (graph, pair) observations at that level. Each bar is split and stacked by the connectivity of the graph the pair was drawn from, blue from graphs at or below a per-panel boundary T and red from graphs above it, with the dashed line at T . The boundary is set per variant to the midpoint of the connectivity range that variant's enumeration reaches (each panel's legend gives its T), so the split stays informative in every panel instead of collapsing to one colour. Bars to the right of the dashed line are wholly red, since a pair of connectivity above T forces its graph above T . To the left the colours mix.

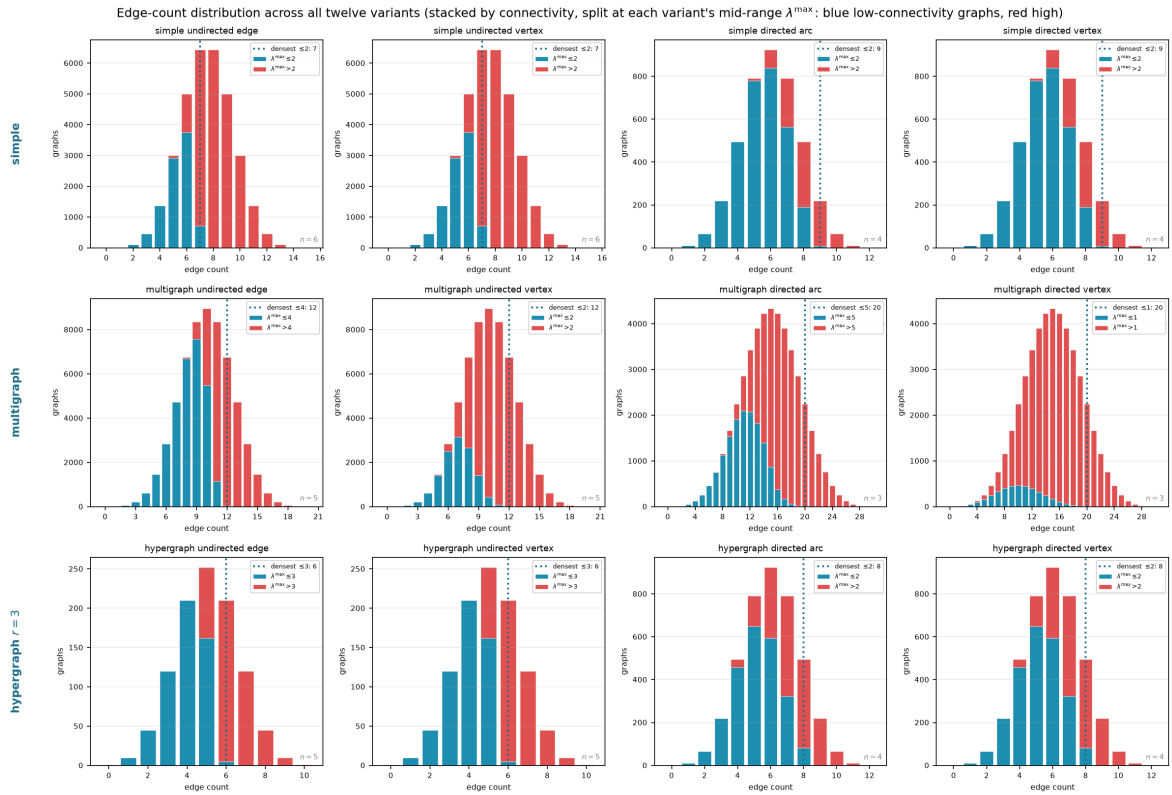


Figure 4.5. Edge-count distribution across all twelve variants, the same enumeration viewed by edge count rather than connectivity. Each bar is split and stacked by graph connectivity, blue for graphs at or below the per-panel boundary T and red above it (each panel's legend gives its T), so the full height is the total number of graphs at that edge count. The dotted vertical line marks the densest graph on the low-connectivity side. The reader can see directly how the two populations separate by edge count and how far low-connectivity graphs sit from the bulk.

Erdős 915: optimal bound over (n, m) , blue where proved, purple where only a search lower bound is known

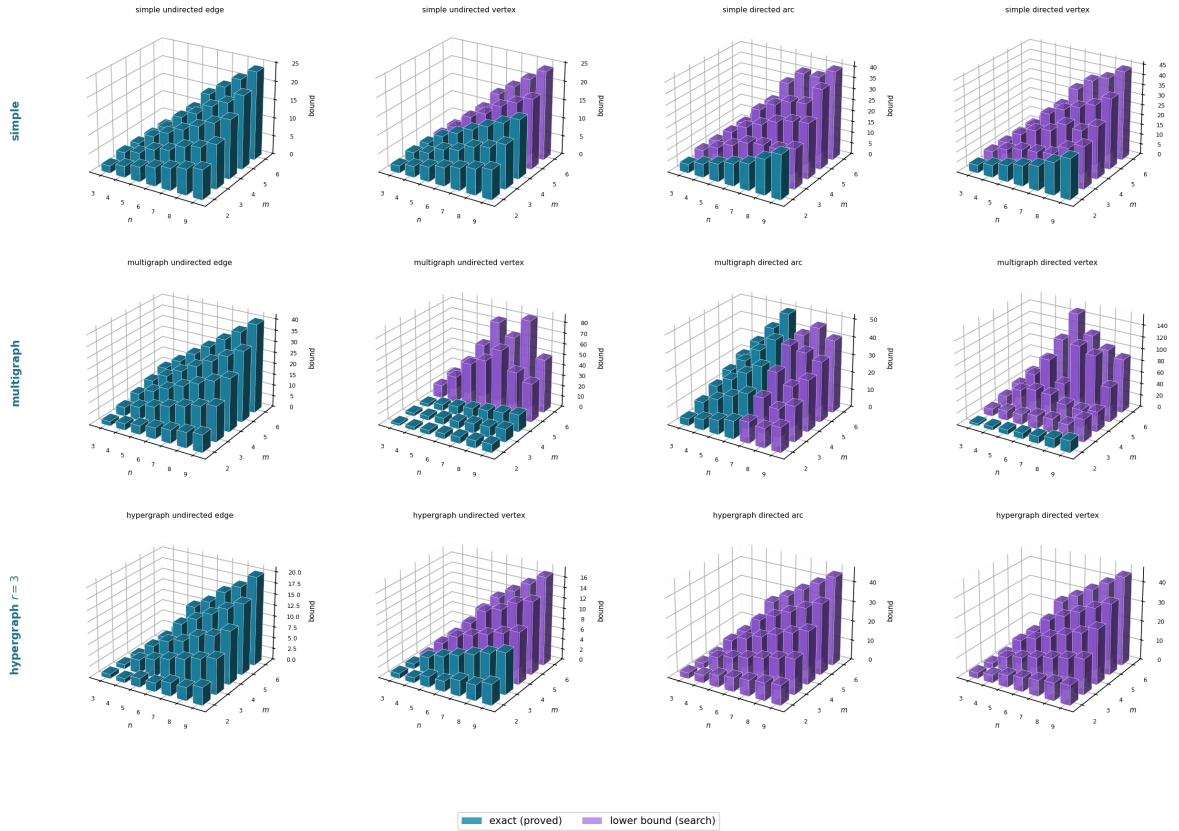


Figure 4.6. The optimal bound as a surface over the (n, m) grid for all twelve variants, every panel drawn from the same camera so the shapes compare directly. Each bar's height is the largest feasible edge count found by `solve` at that vertex count and forbidden threshold m . The single thing to read off is colour: a bar is *blue* exactly where the value is machine-proved and *purple* exactly where only a search lower bound is known, so each panel is a little map of what is settled. The three fully proved variants (Mader's undirected edge cases and the multigraph analogue) are solid blue sheets, rising linearly or quadratically as their formulas predict, while the open variants turn purple precisely where proof runs out. Reading along the m axis shows how the bound scales with the forbidden threshold, and reading along the n axis shows the growth rate for a fixed variant.

safe, but they remain runtime-limited and have not yet returned certificates. And no amount of either replaces the two genuinely structural gaps the thesis leaves open: the backward-arc lemma that would turn [Conjecture 1.13](#) into a theorem, and the vertex-disjoint problem at $m \geq 6$, which has resisted for half a century and which the program can probe but not crack.

4.5 Open problems

- ★ The backward-arc lemma: prove that an extremal non-hub digraph has no arc from B to A , completing the upper bound in [Conjecture 1.13](#) for $m \geq 3$, and with it the directed vertex case.
- ★ The two finite $n = 7$ computations that close the directed multigraph problem at $m = 3$. By the attachment lemma and the conditional odd step ([Lemma A.26](#), [Corollary A.27](#), [Theorem A.28](#)), the entire $m = 3$ problem (value and extremal uniqueness for all $n \geq 8$) reduces along a single induction to two integral statements about multigraphs with multiplicities in $\{0, 1, 2\}$ ([Remark A.29](#)): (a) $L_3^{\text{dir}}(7) = 24$, and (b) every 24-arc feasible multigraph on 7 vertices with maximum degree at most 8 is $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, or the doubled bidirected path P_7 . Both sit at sizes the certifier and the enumerator can reach, but the current implementations have not closed them: the MILP infeasibility check for (a) reaches its time limit, and the exhaustive enumeration for (b) stays within memory yet remains runtime-limited, so this is a finite and machine-checkable gap rather than a structural one. [Conjecture A.22](#) is thereby bypassed for $m = 3$, though it remains open and of independent interest as the natural minimum-degree statement for the directed multigraph problem.
- ★ Extremal uniqueness at odd levels for directed multigraphs with $m \geq 4$. The conditional odd step ([Theorem A.28](#)) and the propagation of [Remark A.29](#) are stated for $m \in \{3, 4\}$. For $m = 4$ the *value* step goes through, but at an odd extremal level the counting no longer forces a vertex of degree $(m-1)k$ (all degrees $\geq (m-1)k+1$ are arithmetically consistent), so the uniqueness statement does not yet propagate. For $m \geq 5$ the same escape climbs one level and even the *value* step stalls: a surplus extremiser could be degree-regular strictly above the deletion bound, so no vertex deletion is forced to land on an extremal multigraph. Closing either hole asks for an arithmetic or structural input the present averaging argument does not supply.
- ★ The undirected vertex-disjoint problem for $m \geq 6$, open since 1974, where even the extremal structure is unknown.
- ★ The hypergraph vertex problem at $m \geq 4$. The problem is now settled at $m = 2$ and $m = 3$ for every uniformity and all n , with $k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor = \ell_m^{(r)}(n)$ ([Theorems A.34](#) and [A.37](#)). The incidence reduction stands for every m , but the rank bound that drives it ([Lemma A.36](#)) leans on the triconnected (Tutte/SPQR) decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$ it would need a 4-connectivity analogue, where no comparably clean decomposition is available. Exhaustive computation gives $k_4^{(3)}(5) = 6 = \lfloor 3 \cdot 4/2 \rfloor$, so the agreement $k_m^{(r)} = \ell_m^{(r)}$ survives

at least one step past the theorem and there is no early counterexample. Where it first breaks for $r \geq 3$ (the analogue of the $m = 5$ graph divergence) is unknown.

- ★ The directed hypergraph variants. The tail-and-heads model has first bounds and a quadratic lower-bound family ([Proposition A.39](#)): a pair-codegree count caps the hyperedges at $(m - 1)n(n - 1)/(r - 1)$, and a bipartite construction reaches $\approx (m - 1)n^2/(4(r - 1))$, so the value is quadratic in n . Its exact order, and the matching upper construction, are unexplored.

The map is filled in where honest mathematics could fill it, and where it could not, the blanks are marked precisely, with a machine standing ready at each of them. That, finally, is the use of building the microscope: not that it answers every question, but that it shows exactly which questions are left, and hands the next reader a clean place to start.

Appendix A

Full Proofs and Computational Transcripts

This appendix holds the arguments deferred from the body: the ones that are either standard machinery or long case-checking, and whose length, not whose difficulty, kept them out of the narrative. The body carries the ideas. Here they are carried out in full. Because graphs are visual objects, the longer arguments here are accompanied by figures: each is drawn rather than merely described, and the prose points the reader to the picture at the moment it is needed.

How to read this appendix. The sections divide into standard machinery and the thesis's own proofs, and a reader chasing one thread of the body need not read all of them. The first two sections, on [Theorem A.1](#), [Theorem A.2](#), and Mader's theorem, are classical results included only so the body can cite them in the exact form it uses, and a reader who already trusts Menger, Gomory–Hu, and Mader may skip them. Three sections carry the load-bearing original arguments. The directed arc value at $m = 2$ supplies the finite base cases the body's induction rests on. The directed multigraph problem at large n is the longest argument here and the one behind the main directed contribution. The hypergraph vertex problem proves the incidence-rank lemma that settles the $m = 3$ hypergraph case. The remaining sections, on the hypergraph edge bounds and the random threshold, each support a single claim of the body and can be read on their own when that claim is reached. Finally, everything the program asserts is reproduced verbatim in the computational transcripts and the self-test section, so the machine proofs can be audited without running anything.

A.1 Menger, Gomory–Hu, and the degree bound

The two classical theorems below are the foundation of everything the thesis measures, and we state them in the form we use them.

Theorem A.1 (Menger [15]). *For any graph G and distinct vertices u, v , the maximum number of pairwise edge-disjoint u – v paths equals the minimum number of edges whose removal separates u from v :*

$$\lambda_G(u, v) = \min\{|C| : C \subseteq E(G), G - C \text{ has no } u\text{--}v \text{ path}\}.$$

The same statement holds for digraphs with arc-disjoint directed paths and directed arc cuts, and, replacing edges by internal vertices, for internally vertex-disjoint paths and vertex cuts.

Theorem A.2 (Gomory–Hu [7]). *Every weighted graph G has a weighted tree T on the same vertex set such that for all u, v the value $\lambda_G(u, v)$ equals the minimum edge weight on the $u-v$ path in T , and each tree edge e corresponds to a minimum cut of G whose capacity is the weight of e .*

The most basic constraint on local connectivity is local: a pair cannot have more independent routes than either endpoint has edges. The constructions below use it constantly.

Lemma A.3 (Degree bound). *For every pair of vertices of a graph G , $\lambda_G(u, v) \leq \min(\deg u, \deg v)$. In a digraph D , $\lambda_D(u, v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint $u-v$ path uses a distinct edge at u , so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct out-arc and enters v by a distinct in-arc. $\square$

A.2 Mader’s theorem in full

The upper bound of [Theorem 1.2](#) rests on three short lemmas about a Gomory–Hu tree T of G . For an edge $e = \{u, v\}$ of G , write $\text{dist}_T(e)$ for the number of tree edges on the unique $u-v$ path in T . Since T is a tree on the very same vertex set, $\text{dist}_T(e) \geq 1$ for every edge. [Figure A.1](#) shows a small graph beside one of its Gomory–Hu trees and marks which edges have $\text{dist}_T(e) = 1$ and which have $\text{dist}_T(e) \geq 2$. It is worth keeping in view, since the three lemmas below are exactly statements about that picture. The first lemma is the one place the proof uses that G is *simple*, and it is exactly where the simple-graph bound improves on the multigraph bound.

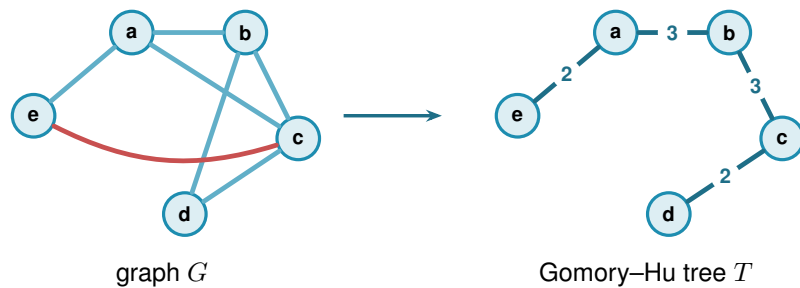


Figure A.1. A simple graph G (left) and a Gomory–Hu tree T of it (right). Both live on the same five vertices, and here T is the path $e-a-b-c-d$ with the marked cut weights. The four edges ea, ab, bc, cd are also tree edges, so each has $\text{dist}_T(e) = 1$: these are the E_1 edges of [Lemma A.4](#), and a *simple* graph admits at most $n-1 = 4$ of them, one per tree edge. The edges ac and bd join tree-distance-2 pairs, and the red chord ec has $\text{dist}_T(ec) = 3$, crossing all three cuts on the path $e-a-b-c$. Reading $\text{dist}_T(e)$ as the number of tree cuts an edge crosses makes $\sum_e \text{dist}_T(e) = \sum_t c(t)$ ([Lemma A.6](#)). Every edge crosses at least one cut, but only the $n-1$ tree edges cross exactly one, which is the surplus that drives [Lemma A.5](#) and ultimately the factor of two in Mader’s bound.

Lemma A.4 (Distance-one edges). *At most $n-1$ edges of a simple graph G satisfy $\text{dist}_T(e) = 1$.*

Proof. An edge $e \in E(G)$ has $\text{dist}_T(e) = 1$ if and only if its endpoints are adjacent in T . The tree

T has exactly $n - 1$ edges, and since G is simple each pair of adjacent tree vertices supports at most one edge of G . $\square$

Lemma A.5 (Sum of tree distances). $\sum_{e \in E(G)} \text{dist}_T(e) \geq 2|E(G)| - (n - 1)$.

Proof. Partition $E(G)$ into $E_1 = \{e : \text{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \text{dist}_T(e) \geq 2\}$. Then

$$\sum_{e \in E(G)} \text{dist}_T(e) \geq |E_1| + 2|E_{\geq 2}| = 2|E(G)| - |E_1| \geq 2|E(G)| - (n - 1),$$

using [Lemma A.4](#) in the last step. $\square$

Lemma A.6 (Double-counting identity). *For any graph G and Gomory–Hu tree T with weights c ,*

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \text{dist}_T(e).$$

Proof. By [Theorem A.2](#) each tree edge t induces a cut of G of capacity $c(t)$, so $c(t) = \sum_{e \in E(G)} \mathbf{1}_{e \text{ crosses } t}$. Exchanging the order of summation,

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \sum_{t \in E(T)} \mathbf{1}_{e \text{ crosses } t} = \sum_{e \in E(G)} \text{dist}_T(e),$$

where the last equality holds because an edge $\{u, v\}$ crosses exactly those tree cuts induced by the tree edges on the unique u – v path in T . $\square$

Proof of [Theorem 1.2](#), upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m - 1$ and let T be a Gomory–Hu tree of G . Each tree edge $t = \{x, y\}$ has weight $c(t) = \lambda_G(x, y) \leq m - 1$, so summing over the $n - 1$ tree edges, $\sum_t c(t) \leq (m - 1)(n - 1)$. Combining with [Lemmas A.5](#) and [A.6](#),

$$2|E(G)| - (n - 1) \leq \sum_{e \in E(G)} \text{dist}_T(e) = \sum_{t \in E(T)} c(t) \leq (m - 1)(n - 1).$$

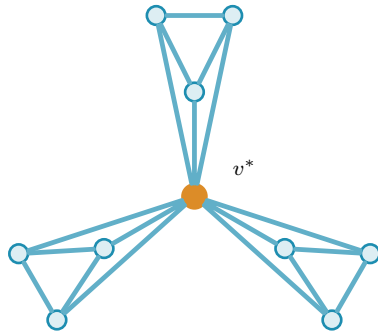
Rearranging gives $2|E(G)| \leq m(n - 1)$, and since $|E(G)|$ is an integer, $|E(G)| \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. $\square$

Note where the factor of two came from: every edge crosses at least one tree cut, but a simple graph can afford at most $n - 1$ edges that cross only one, so on average each edge is charged closer to twice. For multigraphs that refinement is unavailable, since parallel edges between tree-adjacent pairs each have $\text{dist}_T = 1$, and the same chain of inequalities stops at $|E| \leq \sum_e \text{dist}_T(e) \leq (m - 1)(n - 1)$, which is exactly the multigraph bound of [Theorem 1.3](#).

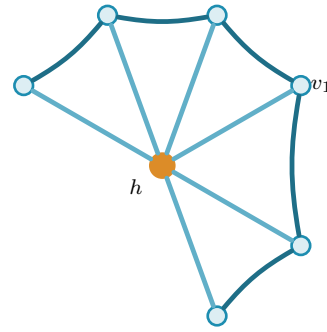
The lower bound is a pair of constructions, drawn side by side in [Figure A.2](#). The first is the shape the bound was guessed from, and works when $m - 1$ divides $n - 1$. The second generalises it to every $n \geq m \geq 2$.

Construction A.7 (Shared-clique graph). For $m \geq 3$ with $(m-1) \mid (n-1)$, take $k = (n-1)/(m-1)$ copies of the complete graph K_m and identify one vertex of each copy into a single shared hub vertex v^* . The result has $k(m-1) + 1 = n$ vertices and $k \binom{m}{2} = \frac{m(n-1)}{2}$ edges. Every non-hub vertex has degree $m-1$, and any pair of vertices contains a non-hub vertex, so $\lambda^{\max} \leq m-1$ by Lemma A.3.

Construction A.8 (Hub-and-spokes graph). For $n \geq m \geq 2$, let $H_{m,n}$ have vertex set $\{h, v_1, \dots, v_{n-1}\}$ with (i) all $n-1$ hub edges $\{h, v_i\}$, and (ii) a spoke graph on $\{v_1, \dots, v_{n-1}\}$ with $\left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor$ edges in which every v_i has spoke degree at most $m-2$. Such a spoke graph exists by Lemma A.9.



(a) Shared-clique graph: three K_4 's glued at one hub v^* ($m = 4$, $k = 3$). Each non-hub vertex has degree $m-1 = 3$.



(b) Hub-and-spokes $H_{m,n}$: the hub edges $\{h, v_i\}$ (light) plus a sparse spoke graph (dark) of spoke-degree at most $m-2$.

Figure A.2. The two extremal families. (a) The shared-clique graph of Construction A.7 works when $(m-1) \mid (n-1)$: every pair contains a non-hub vertex of degree exactly $m-1$, so $\lambda^{\max} \leq m-1$ by the degree bound, while the edge count hits $\frac{m(n-1)}{2}$. (b) The hub-and-spokes graph $H_{m,n}$ of Construction A.8 reaches the same bound for every $n \geq m$: each spoke vertex v_i has total degree $1 + (m-2) = m-1$, and choosing the spoke graph as disjoint cliques K_{m-1} recovers (a). The hub is drawn in orange in both.

Lemma A.9 (Near-regular graphs exist). For $N \geq 1$ and $0 \leq r \leq N-1$ there is a graph on N vertices with $\left\lfloor \frac{rN}{2} \right\rfloor$ edges and maximum degree at most r .

Proof. If rN is even we build an r -regular circulant graph on vertex set $\{0, \dots, N-1\}$: for even r connect each i to $i \pm j \pmod{N}$ for $j = 1, \dots, r/2$, and for odd r (then N is even) use $j = 1, \dots, (r-1)/2$ together with the diameter distance $N/2$, which contributes degree one. Both are r -regular with $rN/2$ edges. If rN is odd, then r and N are both odd: take the $(r-1)$ -regular circulant with distances $1, \dots, (r-1)/2$ and add the near-perfect matching that pairs vertex i with $i + (N+1)/2 \pmod{N}$ for $i = 0, \dots, (N-3)/2$, which covers all vertices but one. The total is $\frac{(r-1)N}{2} + \frac{N-1}{2} = \frac{rN-1}{2} = \left\lfloor \frac{rN}{2} \right\rfloor$ edges, with one vertex of degree $r-1$ and all others of degree r . $\square$

Theorem A.10 (Extremal graphs exist). For all $n \geq m \geq 2$ the graph $H_{m,n}$ has $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ edges and $\lambda^{\max} \leq m-1$.

Proof. The edge count is $(n-1) + \left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, since $(m-2)(n-1)$ and $m(n-1)$ have the same parity. For the connectivity, any two distinct vertices include at least one spoke vertex v_i , whose total degree is $1+(m-2) = m-1$. [Lemma A.3](#) then gives $\lambda(u, v) \leq m-1$. When $(m-1) \mid (n-1)$, choosing the spoke graph as k disjoint cliques K_{m-1} recovers [Construction A.7](#), so the hub-and-spokes family genuinely generalises the shared-clique one. $\square$

Together the upper bound and [Theorem A.10](#) prove [Theorem 1.2](#). The proof also reveals which graphs are extremal: equality forces both inequalities in the chain to be tight.

Theorem A.11 (Characterisation of equality). *Let G be simple with $\lambda_G^{\max} \leq m-1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G . If $m(n-1)$ is even, then every tree edge has weight exactly $m-1$, every tree edge is also an edge of G , and every edge of G joins vertices at tree distance at most 2. If $m(n-1)$ is odd, there is exactly one unit of slack, located in exactly one of three places: one tree edge has weight $m-2$, or one tree edge is not an edge of G , or one edge of G joins vertices at tree distance 3. Everything else is tight.*

Proof. Write the chain of the upper-bound proof as three nonnegative slacks. The first, $S_1 = (m-1)(n-1) - \sum_t c(t)$, counts tree weights below $m-1$. The second, $S_2 = \sum_{e: \text{dist}_T(e) \geq 2} (\text{dist}_T(e) - 2)$, counts edges beyond tree distance two. The third, $S_3 = (n-1) - |E_1|$, counts tree edges that support no edge of G . Retracing the proof, $\sum_e \text{dist}_T(e) = 2|E| - |E_1| + S_2$ and $\sum_e \text{dist}_T(e) = \sum_t c(t) = (m-1)(n-1) - S_1$, so altogether $2|E| = m(n-1) - (S_1 + S_2 + S_3)$. If $m(n-1)$ is even, equality $|E| = m(n-1)/2$ forces $S_1 = S_2 = S_3 = 0$. If $m(n-1)$ is odd the floor absorbs exactly one unit, so $S_1 + S_2 + S_3 = 1$ and exactly one of the three slacks equals one. $\square$

For $m = 2$ the characterisation says the extremal graphs are exactly the spanning trees: every tree edge must carry weight one and lie in G , so G contains its own Gomory–Hu tree and has only $n-1$ edges to spend. For $m = 4$ and $n \equiv 1 \pmod{3}$ it is consistent with the trees of K_4 blocks glued at shared cut vertices, the case of [Construction A.7](#) that the search of [Chapter 3](#) rediscovers.

A.3 The directed arc value at $m = 2$

Three short lemmas feed the induction. The first shows a hub forces a linear bound for every m , and is the proof behind the hub branch of [Conjecture 1.13](#).

Lemma A.12 (Two-hop lemma). *Let D be a simple digraph with $\lambda_D^{\max} \leq m-1$ containing a vertex r with $d^+(r) = n-1$. Then every $v \neq r$ has at most $m-2$ in-arcs from $V \setminus \{r, v\}$.*

Proof. If $u_1, \dots, u_{m-1}$ were distinct in-neighbours of v other than r , the paths $r \rightarrow v$ and $r \rightarrow u_i \rightarrow v$ for $i = 1, \dots, m-1$ would be m arc-disjoint routes, since the u_i are distinct and $d^+(r) = n-1$ supplies every starting arc. That contradicts $\lambda_D(r, v) \leq m-1$. $\square$

Theorem A.13 (Hub upper bound). *If a simple digraph D with $\lambda_D^{\max} \leq m - 1$ has a vertex r with $d^+(r) = n - 1$, then $|A(D)| \leq m(n - 1)$.*

Proof. Split the arcs into those leaving r (exactly $n - 1$), those entering r (at most $n - 1$), and the internal arcs avoiding r . By [Lemma A.12](#) each of the $n - 1$ other vertices receives at most $m - 2$ internal arcs, so there are at most $(m - 2)(n - 1)$ internal arcs, and the total is at most $m(n - 1)$. $\square$

Lemma A.14 (No transitive triangle). *A simple digraph with $\lambda^{\max} \leq 1$ contains no three vertices x, y, z with all of $(x, y), (y, z), (x, z)$ present, since $x \rightarrow z$ and $x \rightarrow y \rightarrow z$ would be two arc-disjoint routes.*

Lemma A.15 (Deletion monotonicity). *For any digraph D and vertex v_0 , $\lambda^{\max}(D - v_0) \leq \lambda^{\max}(D)$, since arc-disjoint paths in $D - v_0$ are arc-disjoint paths in D .*

Proof of Theorem 1.9. Write $M(n) = \max(2(n - 1), \lfloor n^2/4 \rfloor)$. A direct computation shows $2(n - 1) \geq \lfloor n^2/4 \rfloor$ exactly for $n \leq 7$, with equality at $n = 7$, so $M(n) = 2(n - 1)$ for $n \leq 7$ and $M(n) = \lfloor n^2/4 \rfloor$ for $n \geq 7$. The lower bound $\ell_2^{\text{dir}}(n) \geq M(n)$ is witnessed by the directed hub construction ([Construction 1.8](#) at $m = 2$, the bidirected star) and the one-directional bipartite construction ([Construction 1.7](#)).

For the upper bound we show by strong induction on n that $\lambda_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$.

Base cases $2 \leq n \leq 7$. For $n = 2$ there are at most 2 arcs and $M(2) = 2$. For $n = 3$, suppose $|A(D)| \geq 5$: then D misses at most one arc of the complete bidirected triangle, and whichever arc is missing, three of the remaining arcs form a transitive triangle, contradicting [Lemma A.14](#), hence $|A(D)| \leq 4 = M(3)$. For $4 \leq n \leq 7$ the pruned exhaustive search of [Chapter 2](#) walks every simple digraph with $\lambda^{\max} \leq 1$ and reports the maxima 6, 8, 10, 12, matching $M(n)$. The transcript is reproduced below, and the cut-counting certifier independently confirms the values within its reach through the $m = 2$ reduction of [Theorem A.24](#).

Inductive step $n \geq 8$. Let D be a simple digraph on n vertices with $\lambda_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 minimising the total degree $d(v) = d^+(v) + d^-(v)$. Since $\sum_v d(v) = 2a$, averaging gives $d(v_0) \leq 2a/n$. By [Lemma A.15](#) the digraph $D - v_0$ on $n - 1 \geq 7$ vertices satisfies the induction hypothesis, so $a - d(v_0) \leq M(n - 1) = \lfloor \frac{(n-1)^2}{4} \rfloor$. Combining,

$$a \leq \frac{2a}{n} + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \quad \implies \quad a \leq \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

It remains to check that this right-hand side never exceeds $\lfloor n^2/4 \rfloor$, and the two parities behave differently enough to treat them in turn.

For even $n = 2k$ with $k \geq 4$, first simplify $\lfloor (n - 1)^2/4 \rfloor$. Expanding $(2k - 1)^2 = 4k^2 - 4k + 1$

and dividing by 4 gives $k^2 - k + \frac{1}{4}$, whose floor is $k(k-1)$. The bound then reads

$$a \leq \frac{2k}{2k-2} k(k-1) = \frac{k}{k-1} k(k-1) = k^2,$$

and since $\lfloor n^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$ as well, the bound lands exactly on the target with no slack left over.

For odd $n = 2k+1$ with $k \geq 4$, the same simplification gives $\lfloor (n-1)^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$, so the bound is $a \leq \frac{2k+1}{2k-1} k^2$. To compare this fraction with the target, split off its integer part by dividing $(2k+1)k^2$ by $2k-1$. The remainder is exactly k , because $(2k-1)(k^2+k) = 2k^3+k^2-k$ and $(2k+1)k^2 = 2k^3+k^2$ differ by k , so

$$\frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor = \frac{(2k+1)k^2}{2k-1} = k(k+1) + \frac{k}{2k-1}.$$

The leftover fraction $\frac{k}{2k-1}$ lies strictly between 0 and 1 for every $k \geq 1$, so the bound puts a at most an integer $k(k+1)$ plus a positive amount below one. Since a is itself an integer it cannot reach the next integer up, so $a \leq k(k+1)$. The target $\lfloor n^2/4 \rfloor = \lfloor (2k+1)^2/4 \rfloor = k^2 + k = k(k+1)$ agrees, so for odd n it is integrality, not the raw arithmetic, that closes the last fraction of a unit. In both parities $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction. $\square$

Two remarks on the shape of this proof. First, the minimum-degree deletion closes the step on its own, and no separate hub case is needed, because the averaging bound holds whether or not D contains a hub, although [Theorem A.13](#) shows directly that a hub digraph never exceeds the linear branch. Second, for even n the bound is exactly tight against the one-directional bipartite construction, which is $\frac{n}{2}$ -regular in total degree, so every step of the chain is achieved by the conjectured extremiser, and for odd n the slack $\frac{k}{2k-1} < 1$ is absorbed by integrality. The proof is long not because it is deep but because the two regimes of $M(n)$ must be carried through the arithmetic separately, and that bookkeeping is precisely what [Chapter 2](#) hands to the machine at the base cases.

Proof of [Theorem 1.10](#). By Whitney's inequality $\kappa^{\max} \leq \lambda^{\max}$, every digraph feasible for the vertex problem is feasible for the arc problem, so $k_2^{\text{dir}}(n) \leq \ell_2^{\text{dir}}(n)$. Conversely the hub and one-directional bipartite constructions have $\kappa^{\max} = 1$, so they are feasible for the vertex problem and give the matching lower bound. Hence $k_2^{\text{dir}}(n) = \ell_2^{\text{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. $\square$

A.4 Structural propositions on the directed frontier

These are the proved ingredients of the reduction discussed in [Chapter 4](#). The upper bound for $m \geq 3$ is conditional only on the backward-arc lemma stated there.

Proposition A.16 (Mutually unreachable out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$. For any vertex u and distinct out-neighbours v, w of u , there is no directed path from*

v to w in $D - u$.

Proof. If P were a directed v – w path avoiding u , then $u \rightarrow w$ and $u \rightarrow v \xrightarrow{P} w$ would be two arc-disjoint u – w routes: the first uses only the arc (u, w) , the second starts with (u, v) and continues along P , none of whose arcs touch u . This contradicts $\lambda_D(u, w) \leq 1$. The restriction to $D - u$ is not cosmetic, and Figure A.3 draws both the argument and the reason for the restriction: in the digraph with arcs (u, v) , (v, u) , (u, w) every local connectivity is 1, yet $v \rightarrow u \rightarrow w$ reaches w from v through u . $\square$

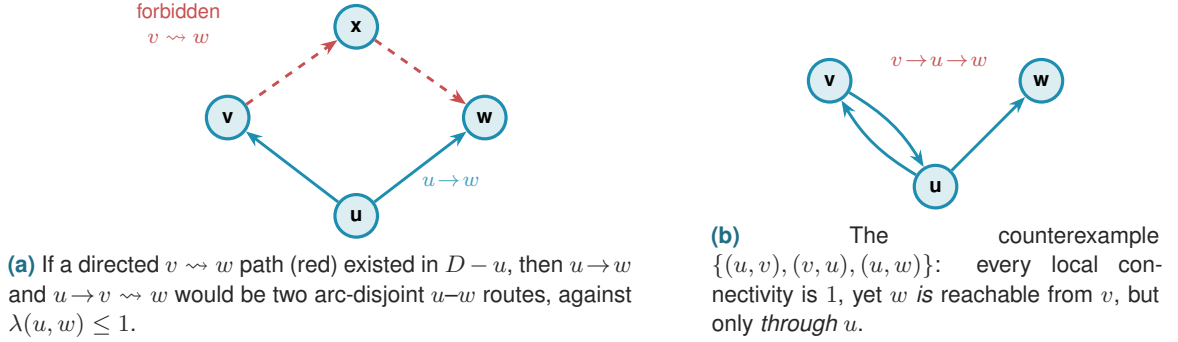


Figure A.3. Why Proposition A.16 must exclude paths through u . (a) For out-neighbours v, w of u in a digraph with $\lambda^{\max} \leq 1$, no $v \rightsquigarrow w$ path can avoid u : such a path would combine with the lone arc $u \rightarrow w$ into two arc-disjoint routes. (b) The restriction to $D - u$ is essential. In the three-vertex digraph $\{(u, v), (v, u), (u, w)\}$ all connectivities equal 1, yet v reaches w via $v \rightarrow u \rightarrow w$, a route that reuses the vertex u , which the proposition allows.

Proposition A.17 (Disjoint second out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$, let $u \in V$, and let $v_1 \neq v_2$ be out-neighbours of u . Then $(N^+(v_1) \setminus \{u\}) \cap (N^+(v_2) \setminus \{u\}) = \emptyset$.*

Proof. A common out-neighbour $w \neq u$ would give the two arc-disjoint routes $u \rightarrow v_1 \rightarrow w$ and $u \rightarrow v_2 \rightarrow w$, contradicting $\lambda_D(u, w) \leq 1$. $\square$

Proposition A.18 (Two-hop budget on bipartite partitions). *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$. If V admits a partition $A \cup B$ with every arc from A to B present, then every $b \in B$ has in-degree at most $m - 2$ from inside B , and consequently the number of arcs inside B is at most $(m - 2)|B|$.*

Proof. Fix $a \in A$ and $b \in B$, and let $b'_1, \dots, b'_d$ be the in-neighbours of b inside B , where $d = \deg_B^-(b)$. The direct arc $a \rightarrow b$ together with the two-step detours $a \rightarrow b'_i \rightarrow b$ are $1 + d$ arc-disjoint a – b routes, all present because $A \rightarrow B$ is complete. Feasibility forces $1 + d \leq m - 1$, so $d \leq m - 2$, and summing over B bounds the internal arcs by $(m - 2)|B|$. $\square$

Combining Proposition A.18 with the best partition gives, for any extremal digraph whose arcs all run from A to B or inside B (a complete $A \rightarrow B$ partition with no arc inside A and no arc from B back to A), the bound $|A(D)| \leq |B|(n - |B| + m - 2)$, since the $|A||B|$ arcs across the partition and the at most $(m - 2)|B|$ arcs inside B are then all there are: under the no-back-arc

hypothesis a route between two vertices of B never leaves B , so the two-hop budget applies to B 's own arcs. Maximised at $|B| = \lceil (n + m - 2)/2 \rceil$, this is $\lfloor (n + m - 2)^2/4 \rfloor$, the second branch of [Conjecture 1.13](#), uniformly in m , and the remaining gap is exactly the backward-arc hypothesis.

A.5 The directed multigraph problem at large n

The body proves $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$ for $n \leq 6$ ([Theorem 2.1](#)). This section records what is known beyond, where the one-directional bipartite construction takes over. Throughout, the scaling reduction of [Chapter 2](#) is used in the following form.

Lemma A.19 (Scaling reduction). *Let D be a directed multigraph with $\mu(u, v) \leq m - 1$ on every ordered pair and $\lambda^{\max}(D) \leq m - 1$. Then $w := \mu/(m - 1)$ takes values in $[0, 1]$, satisfies $\text{maxflow}(s, t; w) \leq 1$ for all $s \neq t$, and $|A(D)| = (m - 1) \sum_{u \neq v} w(u, v)$.*

Proof. Dividing all capacities by $m - 1$ divides every flow value by $m - 1$, and by Menger the maximum flow under μ is $\lambda_D(s, t) \leq m - 1$. $\square$

Lemma A.20 (Arithmetic identity). *For all $n \geq 2$, $\lfloor \frac{(n-1)^2}{4} \rfloor + \lfloor n/2 \rfloor = \lfloor \frac{n^2}{4} \rfloor$.*

Proof. For $n = 2k$ the left side is $k(k - 1) + k = k^2$, and for $n = 2k + 1$ it is $k^2 + k = k(k + 1)$. Both match the right side. $\square$

Proposition A.21 (Bipartite upper bound for even $n \geq 8$, conditional). *Let $n = 2k \geq 8$ be even. Suppose every fractional weighting w' on $n - 1$ vertices with values in $[0, 1]$ and all pairwise maximum flows at most one has total weight at most $\lfloor \frac{(n-1)^2}{4} \rfloor$. Then every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq m - 1$ satisfies $|A(D)| \leq (m - 1) \lfloor \frac{n^2}{4} \rfloor$.*

Proof. By [Lemma A.19](#) it suffices to bound $M := \sum_{u \neq v} w(u, v)$ by $\lfloor n^2/4 \rfloor$ for the scaled weighting w . Pick v_0 minimising the weighted total degree $d(v_0) = \sum_u (w(u, v_0) + w(v_0, u))$. Averaging gives $d(v_0) \leq 2M/n$. Deleting v_0 can only lower maximum flows, so the hypothesis applies to the restricted weighting and $M - d(v_0) \leq \lfloor (n - 1)^2/4 \rfloor = k(k - 1)$. Combining, $M(1 - \frac{2}{n}) \leq k(k - 1)$, hence $M \leq \frac{k}{k-1} \cdot k(k - 1) = k^2 = \lfloor n^2/4 \rfloor$. $\square$

The small cases $n \in \{4, 6\}$ genuinely do not satisfy the conclusion, and must not: there $2(n - 1) > \lfloor n^2/4 \rfloor$ and the double star beats the bipartite weighting, which is why the certified values of [Theorem 2.1](#) sit on the linear branch. For odd n the same deletion gives $M \leq k(k + 1) + \frac{k}{2k-1}$, an overshoot of less than one that integrality no longer absorbs, because fractional weights need not have integer totals. The induction would close if some vertex always had weighted degree at most $(m - 1)k$ before scaling, which is the missing minimum-degree statement.

Conjecture A.22 (Minimum-degree lemma for odd n). *In any directed multigraph D on $n = 2k+1$ vertices ($k \geq 3$) with $\lambda^{\max}(D) \leq m-1$ and $m \geq 3$, some vertex has total degree $d(v) \leq (m-1)k$, counting arcs with multiplicity.*

At $m = 2$ the statement is a proposition rather than a conjecture, through a reduction that also settles the whole multigraph problem at $m = 2$.

Proposition A.23 (Parallel-arc elimination at $m = 2$). *A directed multigraph with $\lambda^{\max} \leq 1$ has no parallel arcs, since two parallel arcs are already two arc-disjoint routes. Hence it is a simple digraph.*

Theorem A.24 (Directed multigraphs at $m = 2$). *For all $n \geq 2$, $L_2^{\text{dir}}(n) = \ell_2^{\text{dir}}(n) = \max(2(n-1), \lfloor \frac{n^2}{4} \rfloor)$.*

Proof. Immediate from [Proposition A.23](#) and [Theorem 1.9](#). □

Proposition A.25 (Minimum degree at $m = 2$, odd n). *In any directed multigraph on $n = 2k+1 \geq 7$ vertices with $\lambda^{\max} \leq 1$, some vertex has $d(v) \leq k$.*

Proof. By [Proposition A.23](#) and [theorem 1.9](#), $|A(D)| \leq \max(2(n-1), \lfloor n^2/4 \rfloor) = \lfloor n^2/4 \rfloor = k(k+1)$, the maximum being the quadratic branch because $n \geq 7$. The average total degree is $\frac{2k(k+1)}{2k+1} < k+1$, and some integer degree is at most k . The hypothesis $n \geq 7$ is needed: on the linear branch the statement can fail, as the bidirected star on three vertices has minimum total degree $2 > 1$. □

For $m \geq 3$ the two extreme regimes of [Conjecture A.22](#) are settled. In the *purely saturated* regime, where every multiplicity is 0 or $m-1$, the scaled weighting is the indicator of a simple digraph with $\lambda^{\max} \leq 1$, its degrees are integers, and the argument of [Proposition A.25](#) applies verbatim after multiplying back by $m-1$. In the *all-unsaturated* simple regime at $n = 7$, $m = 3$, the program certifies by a cut-counting infeasibility that no simple digraph satisfies all two-hop inequalities with every total degree at least 7. The open case is the *mixed* regime, where some multiplicity lies strictly between 0 and $m-1$: there the scaled degrees are genuinely fractional, the averaging bound $\frac{2k(k+1)}{2k+1}$ lands strictly between k and $k+1$, and no integrality rescues it. This is the single identified gap that separates the certified values of [Theorem 2.1](#) from the full conjecture

$$L_m^{\text{dir}}(n) = (m-1) \max(2(n-1), \lfloor \frac{n^2}{4} \rfloor),$$

whose lower bound is unconditional: the double star realises $2(n-1)(m-1)$ and the one-directional bipartite multigraph, every arc at multiplicity $m-1$, realises $(m-1) \lfloor n^2/4 \rfloor$ with $\lambda^{\max} = m-1$.

Lemma A.26 (Attachment lemma). *Let $m \geq 2$ and let D_0 be the one-directional complete bipartite multigraph with parts A and B , $|A| = p \geq 2$, $|B| = q \geq 2$, and $\mu(a, b) = m-1$ for*

every $a \in A$, $b \in B$, with no other arcs. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v (multiplicities at most $m - 1$). If $\lambda_D^{\max} \leq m - 1$, then $d(v) \leq (m - 1) \max(p, q)$.

Figure A.4 sets up the proof: the new vertex v , the four classes its incident arcs fall into, and one of the route patterns that the hypothesis forbids.

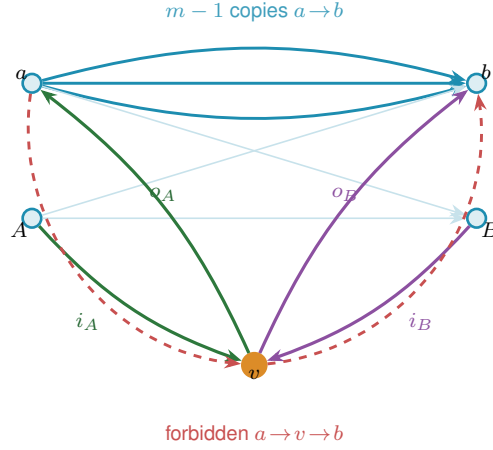


Figure A.4. The attachment lemma (Lemma A.26): the new vertex v glued onto $(m - 1) \cdot B_{p,q}$. Each $a \in A$ sends $m - 1$ parallel arcs to each $b \in B$ (the bent bundle). The arcs at v split into four classes by direction and side: i_A (into v from A), o_A (out of v to A), i_B , o_B . The red dashed route $a \rightarrow v \rightarrow b$ is forbidden: laid on top of the $m - 1$ existing $a \rightarrow b$ arcs it makes an m -th arc-disjoint $a \rightarrow b$ route, so $\lambda(a, b) \geq m$. Ruling out this and two sibling patterns caps $d(v) \leq (m - 1) \max(p, q)$.

Proof. The strategy is to sort the arcs at the new vertex v into four groups by their direction and which side they touch, rule out the combinations that would manufacture an m -th route between two old vertices, and then bound whatever survives. Write $i_A = \sum_a \mu(a, v)$, $i_B = \sum_b \mu(b, v)$, $o_A = \sum_a \mu(v, a)$, $o_B = \sum_b \mu(v, b)$ for the total multiplicities into v from A , into v from B , out of v to A , and out of v to B , so that $d(v) = i_A + i_B + o_A + o_B$. Three route patterns are forbidden outright, because each adds an m -th arc-disjoint route on top of the $m - 1$ parallel arcs of a bipartite pair:

- ☆ $\mu(a, v) \geq 1$ and $\mu(v, b) \geq 1$: the path $a \rightarrow v \rightarrow b$ together with the $m - 1$ arcs $a \rightarrow b$ gives $\lambda(a, b) \geq m$.
- ☆ $\mu(a, v) \geq 1$ and $\mu(v, a') \geq 1$ with $a' \neq a$: the path $a \rightarrow v \rightarrow a' \rightarrow b$ (any $b \in B$) gives $\lambda(a, b) \geq m$.
- ☆ $\mu(b, v) \geq 1$ and $\mu(v, b') \geq 1$ with $b' \neq b$: the path $a \rightarrow b \rightarrow v \rightarrow b'$ (any $a \in A$) gives $\lambda(a, b') \geq m$.

Each bound below rests on one principle. The local connectivity $\lambda(s, t)$ is at least the number of pairwise arc-disjoint s -to- t routes we can exhibit, because any such family of routes is an integral

flow from s to t . Feasibility says $\lambda \leq m - 1$, so the moment we can name m arc-disjoint routes between two vertices we have a contradiction.

Case I: $i_A \geq 1$. Some arc enters v from a vertex $a \in A$. Pattern (1) then forces $o_B = 0$, because any arc $v \rightarrow b$ would complete the route $a \rightarrow v \rightarrow b$, an m -th route on the pair (a, b) laid on top of its $m - 1$ parallel arcs.

Suppose first that $o_A \geq 1$, so v also sends an arc to some vertex of A . Pattern (2) makes this rigid: an arc into v from a and an arc out of v to a different $a' \neq a$ would give the route $a \rightarrow v \rightarrow a' \rightarrow b$ for any b , again an m -th route. So every arc between v and A , in either direction, must touch one single vertex a^* . Counting arc-disjoint routes from a^* to v , there are the $\mu(a^*, v)$ direct arcs, and for each $b \in B$ the two-step route $a^* \rightarrow b \rightarrow v$, which can be run $\min(\mu(a^*, b), \mu(b, v))$ times on disjoint arcs. Different b use different arcs, and none reuse a direct arc, so

$$\lambda(a^*, v) \geq \mu(a^*, v) + \sum_b \min(\mu(a^*, b), \mu(b, v)) = \mu(a^*, v) + i_B,$$

the last equality because $\mu(a^*, b) = m - 1$ is at least $\mu(b, v)$, so each minimum is $\mu(b, v)$ and the sum is i_B . Feasibility caps the left side at $m - 1$, so $\mu(a^*, v) + i_B \leq m - 1$. Every arc at v now touches either a^* or B , so $d(v) = \mu(a^*, v) + i_B + \mu(v, a^*)$, and with the first two summing to at most $m - 1$ and the last at most $m - 1$ we get $d(v) \leq 2(m - 1)$.

Suppose instead $o_A = 0$. With $o_B = 0$ already, v is now a pure sink. The same two-step count applies to every $a \in A$ at once, giving $\lambda(a, v) \geq \mu(a, v) + i_B$, hence $\mu(a, v) \leq (m - 1) - i_B$ for each of the p vertices of A . Summing over A bounds the in-flow from A by $i_A \leq p(m - 1) - p i_B$, so

$$d(v) = i_A + i_B \leq p(m - 1) - (p - 1)i_B \leq p(m - 1),$$

the final step because $(p - 1)i_B \geq 0$.

Case II: $i_A = 0$ and $o_B \geq 1$. This is Case I read backwards. Reversing every arc turns $(m - 1) \cdot B_{p,q}$ into $(m - 1) \cdot B_{q,p}$, swaps in-arcs with out-arcs, and sends pattern (2) to pattern (3), so the same argument applies with the roles of A and B exchanged. It yields either $d(v) \leq 2(m - 1)$, or v is a pure source for which $\lambda(v, b) \geq \mu(v, b) + o_A \leq m - 1$ holds for every b , and summing over the q vertices of B gives $d(v) \leq q(m - 1) - (q - 1)o_A \leq q(m - 1)$.

Case III: $i_A = o_B = 0$. Now v receives nothing from A and sends nothing to B , so its only arcs are the i_B in-arcs from B and the o_A out-arcs to A . For any $a \in A$ the two-step routes $a \rightarrow b \rightarrow v$ over all b give $\lambda(a, v) \geq i_B$, so $i_B \leq m - 1$, and for any $b \in B$ the routes $v \rightarrow a \rightarrow b$ over all a give $\lambda(v, b) \geq o_A$, so $o_A \leq m - 1$. Hence $d(v) = i_B + o_A \leq 2(m - 1)$.

Every case bounds $d(v)$ by either $2(m - 1)$ or $\max(p, q)(m - 1)$. Since $p, q \geq 2$ we have $2(m - 1) \leq (m - 1) \max(p, q)$, so in all cases $d(v) \leq (m - 1) \max(p, q)$. $\square$

Corollary A.27 (Attachment equality). *In the setting of Lemma A.26 with $\max(p, q) \geq 3$, suppose $d(v) = (m - 1) \max(p, q)$. If $q > p$, then v is a pure source with $\mu(v, b) = m - 1$ for every $b \in B$, so $D = (m - 1) \cdot B_{p+1,q}$ with v joining the source side. If $p > q$, then v is symmetrically a pure*

sink at full multiplicity, and if $p = q$, it is one of the two.

Proof. Rerun the cases of [Lemma A.26](#). The mixed cases cap $d(v)$ at $2(m-1) < (m-1)\max(p, q)$. The pure-sink case gives $d(v) \leq p(m-1) - (p-1)i_B$, which reaches $(m-1)\max(p, q)$ only when $p \geq q$, $i_B = 0$ and every $\mu(a, v) = m-1$. The pure-source case symmetrically forces $o_A = 0$ and every $\mu(v, b) = m-1$ when $q \geq p$. Feasibility of the resulting digraph is the bipartite construction itself. $\square$

Theorem A.28 (Conditional odd step, $m \in \{3, 4\}$). *Fix $k \geq 4$ and $m \in \{3, 4\}$, and assume (i) $L_m^{\text{dir}}(2k) = (m-1)k^2$, and (ii) every directed multigraph attaining it is the one-directional complete bipartite multigraph $(m-1) \cdot B_{k,k}$. Then $L_m^{\text{dir}}(2k+1) = (m-1)k(k+1)$.*

Proof. The lower bound is $(m-1) \cdot B_{k,k+1}$. The upper bound is a counting squeeze. If the arc count were even one over the target, then deleting any single vertex would still leave a feasible multigraph on $2k$ vertices, which by (i) cannot exceed $(m-1)k^2$, so every vertex would have to carry a large degree. Yet the average degree is only barely above that floor, so some vertex must sit exactly on it, and deleting that one lands precisely on the even extremiser $(m-1) \cdot B_{k,k}$ of (ii). The attachment lemma then caps how much that vertex could have carried, below what the surplus requires, and the contradiction forces the surplus to vanish. We now make this precise. For the upper bound let D on $n = 2k+1$ vertices have $\lambda^{\max} \leq m-1$ and $|A(D)| = (m-1)k(k+1) + j$ with $j \geq 1$. Deleting any vertex v leaves a feasible multigraph on $2k$ vertices, so by (i) $d(v) \geq |A(D)| - (m-1)k^2 = (m-1)k + j$ for every v . The average degree is

$$\frac{2|A(D)|}{2k+1} = (m-1)k + j + \frac{(m-1)k - (2k-1)j}{2k+1},$$

by direct computation. If $(m-1)k - (2k-1)j < 0$ the average lies below the minimum, a contradiction. Otherwise the correction term is nonnegative and, because $m \leq 4$ and $j \geq 1$, strictly below one: $(m-1)k - (2k-1)j \leq (m-3)k + 1 \leq k+1 < 2k+1$. The average therefore lies in $[(m-1)k + j, (m-1)k + j + 1)$ while every degree is at least $(m-1)k + j$, so some vertex v has $d(v) = (m-1)k + j$ exactly, and $D - v$ has exactly $(m-1)k^2$ arcs. By (ii), $D - v = (m-1) \cdot B_{k,k}$, and [Lemma A.26](#) caps $d(v) \leq (m-1)k < (m-1)k + j$, the final contradiction. Hence $j \leq 0$. $\square$

Remark A.29 (The $m = 3$ chain, sharpened). [Theorem A.28](#) removes [Conjecture A.22](#) from the critical path: the odd case no longer needs a minimum-degree statement, only the extremal characterisation one level down, and for $m = 3$ that characterisation propagates along the same induction in both parities. *Odd levels* ($n = 2k+1 \geq 9$): an extremiser has $|A| = 2k(k+1)$ arcs and every degree at least $2k$ (deletion bound). All degrees $\geq 2k+1$ is impossible since $(2k+1)^2 > 4k(k+1) = \sum_v d(v)$, so some $d(v) = 2k$ exactly, $D - v$ is the even extremiser $2 \cdot B_{k,k}$, and [Corollary A.27](#) (the $p = q$ case) forces v to rejoin as a full-multiplicity pure source or sink: $D = 2 \cdot B_{k+1,k}$ or $2 \cdot B_{k,k+1}$. *Even levels* ($n = 2k \geq 10$): equality in the averaging step forces $2k$ -regularity, hence $D - v$ extremal at $2k-1$ for every v . Odd uniqueness one

level down leaves $2 \cdot B_{k,k-1}$ or $2 \cdot B_{k-1,k}$, and [Corollary A.27](#) with $d(v) = 2k = 2 \max(k-1, k)$, $q > p$, rebuilds exactly $2 \cdot B_{k,k}$. *The seam at $n = 8$:* the same regularity argument runs from $n = 7$, where the branches tie at 24 arcs and several extremisers coexist. On the linear branch the extremisers are richer than the double star alone: every *bidirected spanning tree at full multiplicity $m - 1$* is feasible with $2(n-1)(m-1)$ arcs (the only arcs crossing the bipartition under a tree edge are that edge's own $m - 1$ copies, so every local connectivity is exactly $m - 1$), and exhaustive enumeration (`enumerate_extremal_directed_multigraphs`) confirms that at $n = 4, 5$, $m = 3$ these doubled trees are the *only* extremisers (2 and 3 of them, matching the tree shapes). A deleted vertex of the 8-regular D leaves a 24-arc extremiser with maximum degree at most 8: among doubled trees only the doubled path qualifies (tree degree ≤ 2), the double star (hub degree 24) being excluded. If $D - v$ were the doubled path P_7 , the degree count would force v to attach with weight 4 to each path end, i.e. bidirected at full multiplicity, making D the doubled bidirected cycle C_8 , which is infeasible since the two doubled directions around the cycle give $\lambda = 4 > 2$ between any pair. [Figure A.5](#) draws this collapse of the doubled path into the infeasible doubled cycle. So every deletion is bipartite-type, the attachment equality applies, and $D = 2 \cdot B_{4,4}$ again. The whole $m = 3$ problem therefore rests on two finite integral statements about multigraphs with multiplicities in $\{0, 1, 2\}$: (a) $L_3^{\text{dir}}(7) = 24$, equivalently $M^*(7) = 12$, and (b) every 24-arc feasible multigraph on 7 vertices with maximum degree at most 8 is $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, or the doubled bidirected path P_7 . Both targets sit at sizes the certifier and the enumerator can realistically reach with the strengthened relaxation now wired into `certify_directed_multigraph` (`use_deletion_cuts`: deletion bounds from the certified $M^*(5)$, $M^*(6)$ and the degree-pair inequality $d^+(s) + d^-(t) - w(s, t) \leq n - 1$, all proved valid). For $m \geq 4$ one hole remains: at an odd extremal level the counting no longer forces a vertex of degree $(m-1)k$ (all degrees $\geq (m-1)k + 1$ is arithmetically consistent once $k(m-3) \geq 1$), so the uniqueness statement does not yet propagate through odd levels, although the *value* at odd levels follows from even uniqueness whenever that is available. For $m \geq 5$ the same escape climbs one level: a hypothetical extremiser carrying $j \geq 1$ surplus arcs could have *every* degree strictly above the deletion bound (all degrees $\geq (m-1)k + j + 1$ is consistent exactly when $(2k-1)j \leq (m-1)k - 2k - 1$, which admits $j = 1$ once $m \geq 5$), so no vertex deletion is forced to land on an extremal multigraph and even the *value* step stalls. This is why [Theorem A.28](#) is stated for $m \in \{3, 4\}$.

A.6 The hypergraph bounds

The body's hypergraph claims rest on two theorems: a Menger theorem for Berge routes, which also certifies the helper-point checker of [Chapter 2](#), and a hypergraph Gomory–Hu theorem from the recent literature.

Theorem A.30 (Menger for hypergraphs). *For a hypergraph $\mathcal{H}$ and vertices u, v , the maximum number of hyperedge-disjoint Berge u – v paths equals the minimum size of a set C of hyperedges whose removal separates u from v .*

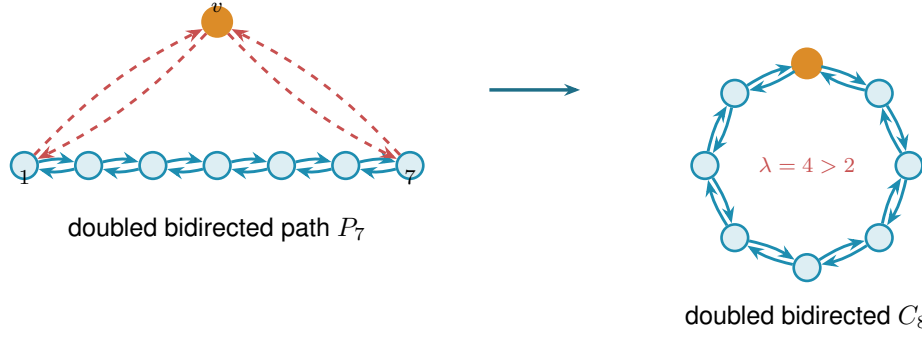


Figure A.5. The seam argument of Remark A.29. A degree count forces a deleted vertex of the 8-regular extremiser D to leave the doubled bidirected path P_7 (left, each path edge a pair of opposing arcs). To reach degree 8 the new vertex v must attach with full multiplicity to the two ends 1 and 7 (red), bidirected. That closes the path into the doubled bidirected cycle C_8 (right), which is *infeasible*: the two directions around the cycle give two arc-disjoint routes each way, so $\lambda = 4 > 2$ between any pair. The contradiction rules out the path branch, forcing every deletion to be bipartite-type and hence $D = 2 \cdot B_{4,4}$.

Proof. Build the flow network of Figure 2.5: for each hyperedge e a helper arc $e^- \rightarrow e^+$ of capacity one, and arcs $x \rightarrow e^-$ and $e^+ \rightarrow x$ of unbounded capacity for each $x \in e$. A Berge path $u, e_1, v_1, \dots, e_k, v$ becomes the directed walk $u \rightarrow e_1^- \rightarrow e_1^+ \rightarrow v_1 \rightarrow \dots \rightarrow v$, and since the e_i are distinct, hyperedge-disjoint Berge paths use distinct helper arcs, so they translate to arc-disjoint flow paths. Conversely an integral maximum flow decomposes into paths, each of which, suppressing the helper nodes, is a Berge path, and no two share a hyperedge because each helper arc has capacity one. Finally, any finite cut of the network consists of helper arcs only, and a set of helper arcs separates u from v exactly when the corresponding hyperedges do. Max-flow min-cut now gives the claim. $\square$

Theorem A.31 (Hypergraph Gomory–Hu, Bérczi et al. [2]). *Every r -uniform hypergraph $\mathcal{H}$ on vertex set V has a weighted tree T^* on V such that $\lambda_{\mathcal{H}}(u, v)$ equals the minimum weight on the u – v tree path, and each tree edge t induces a vertex partition $(S_t, V \setminus S_t)$ with $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$, where $\delta_{\mathcal{H}}(S)$ is the set of hyperedges meeting both sides.*

Proof of Proposition 1.14. Upper bound. Let $\mathcal{H}$ be r -uniform with $\lambda_{\mathcal{H}}^{\max} \leq m - 1$ and let T^* be its Gomory–Hu tree. Every tree weight is a local connectivity, so $\sum_t c^*(t) \leq (m - 1)(n - 1)$. For a hyperedge e let $\text{ST}(e)$ be the smallest subtree of T^* containing the r vertices of e . It has at least $r - 1$ edges, and for each of its edges t the hyperedge e meets both sides of the induced partition, so $e \in \delta_{\mathcal{H}}(S_t)$. Summing over hyperedges and exchanging the order of summation,

$$(r - 1) |E(\mathcal{H})| \leq \sum_e |\{t : e \in \delta_{\mathcal{H}}(S_t)\}| = \sum_t |\delta_{\mathcal{H}}(S_t)| = \sum_t c^*(t) \leq (m - 1)(n - 1),$$

and dividing by $r - 1$ with integrality gives $|E(\mathcal{H})| \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. *Lower bound.* When $(r - 1) \mid (n - 1)$, take a hub h , partition the other $n - 1$ vertices into groups of size $r - 1$, form the star hypertree of hyperedges $\{h\} \cup G_i$, and give each hyperedge multiplicity $m - 1$. Every non-hub vertex lies in exactly $m - 1$ hyperedges, which form a cut isolating it, so by Theorem A.30 every

pair has Berge connectivity at most $m - 1$, while the count is $\frac{(m-1)(n-1)}{r-1}$. For general n under the hypothesis $m - 1 \leq \binom{n-2}{r-2}$, [Theorem A.32](#) below attains the floor exactly, without even needing repeated hyperedges. $\square$

The multiplicities in the lower bound can in fact be avoided: for $r \geq 3$, *simple* r -uniform hypergraphs already attain the same bound, in sharp contrast to the graph case $r = 2$, where simplicity costs the factor that separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$.

Theorem A.32 (Simplicity is free for $r \geq 3$). *Let $r \geq 3$, $n \geq r$, and $2 \leq m$ with $m - 1 \leq \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a simple r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m - 1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is [Proposition 1.14](#), since a simple hypergraph is a multihypergraph. For the lower bound, fix a hub h and let $V' = V \setminus \{h\}$. By [Lemma A.33](#) applied with rank $r - 1$ and degree ceiling $d = m - 1$ on the $n - 1$ vertices of V' , there is an $(r - 1)$ -uniform simple hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m - 1$ and exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Adding h to each hyperedge produces distinct r -sets, so the result $\mathcal{H}$ is simple, and every non-hub vertex w lies in $\deg_{\mathcal{G}^*}(w) \leq m - 1$ hyperedges, which form a cut isolating w . By [Theorem A.30](#), $\lambda_{\mathcal{H}}^{\max} \leq m - 1$. $\square$

Lemma A.33 (Sparse uniform hypergraphs exist). *Let $r \geq 2$, $n \geq r$, and $0 \leq d \leq \binom{n-1}{r-1}$. There exists an r -uniform simple hypergraph on n vertices with maximum degree at most d and exactly $\left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges.*

Proof. Put $e := \left\lfloor \frac{dn}{r} \right\rfloor \leq \frac{n}{r} \binom{n-1}{r-1} = \binom{n}{r}$. By Baranyai's theorem on equitable partitions of the complete r -uniform hypergraph [1], the $\binom{n}{r}$ possible r -sets split into two classes of sizes e and $\binom{n}{r} - e$ such that within each class every vertex is covered either $\lfloor re/n \rfloor$ or $\lceil re/n \rceil$ times. The class of size e has maximum degree at most $\lceil re/n \rceil \leq \lceil d \rceil = d$. $\square$

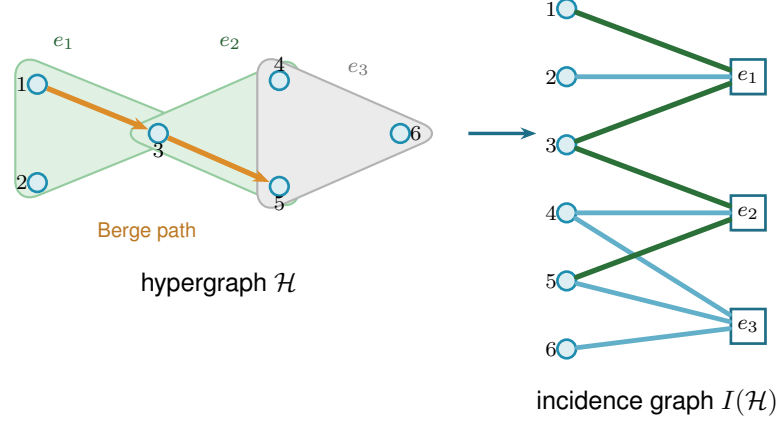
A.7 The hypergraph vertex problem

The whole vertex problem translates into a graph problem through one observation, the one the rest of this section rests on and which [Figure A.6](#) draws in full. Write $I(\mathcal{H})$ for the bipartite *incidence graph* of a hypergraph $\mathcal{H}$: one node per vertex, one node per hyperedge, an edge when the vertex lies in the hyperedge. A Berge path with distinct hyperedges is exactly a path of $I(\mathcal{H})$ between two vertex-class nodes, hyperedge-disjoint Berge paths use disjoint hyperedge-nodes, and internally vertex-disjoint ones use disjoint interior vertex-nodes. Hence

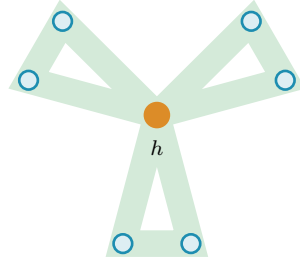
$$\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v),$$

the maximum number of internally disjoint u - v paths in $I(\mathcal{H})$, and the hypergraph vertex problem asks for the maximum number of degree- r nodes on one side of a bipartite graph whose other

side holds n nodes, subject to no two of those n nodes being joined by m internally disjoint paths. [Figure A.6a](#) traces a single Berge path through both pictures, and [Figure A.6b](#) shows the star hypertree of [Theorem A.34](#), whose incidence graph is a tree.



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1-e_1-3-e_2-5$ is highlighted green in both pictures: in $\mathcal{H}$ it threads two hyperedges, in $I(\mathcal{H})$ it is an ordinary path $1-e_1-3-e_2-5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r-1=2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.6. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of [Theorem A.34](#), whose incidence graph is a tree.

Theorem A.34 (Hypergraph vertex value at $m = 2$). *For $r \geq 2$ and $n \geq 1$, the maximum number of hyperedges of an r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 1$ is*

$$k_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree. In particular $k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Proof. We claim $\kappa_{\mathcal{H}}^{\max} \leq 1$ holds if and only if $I(\mathcal{H})$ is a forest. If $I(\mathcal{H})$ contains a cycle $v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell, v_1$ (it alternates classes, so $\ell \geq 2$, all v_i and e_i distinct), then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, e_{\ell-1}, \dots, e_2, v_2$ are two Berge v_1-v_2 paths with disjoint hyperedge sets and disjoint interior vertex sets, so $\kappa(v_1, v_2) \geq 2$. (At $\ell = 2$ this is the repeated pair: two hyperedges sharing two vertices.) Conversely, two such paths between u and v concatenate to a cycle of $I(\mathcal{H})$.

This proves the claim. A forest on $n + q$ nodes has at most $n + q - 1$ edges, while $I(\mathcal{H})$ has exactly rq edges, so $rq \leq n + q - 1$, i.e. $q \leq (n - 1)/(r - 1)$, and q is an integer. The star hypertree has $\lfloor (n - 1)/(r - 1) \rfloor$ hyperedges and a tree as incidence graph (each block vertex lies in one hyperedge, every cycle would need two hyperedges sharing two vertices), so the floor is attained. $\square$

Proposition A.35 (General lower bound). *For $r \geq 3$, $m \geq 2$ and $m - 1 \leq \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \geq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple construction of [Theorem A.32](#) has Berge edge-connectivity at most $m - 1$, and Whitney's inequality $\kappa \leq \lambda$ holds pairwise (an internally vertex-disjoint family is in particular hyperedge-disjoint under our definition), so the same hypergraph is feasible for the vertex problem. $\square$

The $m = 3$ case also closes completely. The engine is a rank bound for graphs in which one side of a partition must stay below local 3-connectivity. The hypergraph theorem then falls out of the incidence translation. Throughout, $\text{rank}(G) = |E(G)| - |V(G)| + 1$ denotes the cycle rank of a connected multigraph, that is, the number of independent cycles it carries. A tree has cycle rank 0, and each edge added beyond a spanning tree closes exactly one new cycle and raises the rank by one, so bounding the rank is the same as bounding how many independent cycles the graph can hold.

Lemma A.36 (Incidence rank lemma). *Let G be a connected multigraph with vertex set $X \cup Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then $\text{rank}(G) \leq |X| - 1$.*

Proof. Induction on $|V| + |E|$. The proof peels G down to a core that cannot exist. Step 1 handles a cut vertex by arguing block by block. Steps 2 and 3 remove a degree-two Z -vertex or X -vertex without changing the bound. What survives is 2-connected with minimum degree at least three, and Step 4 shows that such a graph already violates the connectivity ceiling (iv), so it never arises. *Base* $|V| \leq 2$. A single vertex lies in X by (iii), and $0 \leq |X| - 1$. On two vertices, (i)–(iii) force both into X , parallel edges are internally disjoint paths, so (iv) caps the multiplicity at 2 and $\text{rank} \leq 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex, so it is not yet 2-connected. Break it into its blocks $B_1, \dots, B_c$ with $c \geq 2$, a block being a maximal piece with no cut vertex of its own, that is, either a single bridge edge or a maximal 2-connected subgraph. Two blocks meet in at most one vertex, and any shared vertex is a cut vertex of G . No cycle can pass through two different blocks, so the cycle rank adds up over blocks, $\text{rank}(G) = \sum_i \text{rank}(B_i)$, and it is enough to bound each $\text{rank}(B_i)$ by $|X \cap B_i| - 1$ and sum.

Each block inherits the hypotheses. A bridge block is one edge, so its rank is 0, and by (i) at least one endpoint lies in X , giving $0 \leq |X \cap B| - 1$. Every other block is 2-connected, hence has minimum degree at least 2, so (i) to (iv) hold inside it and the induction hypothesis gives $\text{rank}(B_i) \leq |X \cap B_i| - 1$.

Adding these requires care, because a cut vertex is shared by several blocks and would be counted several times. Write $b(v)$ for the number of blocks containing v , so $b(v) = 1$ except at cut vertices. Counting each X -vertex once for every block it lies in,

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The number of blocks satisfies the block-cut tree identity $c = 1 + \sum_{v \text{ cut}} (b(v) - 1)$, which holds because the blocks and cut vertices form a tree in which each cut vertex v is joined to the $b(v)$ blocks that contain it. Subtracting this count of blocks from the previous display and separating the cut vertices into their X - and Z -parts leaves

$$\text{rank}(G) = \sum_i \text{rank}(B_i) \leq \sum_i (|X \cap B_i| - 1) = |X| - 1 - \sum_{z \in Z \text{ cut}} (b(z) - 1) \leq |X| - 1,$$

where the last inequality holds because each $b(z) - 1 \geq 0$, so the Z -cut sum only pushes the bound further below $|X| - 1$.

Step 2: a Z -vertex of degree two. From now on G is 2-connected, every cut vertex having been handled. Suppose some $z_0 \in Z$ has degree exactly 2. Both its neighbours lie in X by (i), and they are distinct, since two edges to the same vertex would be parallel at a Z -vertex, which (ii) forbids. *Suppress* z_0 by deleting it and joining its two neighbours x, x' with one new edge. This loses two edges and one vertex and gains one edge, so $|E|$ and $|V|$ each drop by one and $\text{rank} = |E| - |V| + 1$ is unchanged. The new edge runs X to X , so $|X|$ is unchanged and (i), (ii), (iii) still hold for the remaining Z -vertices. Replacing the path x, z_0, x' by a direct edge alters no route among the surviving vertices, so every $\kappa(x_1, x_2)$ is preserved and (iv) holds. The graph is strictly smaller, so the induction hypothesis applies and returns the bound for G .

Step 3: an X -vertex of degree two. Now G is 2-connected and, Step 2 being used up, every Z -vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it. Deleting one vertex from a 2-connected graph leaves it connected. Stripping a degree-2 vertex removes two edges and one vertex, so rank drops by exactly one. Each neighbour of x_0 loses one edge, and a neighbour in Z only falls from degree at least 3 to degree at least 2, so (iii) survives, while deleting a vertex cannot raise a connectivity, so (iv) survives. Here $|X|$ drops by one, and $|X| \geq 2$ to begin with, for if $|X| \leq 1$ then by (i) and (ii) every z would send all its edges to one X -vertex without repeats, hence have degree at most 1, against (iii). The smaller graph satisfies the hypotheses, so induction gives $\text{rank}(G) - 1 \leq (|X| - 1) - 1$, that is $\text{rank}(G) \leq |X| - 1$.

Step 4: the remaining case is impossible. Now G is 2-connected with $|V| \geq 3$ and every degree at least 3. First, G is simple: a parallel class lies inside X by (ii), and if $\mu(x, x') \geq 2$ then

a third internally disjoint route exists: walk from x to any third vertex w inside $G - x'$, then from w to x' inside $G - x$. The concatenation visits x only first and x' only last, so it contains an $x-x'$ path with at least one interior vertex, giving $\kappa(x, x') \geq 3$, against (iv). If G is 3-connected, Menger puts every pair at $\kappa \geq 3$, so $|X| \leq 1$ and every z has degree at most 1 by (i), absurd.

Otherwise G is 2-connected but not 3-connected, so it can be taken apart at its 2-vertex separators. The classical triconnected decomposition (Tutte [19], Hopcroft–Tarjan [10]) does this in a canonical way, and Figure A.7 sketches both its shape and the leaf at which the argument bites. It writes G as a *tree of pieces*, where each piece is one of three simple shapes: a cycle of length at least three, a *bond* (two vertices joined by several parallel edges), or a 3-connected simple graph on at least four vertices. Wherever the decomposition cuts G at a separating pair it records the cut by adding a *virtual edge* between those two vertices in each piece, a placeholder that stands in for the whole rest of the graph on the other side. Two properties are all the proof needs from this, and both are visible in the figure. First, a vertex of a piece that touches none of that piece's virtual edges belongs to that piece alone, so it keeps in G exactly the degree it has in the piece. Second, for any virtual edge $\{a, b\}$ the part of G on the far side of the cut contains a genuine $a-b$ path whose interior avoids the piece, so the virtual edge can always be replaced by a real route.

Now G itself is none of the three piece types, since it is not a single cycle (its degrees are at least three), not a bond (it is simple), and not 3-connected (the case just handled), so the tree has at least two leaves. Let L be one, with unique virtual edge $\{a, b\}$. If L is a cycle, a vertex of L off $\{a, b\}$ has degree 2 in G , absurd. If L is a bond, it holds at least two real parallel edges, absurd in a simple graph. If L is 3-connected, then any two of its vertices are joined by three internally disjoint paths inside L , at most one through the virtual edge, which the far-side path replaces: so $\kappa_G(u, v) \geq 3$ for all $u, v \in V(L)$, forcing $|X \cap V(L)| \leq 1$ by (iv). But $|V(L)| \geq 4$ leaves a Z -vertex $z^* \notin \{a, b\}$. Its neighbours lie in $X \cap V(L)$ by (i), at most one vertex, reached by at most one edge since G is simple, giving degree at most 1, which is absurd. $\square$

Theorem A.37 (Hypergraph vertex value at $m = 3$). *For $r \geq 2$, every r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $r \geq 3$ with $2 \leq \binom{n-2}{r-2}$ the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) = \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor = \ell_3^{(r)}(n).$$

Proof. Apply Lemma A.36 to each connected component of the incidence graph $I(\mathcal{H})$, with X the vertex class and Z the hyperedge class: Z is independent, incidences are simple even when hyperedges repeat, hyperedge-nodes have degree $r \geq 2$, components without an X -node are impossible, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \leq 2$ for vertex pairs. Sum rank $\leq |X_c| - 1$ over the C components. The incidence graph $I(\mathcal{H})$ has rq edges (each of the q hyperedges contributes r incidences) and $n + q$ nodes, so $\sum_c \text{rank} = rq - (n + q) + C$, while $\sum_c (|X_c| - 1) = n - C$. The inequality reads $rq - (n + q) + C \leq n - C$, hence $q(r - 1) \leq 2(n - 1) + 2(1 - C) \leq 2(n - 1)$,

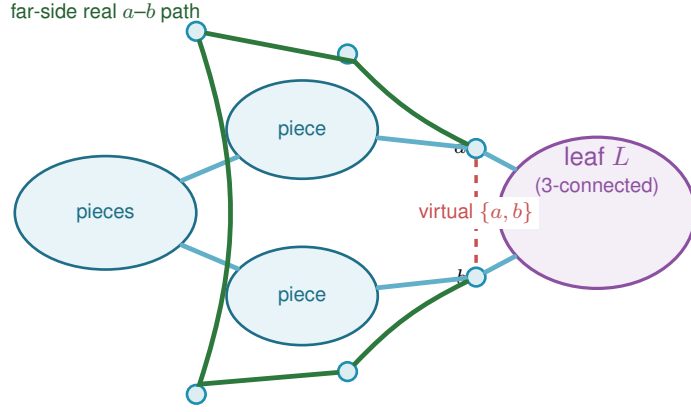


Figure A.7. The triconnected (Tutte/SPQR) decomposition in Step 4 of [Lemma A.36](#), schematic: G splits into a tree of pieces (blobs) glued along virtual edges, and the analysis works at a *leaf* L with its single virtual edge $\{a, b\}$ (dashed red). The far side of the tree supplies a *real* a - b path (green) that internally avoids L . If L is 3-connected, any two of its vertices are joined by three internally disjoint paths inside L , at most one using the virtual edge, and the far-side path replaces it, so $\kappa_G \geq 3$ for every pair in L . With $|V(L)| \geq 4$ that forces a hyperedge-node $z^* \notin \{a, b\}$ of degree at most one, which is the contradiction that closes the lemma.

the last step using $C \geq 1$. The lower bound is [Proposition A.35](#) at $m = 3$. $\square$

Remark A.38 (Edges as gates at $r = 2$, and the boundary of the method). At $r = 2$ the theorem speaks of multigraphs under the hyperedge-as-gate convention, in which parallel edges are distinct one-step routes, giving the cactus value $2(n - 1)$, complementing the body's convention for graphs, where parallel adjacencies count once. The proof of [Lemma A.36](#) leans on the triconnected decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$, the analogous question asks for a rank bound relative to $\kappa \leq 3$, where no comparably clean decomposition exists, and the graph case ($m = 5$, Sørensen–Thomassen) warns that the pattern $k_m^{(r)} = \ell_m^{(r)}$ must fail eventually. Exhaustive computation confirms $k_4^{(3)}(5) = 6 = \lfloor 3 \cdot 4/2 \rfloor$, so the agreement survives at least one step beyond the theorem, and where it first breaks for $r \geq 3$ is open.

Proposition A.39 (Directed hypergraphs: first bounds). *Model a directed r -uniform hyperedge as a tail vertex together with $r - 1$ head vertices, a Berge route entering at the tail and leaving at a head (the program's convention). If every pair of vertices has at most $m - 1$ arc-disjoint directed Berge routes, then*

$$|E(\mathcal{H})| \leq \frac{(m - 1) n(n - 1)}{r - 1},$$

and for $m - 1 \leq \binom{n - \lceil n/2 \rceil - 1}{r - 2}$ there are feasible constructions with $\max_{1 \leq \alpha \leq n - 1} \alpha \left\lfloor \frac{(m - 1)(n - \alpha)}{r - 1} \right\rfloor \approx (m - 1)n^2 / (4(r - 1))$ hyperedges, so the value is quadratic in n .

Proof. Upper bound: a hyperedge with tail u serves the $r - 1$ ordered pairs (u, h) , h a head. If some ordered pair (u, v) were served by m distinct hyperedges, those would be m arc-disjoint one-step routes, so each ordered pair is served at most $m - 1$ times and $(r - 1)|E| \leq (m - 1)n(n - 1)$. Lower bound: split V into tails A , $|A| = \alpha$, and heads B . By [Lemma A.33](#) there is a simple $(r - 1)$ -uniform hypergraph $\mathcal{G}^*$ on B with maximum degree $m - 1$ and $\lfloor (m - 1)|B| / (r - 1) \rfloor$

edges. Give every tail $a \in A$ the hyperedges (a, H) , $H \in \mathcal{G}^*$. No vertex of B is a tail, so every route is a single step and $\lambda(a, b) = \deg_{\mathcal{G}^*}(b) \leq m - 1$, while pairs inside A or starting in B have no routes. $\square$

A.8 The random threshold

The probabilistic ingredients are two Chernoff bounds, quoted in the form we use them.

Theorem A.40 (Chernoff bounds, see [16, Cor. 4.4, Thm. 4.4]). *Let X be a sum of independent $\{0, 1\}$ -valued random variables with mean μ . For $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \exp(-\frac{\min(\delta, \delta^2)\mu}{4})$, and for $\delta \in (0, 1)$, $\Pr[X \leq (1 - \delta)\mu] \leq \exp(-\frac{\delta^2\mu}{2})$.*

Corollary A.41. *Let $X \sim \text{Binomial}(n - 1, p)$ with $p = cm/n$ for a constant $c \in (0, 1)$. Then for all sufficiently large m , $\Pr[X \geq m] \leq e^{-\alpha m}$ with $\alpha = \alpha(c) > 0$.*

Proof. The mean is $\mu = (n - 1)p \leq cm$, so $m \geq (1 + \delta)\mu$ with $\delta \geq \frac{1-c}{c} > 0$, and **Theorem A.40** applies with $\mu \geq cm/2$ for $n \geq 2$, giving $\alpha = \min(\frac{1-c}{c}, (\frac{1-c}{c})^2) \frac{c}{8}$. $\square$

Proof of Theorem 1.15. Recall the hypotheses: $m = m(n) \geq 2$ with $m/\ln n \rightarrow \infty$ and $m = o(n)$, and $p = cm/n$ for a constant $c > 0$.

Case $c > 1$. The edge count $|E|$ of $G(n, p)$ is binomial with mean $\mu = \binom{n}{2}p = \frac{cm(n-1)}{2}$. With $\delta = 1 - \frac{1}{c}$ the lower-tail bound of **Theorem A.40** gives

$$\Pr\left[|E| \leq \frac{m(n-1)}{2}\right] \leq \exp\left(-\frac{(c-1)^2 m(n-1)}{4c}\right) \rightarrow 0,$$

so with high probability $|E| > \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and the contrapositive of Mader's theorem (**Theorem 1.2**) forces a pair with $\lambda \geq m$. Hence $\Pr[\lambda^{\max} \geq m] \rightarrow 1$.

Case $c < 1$. Each degree is $\text{Binomial}(n - 1, p)$, so by **Corollary A.41** and a union bound over the n vertices,

$$\Pr[\Delta(G) \geq m] \leq n e^{-\alpha m} = e^{\ln n - \alpha m} \rightarrow 0,$$

since $m/\ln n \rightarrow \infty$. By the degree bound (**Lemma A.3**) $\lambda^{\max} \leq \Delta(G)$, so $\Pr[\lambda^{\max} \geq m] \rightarrow 0$. $\square$

Remark A.42 (The vertex and directed analogues). Below the threshold, $\kappa^{\max} \leq \lambda^{\max}$ gives the vertex statement for free, and in $D(n, p)$ the out-degree version of the same union bound applies. Above the threshold, the minimum degree exceeds m with high probability by the lower-tail bound. Mader's theorem alone does not carry this to the vertex problem, since $\kappa^{\max} \leq \lambda^{\max}$ runs the wrong way to deduce $\kappa^{\max} \geq m$ from the edge bound. What supplies it instead are the theorems of Bollobás [3] for $G(n, p)$ and its directed analogue [6], which state that in this regime the global connectivities equal the minimum degree (respectively minimum semi-degree) with high probability, so $\kappa^{\max}$ and the directed $\lambda^{\max}$ also cross m at $p^* = m/n$. The growth

hypothesis $m/\ln n \rightarrow \infty$ is what lets the union bound beat the factor n . For fixed m , the bound is vacuous and finer tools (Poisson approximation of low-degree vertex counts) would be needed.

A.9 Computational transcripts

The cut-counting optimisation formulation that certifies [Theorem 2.1](#) is given in [Chapter 2](#). Here are the solver transcripts that back it, together with the exhaustive-search output that settles the directed base cases. The complete commented source accompanies the document as a single self-contained file, `erdos915_unified.py` in the `program/` directory. Its core runs on `numpy` and `scipy` alone, the certifier adds `pulp`, and `matplotlib` and `networkx` are optional and used only for the figures. An optional compiled helper accelerates selected small hot paths but is not needed for correctness.

```

1 Cut-MILP certification of the directed multigraph problem
2 M*(n) is the scaled optimum; L_m^dir(n) = (m-1) M*(n).
3
4 n=3: status=OPTIMAL, M*=4.000, target=2(n-1)=4, proof=True, time=0.0s, L_3^
    ↳ dir(n)=8
5 n=4: status=OPTIMAL, M*=6.000, target=2(n-1)=6, proof=True, time=0.3s, L_3^
    ↳ dir(n)=12
6 n=5: status=OPTIMAL, M*=8.000, target=2(n-1)=8, proof=True, time=5.9s, L_3^
    ↳ dir(n)=16
7 n=6: status=OPTIMAL, M*=10.000, target=2(n-1)=10, proof=True, time=1259.9s,
    ↳ L_3^dir(n)=20

```

Listing A.1. Solver output: $M^*(n) = 2(n-1)$, optimal with zero gap, for $n = 3, 4, 5, 6$.

The cut-counting optimisation closes the gap up to $n = 5$ inside a short budget and reaches $n = 6$ in about twenty minutes, after which the branch count defeats it. The base cases of the directed induction ([Theorem 1.9](#)) run to $n = 7$, and those are settled instead by the pruned exhaustive search over all simple digraphs with $\lambda^{\max} \leq 1$. That search is exact and complete at these sizes, and it agrees with $\ell_2^{\text{dir}}(n) = M(n)$ throughout.

```

1 Exhaustive search over all simple digraphs with lambda^max <= 1 (directed,
    ↳ m=2).
2 Branch and bound; reported max arc count = ell_2^dir(n) = max(2(n-1), floor
    ↳ (n^2/4)).
3
4 n=2: max= 2, formula= 2, proved (complete), nodes=5, 0.0s
5 n=3: max= 4, formula= 4, proved (complete), nodes=22, 0.0s
6 n=4: max= 6, formula= 6, proved (complete), nodes=537, 0.0s
7 n=5: max= 8, formula= 8, proved (complete), nodes=17823, 0.1s
8 n=6: max=10, formula=10, proved (complete), nodes=788101, 9.1s
9 n=7: max=12, formula=12, proved (complete), nodes=45122129, 718.1s
10
11 The cut-MILP certifier of the multigraph problem closes n <= 6 (n = 6 in
    ↳ about
12 21 minutes); the directed m = 2 base cases through n = 7 that the induction

```

13 | requires are settled by this exhaustive search instead.

Listing A.2. Exhaustive-search output for the base cases $2 \leq n \leq 7$: the maximum arc count equals $M(n) = 2, 4, 6, 8, 10, 12$, proved complete at each n .

A.10 How the program checks itself

The invariants are checked by the self-test at the foot of `erdos915_unified.py`, run simply with `python erdos915_unified.py`. It confirms the checker against the proven $m = 2$ values 2, 4, 6, 8, the named constructions against their predicted sizes and connectivities, the Gomory–Hu shortcut against brute-force max-flow, the Berge connectivities of the hypergraph checker, the Monte Carlo appearance threshold and Whitney’s inequality on every sample, the cut-counting optima for small n , the search reaching the certified optimum, and the driver returning the proven value or a feasible witness in each mode. Every check passes. A single companion script, `make_figures.py`, regenerates every figure in the thesis from the same file, with the mapping listed in `program/README.md`.

Appendix B

Supplementary Figures

This appendix collects the supplementary figures that the program produces but that the chapters reference rather than reproduce in full. Each one is named where it belongs in the body, and the file name printed in its caption matches the output of `make_figures.py` and the figure table in `program/README.md`. Figures already shown in the running text are not repeated here.

B.1 A gallery of extremal graphs

The densest feasible graphs the program finds are not abstractions. [Figure B.1](#) draws a spread of them, one extremal graph per panel across the matrix variants, each annotated with its extremal value and the size of its automorphism group $|\text{Aut}|$, computed as $n!$ divided by the number of distinct labellings of that shape. A multiplicity above one is shown by a $\times k$ label rather than parallel lines, and directed arcs carry arrowheads. The shapes are exactly the named extremisers the body argues about, the tree and Mader graph for the simple undirected problem, the star at full multiplicity for the multigraph, the bidirected star for the simple directed problem, and the double star for the directed multigraph, recovered here by enumeration rather than by hand.

B.2 Search traces across the variants

[Figure 3.2](#) in [Chapter 3](#) follows one cooling run in detail. The runs below repeat that experiment for the other variants. Each plots every visited graph by its binding connectivity (horizontal) against its size (vertical), colouring each point by the step at which it was visited, from the hot early exploration in warm tones to the converged tail in blue. In every case the walk strays past the feasibility boundary while hot and is driven back onto the densest feasible graph as the temperature falls, exactly as in the body figure. The seeds are fixed, so each picture reproduces verbatim.

B.3 The variant landscape at $m = 6$ and in three dimensions

This appendix shows the per-graph connectivity distribution at the fixed forbidden threshold $m = 6$, alongside a three-dimensional view of the appearance threshold ([Figure B.8](#)). They confirm that raising the forbidden threshold relocates the feasibility boundary without changing the qualitative picture. The pooled pair-connectivity and edge-count grids of [Chapter 4](#) ([Figures 4.4](#) and [4.5](#)) already split each variant at its own mid-range connectivity, so they carry no separate

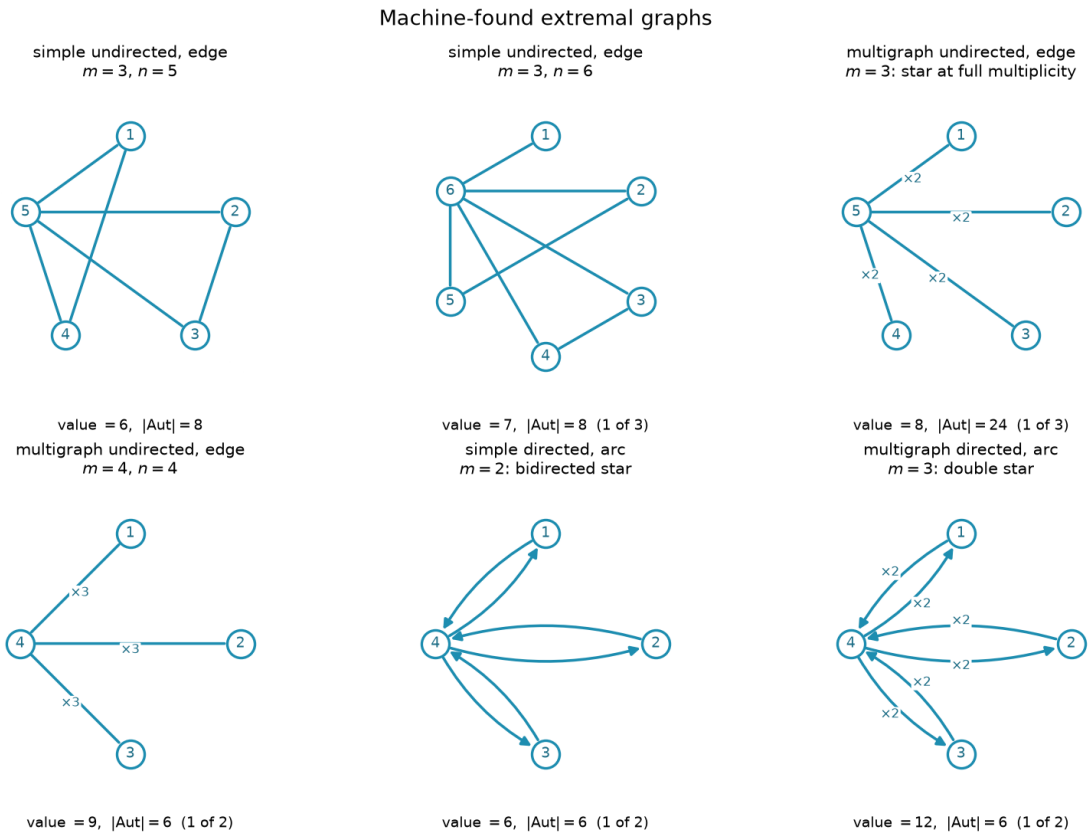


Figure B.1. Machine-found extremal graphs across the matrix variants. Each panel is one densest feasible graph at the stated n and m , drawn from the enumeration output, with its extremal value and automorphism count $|\text{Aut}|$ below it. Edge or arc multiplicities appear as $\times k$ labels, and a note “1 of k ” marks a variant with several non-isomorphic extremisers, of which one is drawn. File `extremal_graphs_gallery.png`.

Search trace: simple undirected, $n = 7, m = 3$

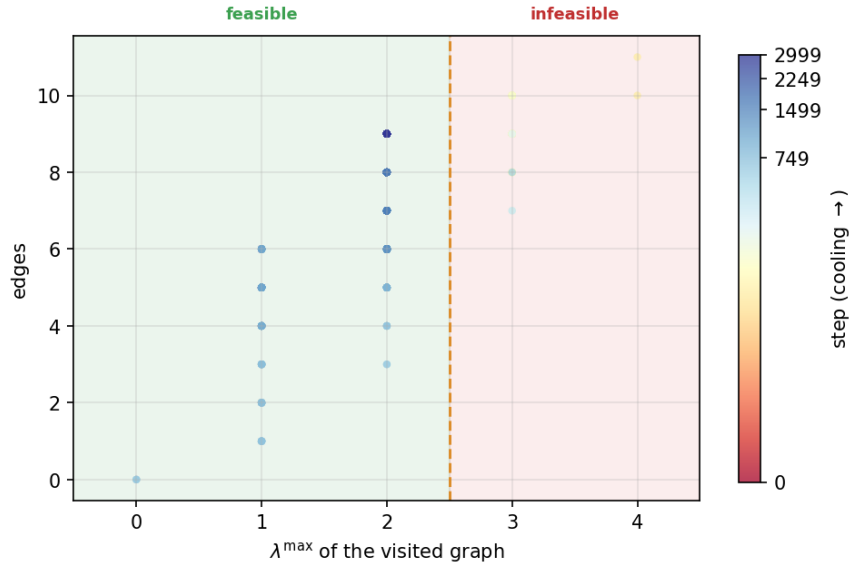


Figure B.2. Search trace for the simple undirected edge problem at $n = 7, m = 3$. File `trace_simple_undirected_n7_m3.png`.

Search trace: simple directed, $n = 5, m = 3$

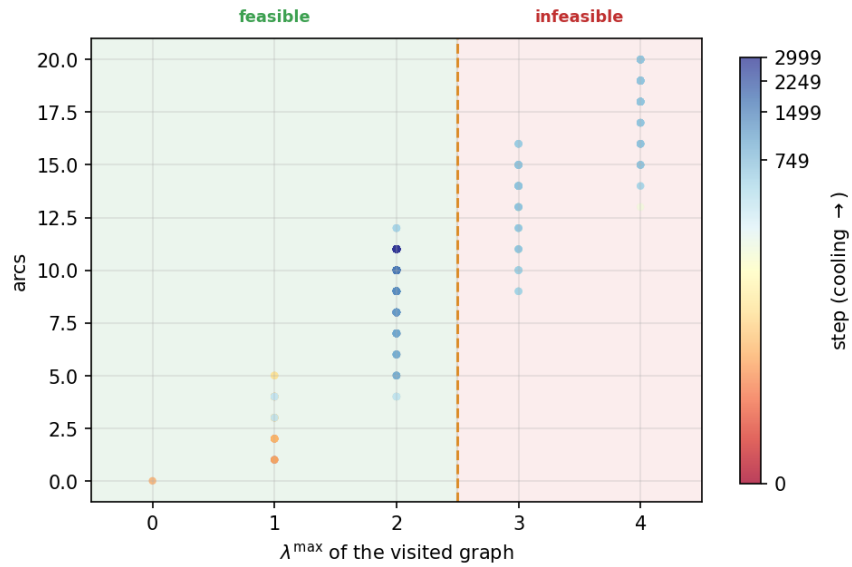


Figure B.3. Search trace for the simple directed arc problem at $n = 5, m = 3$. File `trace_simple_directed_n5_m3.png`.

Search trace: multi undirected, $n = 5$, $m = 3$

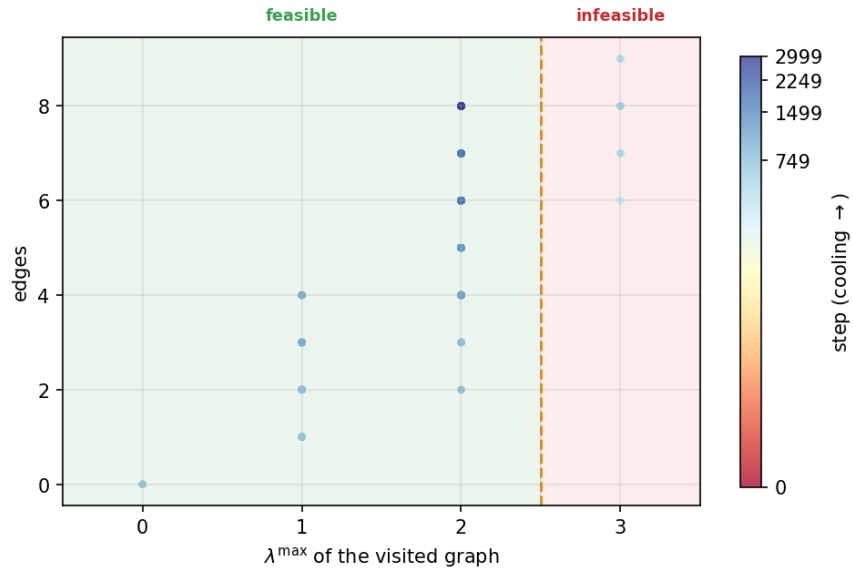


Figure B.4. Search trace for the multigraph undirected edge problem at $n = 5$, $m = 3$. File `trace_multi_undirected_n5_m3.png`.

Search trace: multi directed, $n = 5$, $m = 3$

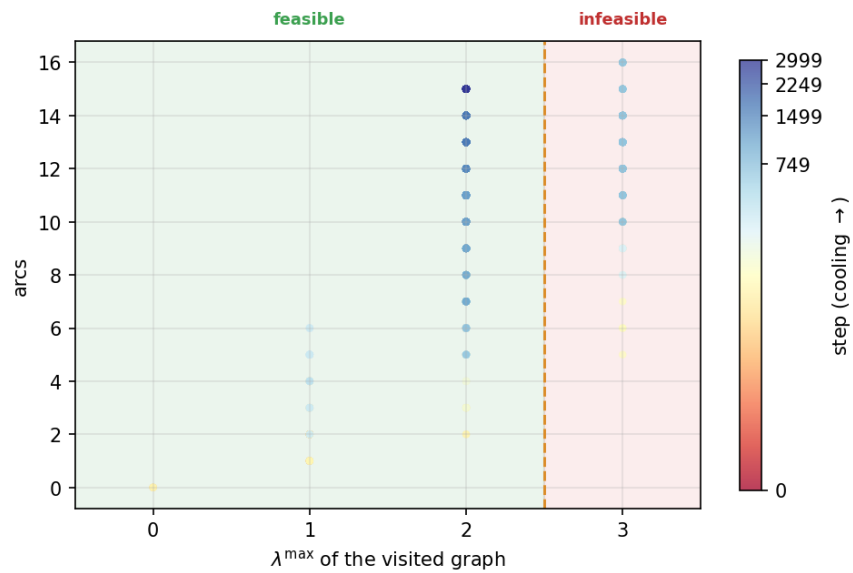


Figure B.5. Search trace for the multigraph directed arc problem at $n = 5$, $m = 3$. File `trace_multi_directed_n5_m3.png`.

Search trace: multi directed vertex-sep, $n = 4$, $m = 3$

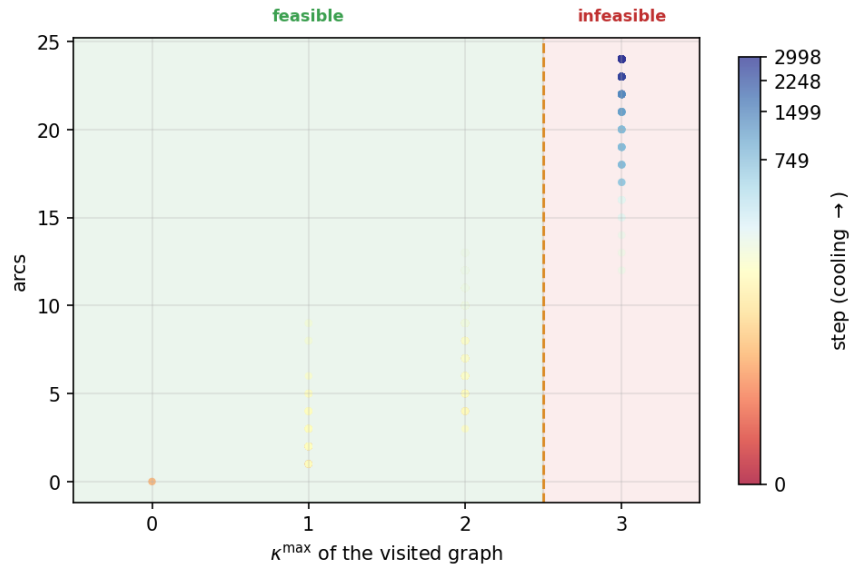


Figure B.6. Search trace for the multigraph directed *vertex*-disjoint problem at $n = 4$, $m = 3$, where the binding quantity is $\kappa^{\max}$. File `trace_multi_directed_n4_m3_vertex.png`.

$m = 6$ companion.

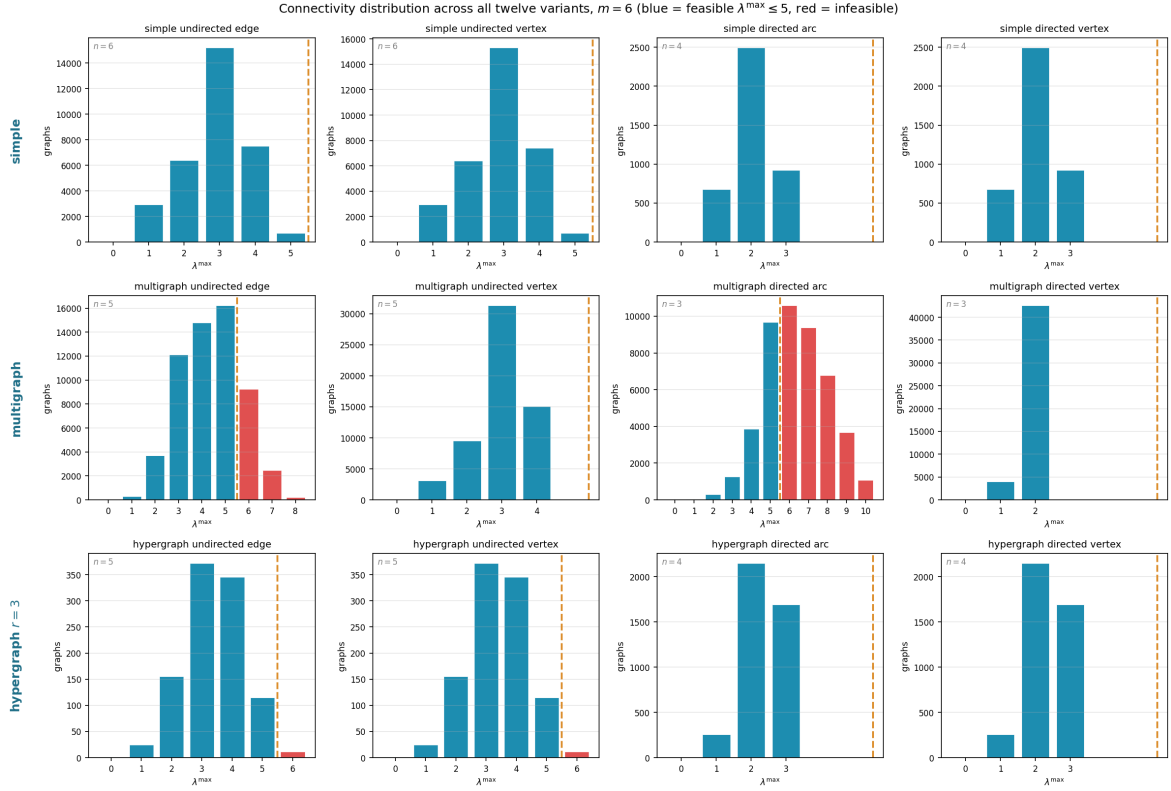


Figure B.7. How the full small- n enumeration distributes by binding connectivity $\lambda^{\max}$ across all twelve variants at the fixed forbidden threshold $m = 6$. Blue bars are feasible graphs ($\lambda^{\max} \leq 5$), red bars infeasible ones, and the dashed line marks the feasibility boundary. At this higher threshold most of the small graphs the enumeration reaches are feasible, so the blue mass dominates: raising m moves the boundary out past the bulk of the distribution. File `conn_dist_m6.png`.

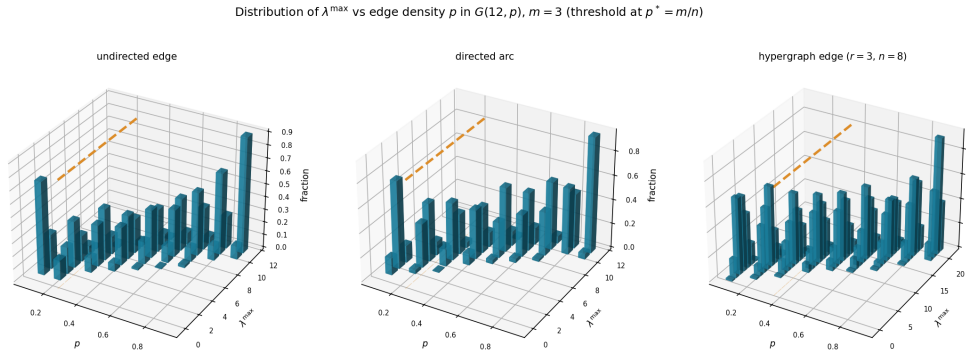


Figure B.8. The appearance threshold $p^* = m/n$ as a three-dimensional bar chart, for three representative variants (simple undirected edge, simple directed arc, three-uniform hypergraph edge) at $n = 12$, $m = 3$. Each panel plots the fraction of $G(n, p)$ samples (height) reaching each binding connectivity $\lambda^{\max}$ (depth) as the density p varies (width), so the mass migrates from low to high connectivity as p crosses the proved threshold $p^* = m/n$ of [Theorem 1.15](#). File `threshold_3d.png`.

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